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PROJECT DELIVERY REPORT

Controlling Differential Power Processors in Solar Powered Applications

THE DESIGN AND IMPLEMENTATION OF MODERN STATE-SPACE METHODS TO SERIAL PARTIAL POWER CONVERTERS

STUDENT ENGINEER

Jared Saunders
n11034459

SUPERVISOR

Assoc Prof Geoff Walker
School of Electrical Engineering & Robotics

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Executive Summary

This project designed, built and field-tested a controller for the ‘Power Shuffler’, and showed on a real solar string that it recovers power a shaded string would otherwise lose. Shading one panel in a series string limits the current of the whole string, and the bypass diodes used to protect it simply discard the shaded panel’s power. The shuffler is a serial partial power converter (S-PPC), a type of differential power processor, that instead moves the difference in power between series-connected modules to balance their voltages. A sister report covers the shuffler hardware; this report covers its control system, from the analytical model through to closed-loop validation on the bench and in the field.

Each shuffler is an interleaved buck-boost converter built from decoupling capacitors, a coupled inductor and two half-bridges. Each shuffler shares PV modules with its neighbours, so their dynamics are coupled and cannot be described by a set of single-input single-output transfer functions. The controller is therefore built on a multiple-input multiple-output state-space model with three states (PV voltage differential, output current and circulating current) and two inputs: common-mode duty (the average of the two half-bridge duties, which shuffles power between modules) and differential duty (their difference, which controls the circulating current). The development process followed a unique spin on a systems engineering approach. The analytical model set the controller architecture. A Simulink model of a four-module string validated these control loops, providing requirements drove the choice of sensors and microcontroller. The same C control core was then implemented inside a hardware-accurate QSPICE model, adding sensor noise and sampling delay, before going into the firmware unchanged. The final product is a single board with two shufflers, a hall effect current sensors and power monitoring voltage sensing and an STM32G474RE microcontroller, housed in a 3D-printed enclosure with a Raspberry Pi 4B that streams telemetry to a timeseries database.

On the bench (NGE103 supply, DL3021 electronic load) the closed loop held the module-voltage differences within $\pm 0.010\,\mathrm{V}$ and the circulating current within $\pm 0.07\,\mathrm{A}$ across 22 settled test segments at 200–400 kHz. The 44 Hz telemetry is too slow to resolve the step response on hardware, so it is taken from the simulation models. For a 1 V step at 30 % gain and 200 kHz, QSPICE gives a rise time of $6.5\,\mathrm{ms}$ and a 2 % settling time of $11.9\,\mathrm{ms}$ with no overshoot, corroborated by the analytical model ($7.6\,\mathrm{ms}$ and $13.6\,\mathrm{ms}$). Of the switching frequencies tested, 300 kHz was the best compromise; 400 kHz gave marginal gains at higher losses with almost no change in response from gain increases.

The most significant result is from the solar test: three 80 W panels without bypass diodes in series into a 61 V constant-voltage load. With the top module shaded to almost zero current (a shirt over about a third of the panel), a plain series string would have delivered no power at all. With the shufflers running, the string delivered up to $99\,\mathrm{W}$ during the shading event and $10.6\,\mathrm{Wh}$ over 36 minutes of testing, nearly all of it energy that would otherwise have been lost.

The main limitation is scope: one board with two shufflers that are hard-wired together, so they cannot be driven independently and boards cannot be stacked, which leaves scalability untested. The top-level INA239 (U10) does not work, so V_{M1} was inferred from the electronic load; the gains have to be re-derived in simulation whenever the switching frequency changes; and poor weather on the test day meant the closed loop was only partly exercised against real panel dynamics. The recommended next steps are a stackable board with independently driven shufflers, to test strings of more than three modules, and a study of other applications for S-PPCs that may offer more impact to industry than solar.

Abstract

Solar is one of the fastest-growing sources of electricity, yet a solar string is only as strong as its weakest panel. When a panel in a series string is shaded it limits the current of the whole string, and the bypass diodes used by industry to protect it simply skip the shaded cells and leave their energy on the table. To extract this energy a device coined the 'Power Shuffler' has been built: a bidirectional interleaved buck-boost converter that equalises the module voltages by moving current up and down the string. The complexity of modeling the dynamics of the shuffler stems from how physical states are coupled, not just by the coupled inductor itself but also through connection to neighbouring shufflers, so the controller is a multiple-input multiple-output state-space design, developed through a systems-engineering V-model: an analytical model, a high-level Simulink controller architecture, controller component selection, a combined controller-and-hardware simulation in QSPICE, and finally firmware tested on the bench and in the sun. Even in open loop at a fixed 50 % duty the shuffler returned power from a partially shaded string. In closed loop, a PI gain matrix at 10 % of the simulated gains held the module-voltage differences within $\pm 0.010\,\mathrm{V}$ on the bench, below the sensing noise, and simulation shows no overshoot at the deployed gains with 2 % settling in 31–41 ms (18–21 ms at 20 %). In the solar test, with one of three 80 W modules shaded to almost zero current, a plain series string would have delivered nothing; the shufflers delivered up to 99 W and 10.6 Wh over 36 minutes. The controller is stable and extracts power during partial shading events. The next step is a stackable board to test strings of more than three modules.

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1 Introduction

Photovoltaic (PV) solar is the fastest-growing energy source globally [1] and Australia's leading renewable resource. Rooftop and large-scale solar together supplied 19.6 % of Australia's electricity in 2024, more than wind (13.4 %) or hydro (5.5 %) [2]. Australia added 6.8 GW of renewables in 2025, and renewables now supply 43.7 % of generation in the National Electricity Market (NEM) [3]. To collect as much energy as possible, PV arrays are paired with maximum power point trackers (MPPTs), which adjust the power converter so that the array operates at its maximum power point (MPP). When a series string of PV modules is partially shaded, the bypass diodes of the shaded modules conduct, the MPP drops and the power–voltage curve develops multiple local maxima, turning MPPT into a global optimisation problem. The power shuffler addresses this by creating new paths for current to move up and down the string, equalising the module voltages so that the whole string is pinned to a single MPP. Because shufflers are partial power processors, they only need to be rated for the mismatch power rather than the full string power [4, 5]. This project revives QUT's earlier work on the shuffler [6].

The aim of this project is to design and implement a controller for the power shuffler. The research question that guided the work is: how can a controller capture the coupled dynamics of multiple interleaved buck-boost converters and command their currents so that the module voltages are balanced and maximum power is extracted despite shading?

The project has two parts. A sister report covers the electrical design of the shuffler, including the converter topology, QSPICE hardware simulations, circuit board design, manufacturing and hardware validation. In contrast, this report covers all aspects of the control system architecture: analytical solutions, Simulink models, controller component selection, gain refinement, firmware deployment and finally validation of the closed loop on the bench and out in the field. The final product is a single board carrying two shufflers, tested with a three-module string.

The report begins with a literature review of partial shading, shuffler concepts and their control (section 2), which identifies the gap this project addresses. The methodology (section 3) follows the design pipeline from power-system modelling to a fully integrated system working in the field. The raw results from real-time telemetry and simulation are then analysed and synthesised (section 4) to quantify the performance of the controller and the energy it returns, and to derive a dynamics model that applies to any string of N PV modules.

Objectives.

- Derive an analytical model of the interleaved buck-boost converter dynamics (section 3.3 and appendix C).
- Design a control system architecture (sections 3.4 and 3.6).
- Select the controller's sensors and microcontroller (section 3.5).
- Validate the controller experimentally on a real PV system (sections 4.5 and 4.6).
- Create a self-contained product that showcases the controller's autonomy (section 3.7).

2 Literature Review

This review moves from the problem to the gap in four steps: how partial shading defeats a conventional maximum power point tracker (section 2.1), the differential power processing converters proposed to recover the lost energy (section 2.2), how converters of this kind have been controlled (section 2.3), the modelling, simulation and test methods used to develop them (section 2.4), and the gap this project fills (section 2.5).

2.1 Partial shading and MPPT

At its core a solar controller is a maximum power point tracker (MPPT): a buck or boost stage whose duty is adjusted so that the array operates at its point of highest power. The MPPT literature is mature. Esram and Chapman catalogue 19 tracking methods [7], and Sarvi and Azadian update the field into classical, bio-inspired and machine-learning classes [8]. Most of these methods assume that the power–voltage (P–V) curve of the array has a single peak.

Partial shading breaks that assumption [9, 10, 11]. Every module in a series string carries the same current, so a shaded module either limits the string current to what it can produce or, once its bypass diode conducts, drops out of the string altogether. Each combination of conducting and bypassed modules gives the P–V curve its own peak. Figure 1 shows this for the four-module string model built in Semester 1: with one module at 500 W/m^2 the string offers 1656 W with that module bypassed or about 1240 W with all four conducting, and under graduated shading four separate peaks appear, with the global maximum of 914 W at only 78 V. A conventional perturb-and-observe (P&O) tracker climbs whichever peak it starts nearest and is trapped there [12]. Even a tracker that finds the global maximum leaves energy behind: table 1 compares each string's global maximum with the sum of its modules' individual maximum power points, and the mismatch loss reaches 34 % under graduated shading.

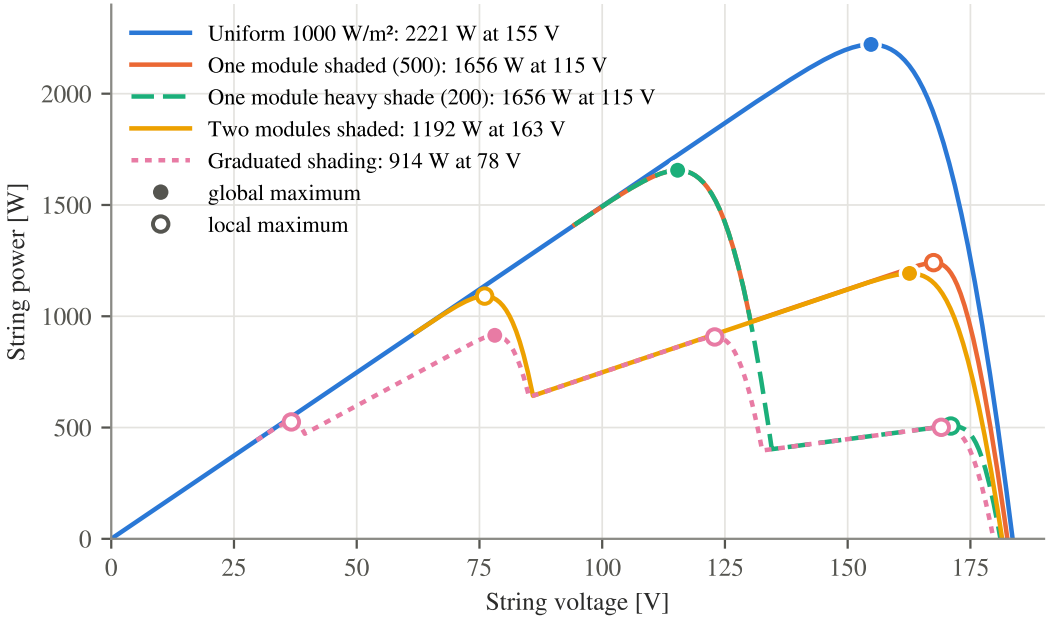


Figure 1: Why partial shading defeats a string MPPT: P–V curves of a four-module string (AIKO Neostar A555, single-diode model with bypass diodes) under five irradiance patterns. Filled markers are the global maxima, hollow markers the local maxima a hill-climbing MPPT can be trapped on (Semester 1 string model, regenerated by `tools/make_pv_shading.py`).

Array reconfiguration (series-parallel, bridge-linked and total-cross-tied) reduces this mismatch loss but cannot remove it [13]. Bypass diodes protect the shaded cells, but at the cost of a multi-peaked

Table 1: Mismatch loss of the shaded strings in fig. 1: string global maximum against the sum of the modules' individual maxima (the power a lossless per-module tracker could extract).

Scenario	Irradiance [W/m^2]	Module MPPs [W]	String MPP [W]	Loss
Uniform	1000, 1000, 1000, 1000	2221	2221	0.0 %
One module shaded	1000, 1000, 1000, 500	1944	1656	14.8 %
One module heavy shade	1000, 1000, 1000, 200	1775	1656	6.7 %
Two modules shaded	1000, 1000, 500, 500	1667	1193	28.5 %
Graduated shading	1000, 800, 500, 200	1388	914	34.1 %

P–V curve and the energy of every bypassed module. Recovering that energy requires moving power between the modules themselves.

2.2 Existing shuffler concepts: DPP and S-PPC

The power shuffler is part of the family of differential power processors (DPP) [4], whose reason for being is to move differences in power around a system, effectively balancing out either voltage or current. The serial partial power converter (S-PPC) specialises in moving the power differentials of modules that are connected in series [14].

The idea has a long history at QUT: Walker, Xue and Sernia first proposed per-module "shuffling" converters that let each module in a string operate at its own maximum power point [6]. Hui's later QUT work documents why this matters outside the lab [15]. Shufflers of a very different design, bearing little resemblance to this one, were fitted to ATN's solar car *Priscilla* for the 2019 Bridgestone World Solar Challenge, where the curvature of the 1.25 kW array and obstacles passing overhead shade it unevenly while driving, and every watt lost to mismatch counts in the race. Rushed into the car under tight timelines with limited testing, the devices did not perform reliably and ultimately failed during the challenge, and the follow-on work was redirected to making them operate safely and reliably. It is a clear reminder that protection and safety features have to be designed into a shuffler from the start (section 3.8). Kasper et al. classify panel-integrated DC–DC concepts and identify the buck-boost as the most promising [4, 16]. Olalla et al. formalise the modelling of DPP architectures and show near-ideal energy extraction with converter ratings that scale with the mismatch rather than the string power [17, 5]. Jeong et al. review DPP architectures [18] and demonstrate a four-unit segmented DPP experimentally [14]; Li and Cobos give a framework for comparing the VA ratings of partial power converters [19]; and dos Santos et al. present a practical two-switch S-PPC [20]. A common thread runs through these works: their focus is the power stage and the architecture, and the control is either not discussed or kept simple and classical, with a PI loop on a single unit.

The S-PPC used in this project is the interleaved buck-boost converter: two half-bridge legs joined through a coupled inductor to the midpoint between two series modules (fig. 3). At a duty of $D = 0.5$ the midpoint sits at half the pair voltage; moving D away from 0.5 moves current from one module to the other, and in steady state the current circulating between the two phases should be zero.

2.3 Controlling the interleaved buck-boost

A single buck-boost converter is described by one second-order ordinary differential equation (ODE), and classical frequency-domain design works well for it. Interleaving changes this: the second phase and the coupled inductance couple the states, so they cannot be controlled independently [21]. Kabalci and Boyar add an extended Kalman filter to an interleaved boost converter to handle these coupled states [22], and the behaviour of coupled inductors in interleaved converters is treated in [23, 24].

State-space methods handle this coupling naturally. Batzelis et al. give a six-state representation of irradiance-driven PV dynamics that includes the boost converter, the PI integrator and the MPPT

reference [25]; this is the modelling style adopted here, extended with the shuffler's winding dynamics. Other state-space MPPT work reports that LQR and LQR-PI controllers outperform P&O [26, 27, 28].

2.4 Modelling, simulation and test methods

Switching converters are normally designed for control with averaged models. State-space averaging replaces the switched circuit with a continuous-time model whose inputs are the duty cycles [29], and Kassakian et al. develop averaged and linearised models of the buck-boost family as the basis for its control design [30]. Averaging discards the switching ripple, so the model is valid only well below the switching frequency, and it says nothing about dead time, sampling delay or sensor noise.

The simulation tools follow from those two limits. Simulink is the standard environment for model-based design: the controller and an averaged plant run together in one model, which makes it the natural tool for model-in-the-loop (MIL) verification on the left of the V-model [31, 32]. QSPICE is a switching-level SPICE simulator with built-in C++ compilation, so compiled C code can run as a block inside the circuit [33]. That allows the firmware's own control core to be tested against real switches, drivers, dead time and sensing models before it reaches hardware, closing the gap the averaged model leaves.

For experimental testing, MPPT performance is standardised: EN 50530 defines dynamic test sequences of irradiance ramps for measuring tracking efficiency, and Li et al. use them to compare MPPT algorithms experimentally [34]. Riquelme-Domínguez and Martínez evaluate MPPT algorithms on state-space models under several such dynamic test procedures [12]. These procedures need a programmable PV simulator. This project draws on the same approach, repeatable steps on the bench with programmable supplies and an electronic load followed by validation on a real string outdoors, but a full EN 50530 campaign is outside its scope.

2.5 The gap

Past controllers have been able to rely on classical methods such as PI because each controls a single converter. Multiple-input multiple-output (MIMO) control of a string of coupled shufflers has had very limited exploration, and control of the interleaved buck-boost converter is itself a niche topic. To the author's knowledge, no published work models an N-module string of interleaved buck-boost DPP converters as one MIMO plant, identifies its DC gain matrix and closes a decoupling integral controller on real hardware, with the module-voltage differences and circulating currents as outputs. This project does that, carrying the design through analytical, Simulink, QSPICE, firmware, bench and solar phases, and reports the same metrics for every phase on one set of test cases.

3 Methodology

3.1 Controller Design Pipeline

The methods applied to the project correspond to different stages of the V-model as seen in fig. 2. It is how systems engineers prefer to map out the development throughout a product's lifecycle [32]. As a result of the literature review the research problem has been identified and the research question articulated. The first step in the design process is to determine the target use-case and application of the product through background research resulting in clear objectives. To determine a set of system requirements for a controller the plant is modelled. Then the controller algorithms are modelled in Simulink. Based on this abstract model components can be selected for a controller PCB. Knowing the controller architecture, especially the microcontroller and sensors, opens the door for firmware development. The hardware and controller SIL simulations can be combined together to provide a theoretical approximation of the experiments. Testing the system in the electronics lab. Testing outdoors with a string of 3 PV modules is the core objective and ultimate validation of the system. Table 2 outlines the connections between the overarching methodology for this project and conventional

industry standards.

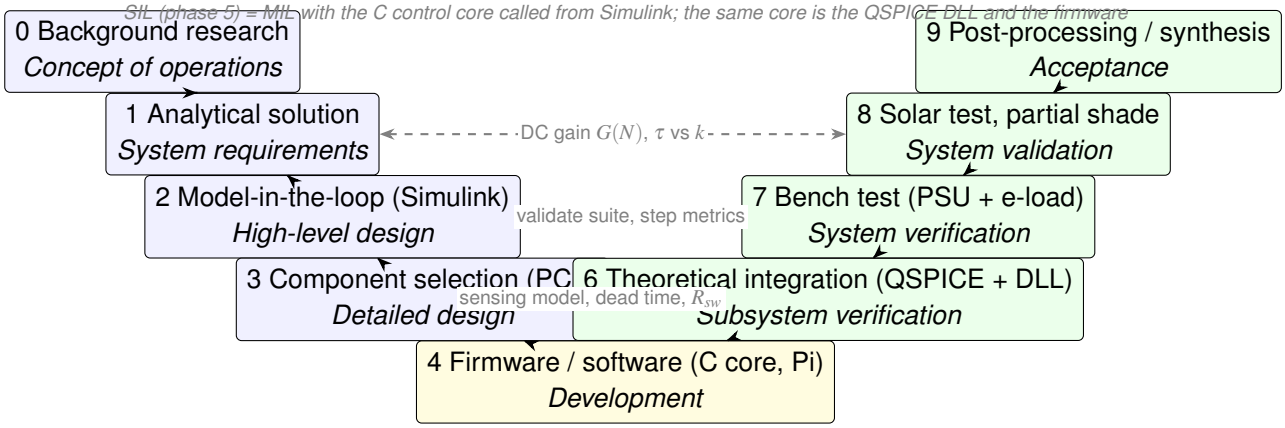


Figure 2: Controller design pipeline mapped onto the systems-engineering V-model. Dashed links are the verification pairs used in section 4.7.

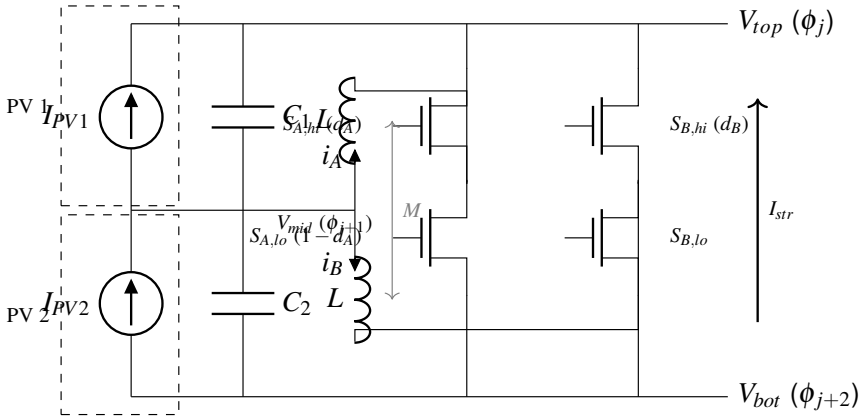
Table 2: Development phases and their V-model counterpart.

Phase	Pipeline	V-model	Primary artefact
0	Background research	Concept of operations	proposal, literature review
1	Analytical solution	System requirements	`nstring.py`, notebooks, appendix C
2	Model-in-the-loop	High-level design	`power_shuffler_sim.slx` (4 modules)
3	Component selection	Detailed design	KiCad Version 0 PCB, BOM
4	Firmware / software	Development	`6_RTOS` (C), `7_Pi` (Python, Docker)
5	Software-in-the-loop	Unit verification	C Caller blocks, golden replay
6	Theoretical integration	Subsystem verification	QSPICE integrated model + control DLL
7	Bench test	System verification	16 Oct 2026, 9 windows (table 23)
8	Solar test, partial shade	System validation	20 Oct 2026, 11 windows (table 23)
9	Post-processing / synthesis	Acceptance	`8_Results/analysis` (`make all`)

- ▷ Why this method fits: each phase has a pass/fail artefact that the next phase consumes (model → gains → params.h → bench), so faults are isolated at the level they are introduced; the C control core is the *same source file* in SIL, QSPICE and firmware, so the thing simulated is the thing deployed.
- ▷ Stakeholders: weekly supervisor meetings (Assoc Prof Walker); hardware teammate Son Nguyen owns the power stage, QSPICE schematic and PCB layout; the hand-over interface is the QSPICE schematic and the sensor set; two-person division control vs power electronics.
- ▷ Data collection: every bench/solar run is telemetry at 44 Hz into InfluxDB plus the e-load log at 5 Hz; Grafana screenshots define the test windows; the analysis re-exports the windows and computes every metric from the same CSVs (section 4.7).

3.2 Background

- ▷ Basic function: the midpoint of the coupled inductor is tied to the node between the two modules; each half-bridge switches its winding between the top and bottom rails at duty d_A, d_B. The *shuffled* current i_m (sum of the windings) moves charge between the two modules; the *circulating* current i_c (difference) flows only around the two legs and is pure loss, so it is regulated to zero.
- ▷ Target system: 2 shufflers, one controller, a telemetry system and user interface, the foundation for scaling, 3 PV modules in a string.
- ▷ Early designs were rated to the highest-rated commercial panel (AIKO Neostar 555 W, V_{mp} 38.7 V, I_{mp} 14.35 A) so the controller would not be the limiting element; the carpark experiment used 80 W KAUSSP80 panels for safety and availability, and the models carry both parameter sets (PV80 / BENCH / MIL-4).

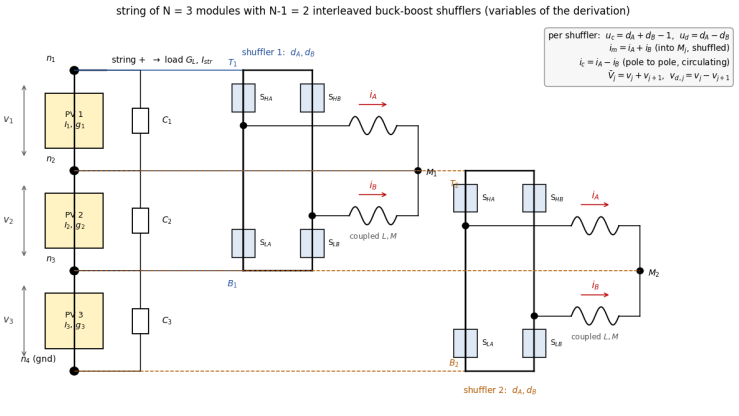


Averaged variables: $u_c = d_A + d_B - 1$ (shuffle), $u_d = d_A - d_B$ (circulate); $i_m = i_A + i_B$ shuffled, $i_c = i_A - i_B$ circulating.
Equilibrium: $d_A = d_B = D$, $V_{C2}/V_{C1} = D/(1-D)$; at $D = 0.5$ the midpoint sits at half the pair voltage.

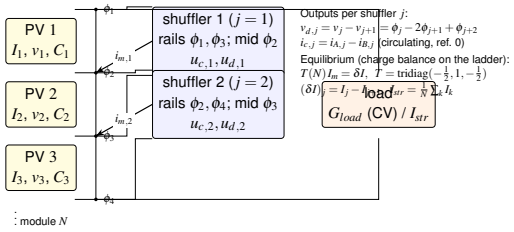
Figure 3: One interleaved buck-boost power shuffler between two series PV modules (phase A and phase B half-bridges on a coupled inductor; C_1, C_2 decouple the modules from the string).

▷ Control variables that make the problem MIMO: per shuffler $u_c = d_A + d_B - 1$ and $u_d = d_A - d_B$; outputs $v_d = V_{M,j} - V_{M,j+1}$ and i_c. With $N-1$ shufflers every u_c moves two module voltages, so the plant is a coupled ladder (fig. 4).

3.3 Analytical Solution



(a) Derivation drawing: $N = 3$ modules, $N-1 = 2$ shufflers, node potentials ϕ_k, per-shuffler variables.



(b) Ladder abstraction used for the equilibrium and the DC gain.

Figure 4: The N-module problem. Key physical parameters: $L \pm M$ (winding inductance in each mode), R_m, R_c (mode resistances incl. the switching loss R_{sw}), C_k (module decoupling), G_{load} (CV load conductance).

Single shuffler (Semester 1 model, appendix C.1)

▷ State $x = [I_{L1}, I_{L2}, V_{C1}, V_{C2}]^T$, control $u = [d_A, d_B]^T$, exogenous $[I_{PV1}, I_{PV2}, I_{load}]^T$; 9 valid switch states, 4 dual-switch states averaged with dwell fractions $d_A d_B$, $d_A(1-d_B)$, $(1-d_A)d_B$, $(1-d_A)(1-d_B)$.
▷ Equilibrium: $D_A = D_B = D$, $\bar{V}_{C2}/\bar{V}_{C1} = D/(1-D)$, $D = (I_{load} - I_{PV1})/(I_{load} - I_{PV1} + I_{PV2})$.
▷ Stability: without parasitics $\sigma(\tilde{A}) = \{0, 0, \pm j\omega_0\}$, $\omega_0^2 = 2D^2/(LC_1) + 2(1-D)^2/(LC_2) \Rightarrow 5.56\,\mathrm{kHz}$ undamped resonance; with the coupled inductor and winding resistance $-67 \pm j27\,000\,\mathrm{s}^{-1}$ ($\zeta \approx 0.0025$). This is why the early classical designs did not work.

- ▷ Controllability: differential mode needs $V_\Sigma \neq 0$ (active bus), common mode needs $\bar{I} \neq 0$ (no DCM), duty unsaturated. Observability: module voltages alone give rank 3 of 4 (the differential current is hidden) $\Rightarrow$ per-phase current sensing is mandatory, which fixed the sensor set (section 3.5).
- ▷ Change of thinking: three cascaded SISO loops treated the states as independent; the coupling makes it a MIMO problem (LQR + integral regulator first, then the decoupling PI that was deployed).

N-module string model (deployed, appendix C.2)

With $x = [\phi, i_m, i_c]$, $u = [u_c, u_d]$, $y = [v_d, i_c]$ the averaged model is three equations per shuffler j and one nodal network:

$$(L-M)\,\dot{i}_{m,j} = \bar{V}_j\,u_{c,j} + v_{d,j} - R_m\,i_{m,j} \tag{1}$$

$$(L+M)\,\dot{i}_{c,j} = \bar{V}_j\,u_{d,j} - R_c\,i_{c,j} \tag{2}$$

$$C_n\,\dot{\phi} = -G_n\,\phi + S\,i_m + w, \qquad v_{d,j} = \phi_j - 2\phi_{j+1} + \phi_{j+2} \tag{3}$$

- ▷ Equilibrium: $D_j/(1-D_j) = v_{j+1}/v_j$; each shuffler carries a weighted sum of all mismatches, $T(N)\,I_m = \delta I$ with $T = \mathrm{tridiag}(-\frac{1}{2}, 1, -\frac{1}{2})$ and $T^{-1}_{jk} = \frac{2}{N}\min(j,k)\,(N - \max(j,k))$; $I_{str} = \frac{1}{N}\sum_k I_k$; open loop at $D = 0.5$: $v_{d,j} = -R_m i_{m,j}$.
- ▷ Difference between the modes (call them the *voltage-control* mode $u_c \to v_d$ through $L - M = 2.05\,\mu$H, and the *current-control* mode $u_d \to i_c$ through $L + M = 13.75\,\mu$H): they are decoupled to first order, which is what the diagonal structure of $G(N)$ exploits.
- ▷ DC gain identification (closed loop): $G(N) = \begin{bmatrix} -(R_m I + Z_e)^{-1} Z_e\,\mathrm{diag}(\bar{V}_j) & 0 \\ 0 & \mathrm{diag}(\bar{V}_j/R_c) \end{bmatrix}$, $Z_e = -D_v G_n^{-1} S$; the voltage block is $\approx -\bar{V}$ on the diagonal with $O(R_m g)$ neighbour terms.
- ▷ Deployed controller: $M = G^{-1}\mathrm{diag}(k_I)$, $u = k\,(M_p e + M\xi)$, $\xi \mathrel{+}= T_s e$, $M_p = M\,\mathrm{diag}(k_{p,v}, k_{p,i})$ with $k_{I,v} = 2\pi \cdot 150$, $k_{I,i} = 2\pi \cdot 400$, $T_s = 100\,\mu$s, $f_c = 2$kHz; design time constants $\tau_v = 1/(k\,k_{I,v})$, $\tau_i = 1/(k\,k_{I,i})$: at $k = 1$ $\tau_v = 1.06$ ms, $\tau_i = 0.40$ ms; at $k = 0.3$ 3.5 / 1.3 ms; at $k = 0.1$ 10.6 / 4.0 ms.
- ▷ The gain scale k (`kscale`) is the single run-time knob: it scales bandwidth without changing the decoupling.
- ▷ Where to put phase portraits, step response and root locus: keep one figure here (fig. 5) and the rest in section 4.2 and appendix C. **[note]** Your call; the body is page-limited.

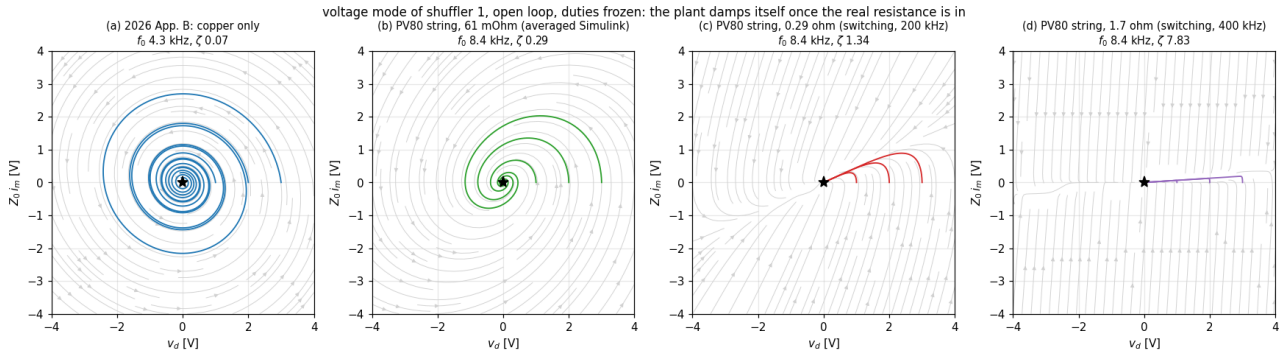
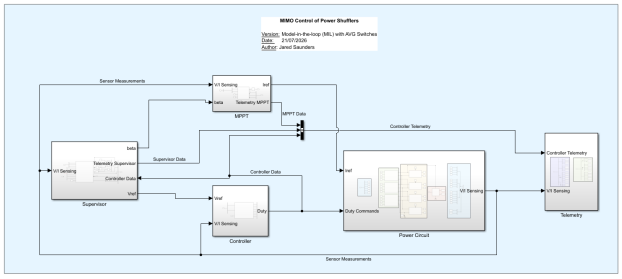


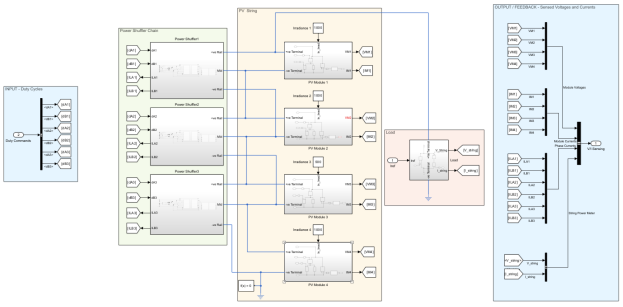
Figure 5: Open-loop phase portraits of the voltage mode in the $(v_d, Z_0 i_m)$ plane: (a) 2026 App. B copper-only model ($\zeta \approx 0.07$), (b) PV80 plant with the 61 mΩ averaged resistance (focus, $\zeta \approx 0.3$, 7–9 kHz), (c) 0.29 Ω switching plant at 200 kHz and (d) 1.7 Ω at 400 kHz (over-damped nodes). The plant damps itself once the real switching resistance is included.

3.4 Model-in-the-Loop (MIL)

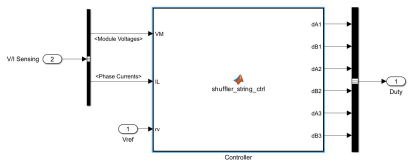
- ▷ Model: 4 AIKO-class modules (single-diode Simscape solar cells), 3 shufflers with *averaged switches* (CVS $= d\,V_{bus}$, CCS $= d\,i_L$), L_{self} 7.9 μH, M 5.85 μH, R_w 1.85 mΩ, R_{phase} 61 mΩ, C 100 μF; controlled current-sink load driven by the MPPT I_{ref}; T_s 50 μs; supervisor and MPPT at 1 ms; variable-step solver, max step 100 μs.



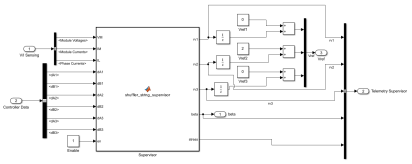
(a) Top level: Controller, Supervisor, MPPT, Power Circuit, Telemetry.



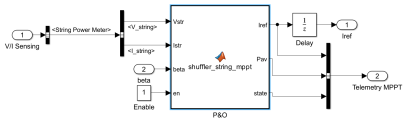
(b) Inside the plant: 4 modules, 3 averaged-switch shufflers.



(c) Controller block.



(d) Supervisor block.



(e) MPPT (P&O) block.

Figure 6: Simulink MIL model (`4_Simulink/1_MIL_Model_In_The_Loop`).

- ▷ Pseudocode for controller, supervisor and MPPT: appendix D (Listings 1–4); keep the description here to three sentences each.
- ▷ Validation: the 7-test suite (uniform, decoupling, half shade, shade with references, deep shade, MPPT uniform, MPPT shade) with pass criteria $|v_d - r_v| < 0.25\,\text{V}$, $|i_c| < 1\,\text{A}$, MPPT $\geq 98.5\,\%$ of the ceiling, $|i_{diff}| \leq 10\,\text{A}$ in protection; all 7 PASS (fig. 29, appendix G).
- ▷ SIL is simply an extension of MIL: the MATLAB Function blocks are replaced by C Caller blocks running `6_RTOS/ctrl_core`; a bit-identical double run ("golden replay") is the acceptance test. This is the cornerstone of the gain refinement pipeline because the C that passes here is what the QSPICE DLL and the firmware compile.

3.5 Component Selection

- ▷ High-level control data flow (fig. 7): user ↔ Pi (SSH, Grafana, params page, IR) ↔ Nucleo (UART telemetry/commands) → PWM → plant; sensors back to the Nucleo (ADC, SPI); the Pi never sits in the control loop.
- ▷ Microcontroller: STM32G474 chosen for its very high clock (170 MHz), price, 5 ADCs with 12-bit resolution, the HRTIM (184 ps resolution, hardware dead time, complementary outputs) and many timers; ADC and PWM functionality were the driving factors. STM32CubeMX was used to allocate peripherals (table 3).
- ▷ INA239 chosen because the INA229 (better ADC resolution) was out of stock; both share one SPI bus, so the voltage sensing is three devices on SPI3 with separate chip selects. Shunt inputs are tied off: only bus voltage and die temperature are used.
- ▷ TMCS1126 hall sensor: galvanically isolated, 500 kHz bandwidth, 100 mV/A, ± precision; safe to put in series with the windings.
- ▷ Raspberry Pi 4B: the safest way to add telemetry, remote control, state display, connectivity and 5 V/3V3 distribution without adding complexity to the PCB; it hosts the web services and flashes the Nucleo, so the laptop never plugs into the system.
- ▷ Routing choices (fig. 8): ADC traces take the most direct path and are via'd to the bottom layer to stay away from the switching nodes; SPI forms a bus up and down the board; PWM and chip selects went wherever convenient. Show the KiCad drawing if space allows.
- ▷ **[note]** U10 (top INA239, CS3) sits behind an ISO7762 wired backwards on Version 0, so V_{M1} is taken from the e-load readback ($V_{str,pi} - V_{M2} - V_{M3}$) – state this as a known hardware limitation here, not as a surprise in the results.

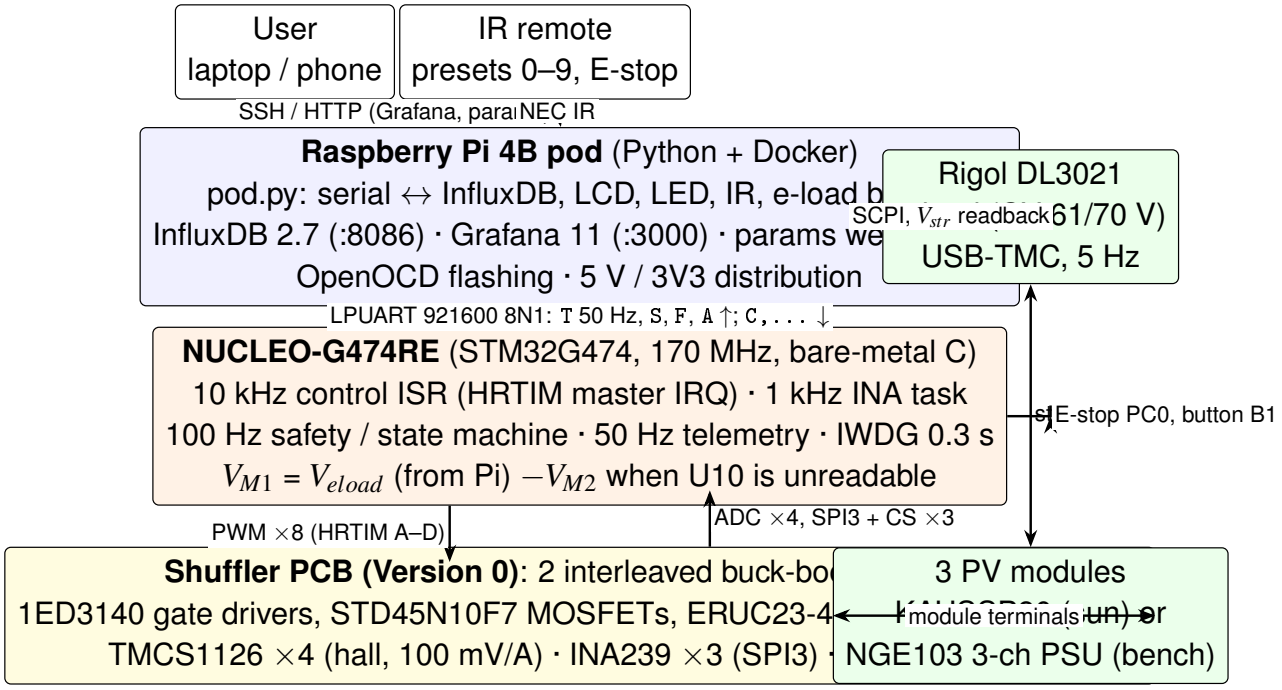
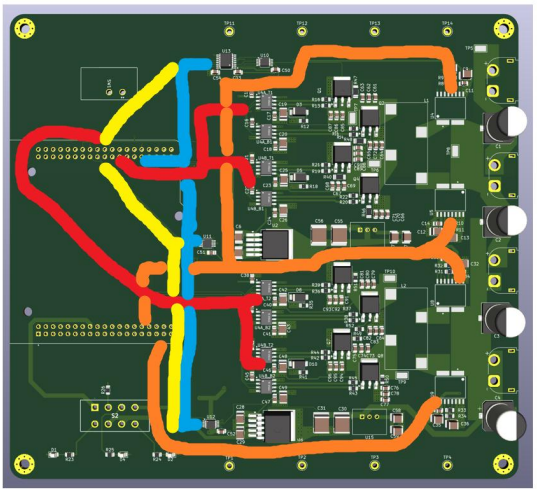
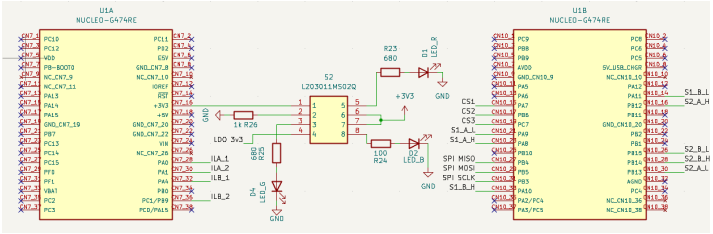


Figure 7: System architecture and data flow: the Pi pod is the user interface, logger and flasher; the Nucleo closes the loop; the PCB carries the power stage and sensors.



(a) Routing on the Version 0 board: PWM (red), SPI (blue), chip selects (yellow), ADC (orange), top and bottom layers.



(b) Nucleo header pinout used by the PCB.

Figure 8: Controller selection and routing (microcontroller NUCLEO-G474RE [35], hall sensor TMCS1126 [36], power monitor INA239 [37], microcomputer Raspberry Pi 4B [38]).

Table 3: Peripheral configuration summary (from the firmware; full detail in appendix E).

Peripheral	Role	Configuration
HRTIM1 A–D	8 PWM outputs, 4 half-bridge legs	5.44 GHz base; PER 27200 / 18133 / 13600 for 200 / 300 / 400 kHz; B legs reset at PER/2 (180°); dead time 100 ns; duty = CMP1 preloaded; pins PA8–11, PB12–15
ADC1, ADC2 + DMA1	4 hall currents	12-bit, 6.5-cycle sample, 4 HRTIM triggers per period (PER/8 + kPER/4), 160-sample circular DMA = one 100 μs tick; 80-sample box average per channel
SPI3 + CS PA7/PB6/PC7	3 × INA239 (bus V, die T)	1.33 MHz mode 1, 1 kHz round robin, 3.125 mV/LSB
LPUART1 (ST-LINK VCP)	Telemetry / commands to the Pi	921600 8N1, `T` frame at 50 Hz, 1-byte RX IRQ ring buffer
HRTIM master IRQ	Control tick	repetition every 20/30/40 periods ⇒ 10 kHz ISR, NVIC priority 1
EXTI0 (PC0), B1	E-stop, enable toggle	falling edge, priority 0
IWDG	Watchdog	0.3 s, kicked by the 100 Hz safety task

3.6 Gain Refinement Pipeline

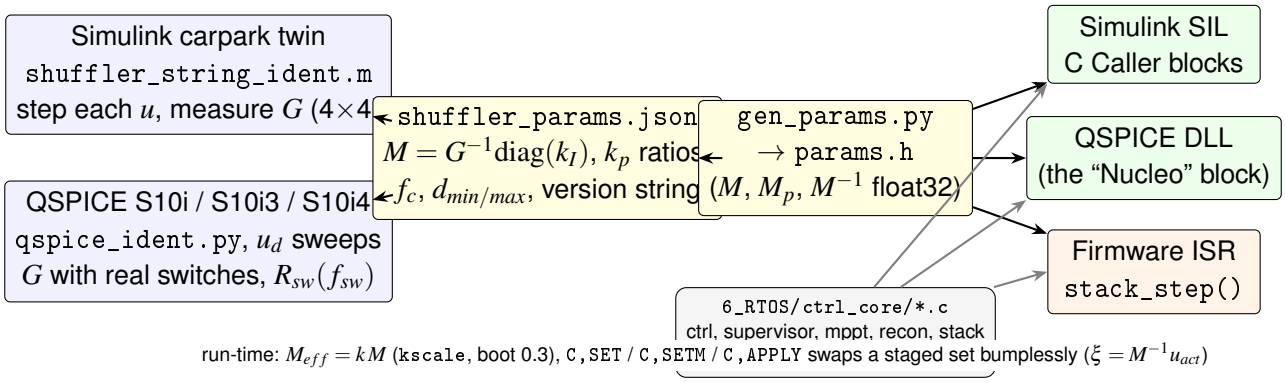
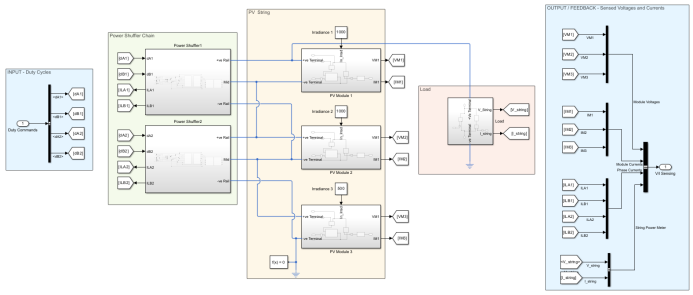


Figure 9: Gain pipeline: identification in Simulink or QSPICE → `shuffler_params.json` → `params.h` → the same C core in SIL, the QSPICE DLL and the firmware.

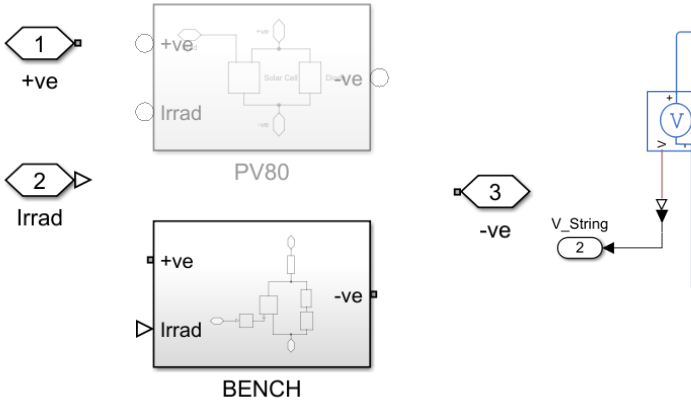
- ▷ Simulink was updated to reflect the experiments: power sources (NGE103 channels or KAUSSP80 cells), DL3021 CV load, plant reduced to 3 modules, a sensing block injecting INA round-robin (1 kHz, 3.125 mV LSB), 12-bit hall ADC, optional noise (2 mV / 20 mA rms) and the Pi-side V_{M1} path at 2 Hz.
- ▷ The control architecture was integrated into the hardware teammate's QSPICE: a controller component wraps the *unchanged* C core as a DLL (sampling at 10 kHz, INA hold at 1 kHz, hall quantisation) and drives HRTIM-style comparators; sensors are modelled (TMCS1126 500 kHz pole, reversed B phases).
- ▷ Gains produced by Simulink vs QSPICE (table 4): the voltage block agrees ($v_d/u_c \approx -46$ V per unit on both), but the current block differs by 5× (742 vs 142 A per unit at 200 kHz) because the switching plant has an effective resistance R_{sw} from pole-capacitance commutation in the dead time, $R_{sw} = 64 C_p f_{sw}^3 (L+M)^2$ ⇒ 0.29 / 0.80 / 1.70 Ω at 200 / 300 / 400 kHz. The Simulink-bench set was therefore rejected for hardware; QSPICE is the arbiter, Simulink the source.
- ▷ How to run the sims and change the switching frequency: `run_scenarios.m` (Simulink) and `run_scenarios.py` (QSPICE) take setup × kscale; f_{sw} is a knob (Tsw) in QSPICE and a compile define (`HW_PWM_KHZ`) in firmware; a new f_{sw} needs a new identification (S10i3 / S10i4) and gain set (appendix F).
- ▷ How Simulink, QSPICE and firmware share scripts: one JSON, one `params.h`, one `ctrl_core`; version strings travel in the telemetry (`pver`) so every logged sample knows its gain set.

3.7 Firmware / Software

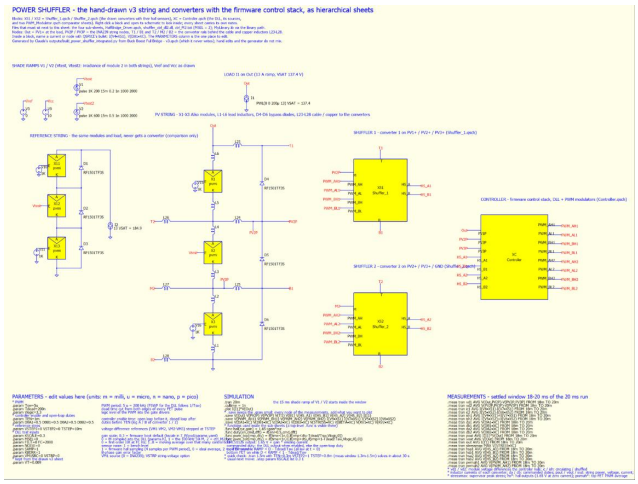
- ▷ What firmware was created: a bare-metal C application on the STM32G474 (CubeMX HAL, CMake + Ninja, arm-none-eabi, `-O2`) with the portable control core (`ctrl_core`: ctrl, supervisor, mppt, recon, stack), HRTIM/ADC/SPI/UART drivers, a supervisory state machine and a line-based telemetry/command protocol



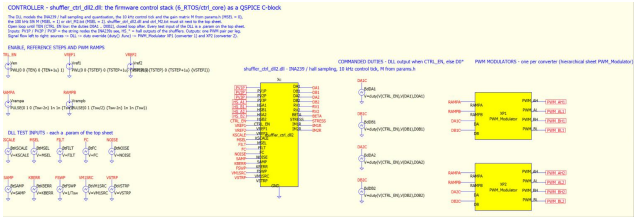
(a) Carpark Simulink twin: 3 modules, 2 shufflers, sensing block (noise + delay), DL3021 CV load.



(b) Power-source (PV80 / BENCH) and load (CV / CC) variants.



(c) QSPICE integrated model, top sheet: PV string, two shufflers, controller.



(d) Controller sheet: the firmware control DLL ("the Nucleo") and the HRTIM-style PWM modulators.

Figure 10: The two identification plants: the Simulink carpark twin (averaged switches, 61 mΩ) and the QSPICE integrated model (real switches, drivers, dead time, hall and INA sensing models).

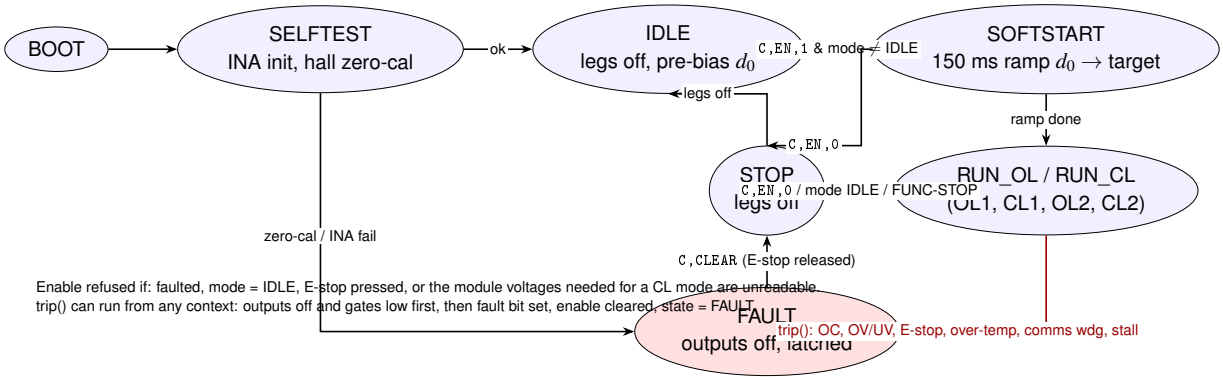


Figure 11: Supervisory state machine of the firmware (`6_RTOS/App/app.c`).

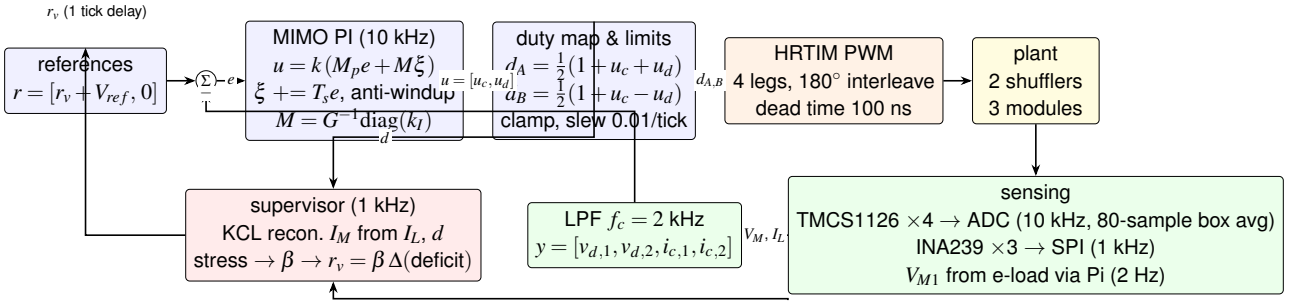


Figure 12: Deployed control architecture as implemented in the 10 kHz ISR and the 1 kHz supervisor.

(appendix E). **[note]** The slides say "FreeRTOS"; the shipped firmware is a superloop + interrupts (FreeRTOS was planned, CubeMX has no RTOS). Describe it as "real-time bare-metal scheduling" or add the RTOS as future work.

- ▷ Scheduling (what is prioritised): 10 kHz control ISR (highest after SysTick/E-stop) → 1 kHz INA polling → 100 Hz safety and state machine → 50 Hz telemetry → command parsing; the safety path (OC latch in the ISR, E-stop EXTI, IWDG) never depends on the loop.
- ▷ Key peripheral parameters: table 3; ADC reads are triggered on the PWM edges four times per period and box-averaged over the whole tick, which is a comb filter with nulls at f_{sw} and its harmonics (fig. 13).
- ▷ Pseudocode of the ISR, soft-start, supervisor and state machine: appendix D.
- ▷ Pi: Docker (InfluxDB 2.7 + Grafana 11), `pod.py` (serial ↔ Influx, LCD, LED, IR), `eload.py` (DL3021 SCPI at 5 Hz, forwards V_{str} at 2 Hz), `params_web.py` (port 8080), OpenOCD flashing, `sync.sh` deployment, systemd services.
- ▷ Validation of physical signals with the oscilloscope: dead time and 180° interleave at 200/300/400 kHz, ADC trigger alignment (fig. 108).

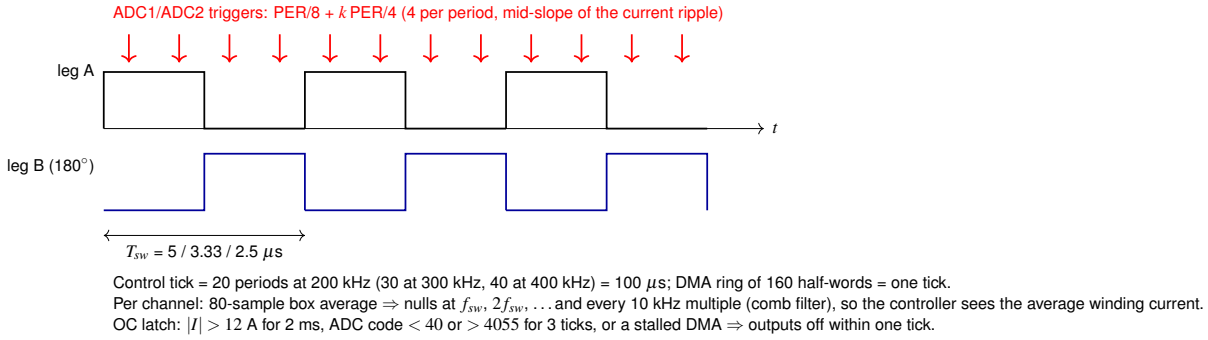


Figure 13: Hall-current sampling scheme: four HRTIM-triggered ADC conversions per switching period, box-averaged over one control tick.

3.7.1 Experimental setup and test plan

- ▷ What we test for: (i) open-loop sanity: $V_{M2}/(V_{M2} + V_{M3}) = 1 - d$, $v_d = -R_m i_m$, duty-step gain $-\bar{V}$; (ii) closed-loop regulation: $v_d \to 0$, $i_c \to 0$, steady-state error and noise vs k and f_{sw}; (iii) disturbance rejection: PSU/e-load steps (bench), shade (sun); (iv) stability margin: raise k until the loop misbehaves; (v) energy: delivered vs returned power under shade.
- ▷ Differences bench vs solar: PSU channels are stiff current limits (no I–V curve, no irradiance), so the bench exercises the loop and the gain sets per f_{sw}; the sun exercises the plant the controller was designed for but only at 200 kHz and at the irradiance the day gave (peak sun 15:04–15:13, minimal sun after 15:48).
- ▷ Test matrix: table 23 (21 windows); the pre-flight checklist and power-up order are in the repository (`0_Documentation/experiment_checklist.md`).

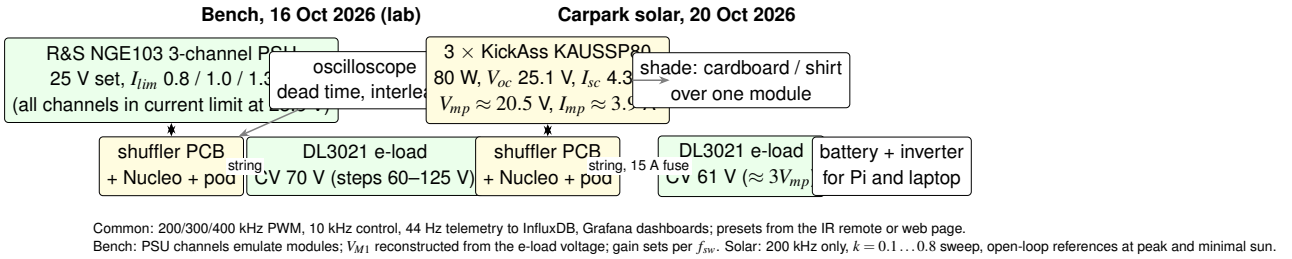
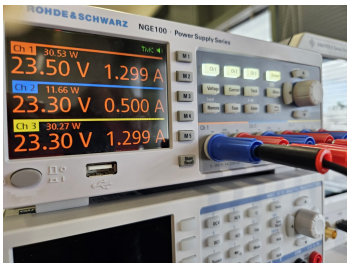
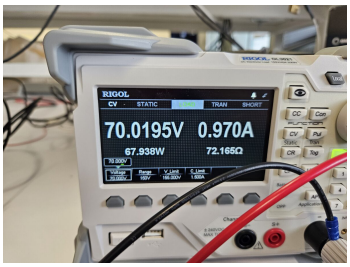


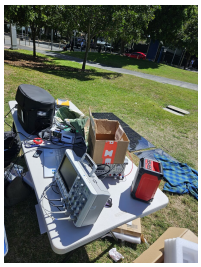
Figure 14: Bench and solar experimental set-ups.



(a) NGE103 PSU: 23.30 V / 1.299 A, 23.30 V / 0.500 A, 23.30 V / 1.299 A.



(b) DL3021 e-load in CV: 70.02 V, 0.970 A, 67.9 W.



(c) Carpark string on 20 Oct 2026.

Figure 15: Photographs of the two set-ups (more in appendix M).

- ▷ Metrics and how they are computed (one definition for every phase, `stepmetrics.py`): initial/final levels (medians), $|\Delta|$, 10–90 % rise, peak value/time, overshoot, undershoot, 2 %/5 %/absolute-band settling, steady-state error, IAE, fitted τ or (ζ, ω_n); noise = MAD of the pre-event window; steady state = mean, std, pk-pk, RMS, dominant frequency (Welch); energy = E_{load}, $E_{shuffled}$, $E_{returned} = \int (I_{str} - \max(\min_k I_k, 0)) V_{str}\, dt$.
- ▷ Hardware resolution caveat (state it in the methodology, not only in the results): telemetry is 44 Hz (11–14 Hz on three 200 kHz bench runs) while loop time constants are 1–10 ms, so hardware rise/settling times are bounds ("$\leq$ 23 ms"); hardware resolves steady-state error, noise, second-scale disturbances, duty-step amplitudes and energy.

3.8 Risks

- ▷ Register: table 10 (appendix B) kept from the proposal and updated; the ones that materialised: scope creep (PCB revisions and features compromised, one board only), firmware failure (mitigated by E-stop, fault latching and state telemetry: faults 0x10 and 0x2000 tripped three times on 16 Oct when gains were applied live, table 20), electrical safety (ratings, 15 A string fuse, 50 V DC class), safe outdoor testing (sun, fire, rain), data security, inaccurate model (the 61 mΩ averaged plant over-estimated the current-loop gain 5×; caught in QSPICE before hardware), controller faults, sensor faults (U10 top INA unusable; VM1 from the e-load).

3.9 Ethics

- ▷ Electrical safety: QUT HSE electrical workspace protocols, PPE, 15 A fuse, hardware and software E-stop, OC/OV/UV/over-temperature envelopes in firmware, never left energised unattended.
- ▷ Data security / privacy: authentication via the Raspberry Pi (locally hosted services, no cloud); only sensed data and internal states are streamed; credentials excluded from version control.
- ▷ Integrity: AI assistance (Claude) was used for derivation checking, scripting and the results pipeline; every result is reproducible from the repository (`make all`) and the derivations are hand-checkable (progress report §4.5 policy carried forward). **[note]** Decide how much of the AI-assistance statement to repeat here.

3.10 Sustainability

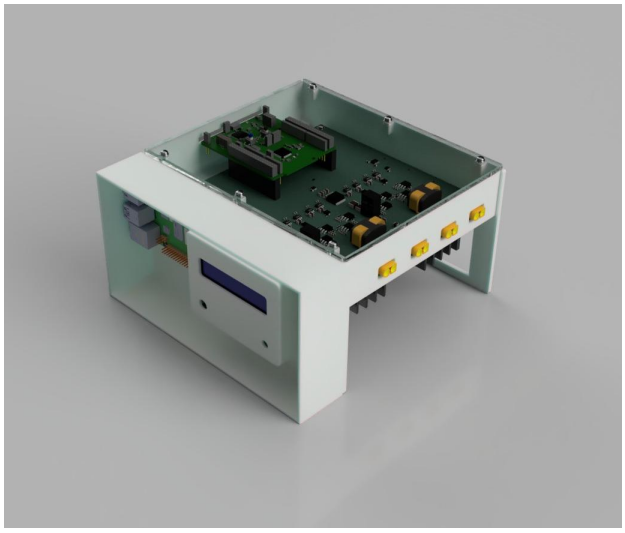
- ▷ Work performance through constant collaboration (weekly supervisor meetings, shared GitHub, hand-over interface with the hardware teammate).
- ▷ Longevity of the project through documentation, videos and GitHub (READMEs per folder, design log,

telemetry contract, experiment checklist, regenerable results).

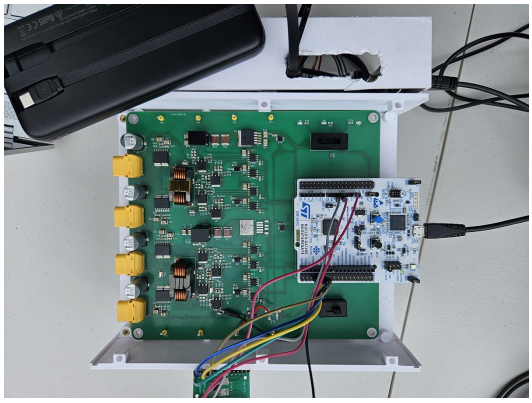
▷ Enclosure / thermal considerations: heatsink and INA die temperatures logged (appendix L); the shuffler processes only the mismatch power, so component ratings and losses scale with shading (UN SDG 7.3).

4 Results, Analysis and Discussion

4.1 Proposed Solution



(a) CAD render of the assembled power shuffler.

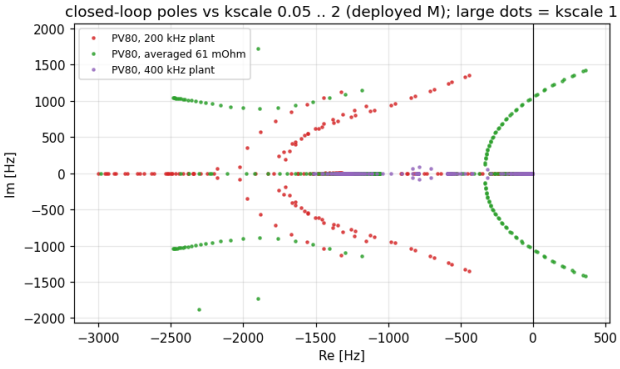


(b) The built unit on 20 Oct 2026: PCB, Nucleo, Pi pod, LCD, heatsink.

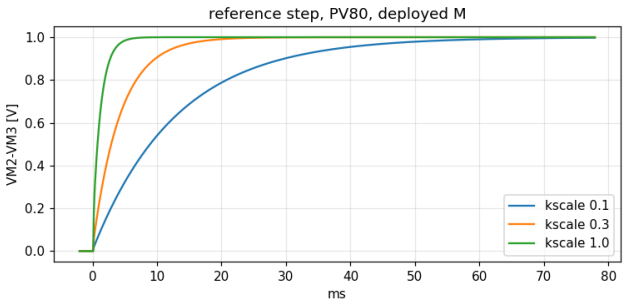
Figure 16: The complete power shuffler device.

▷ One paragraph: what was delivered against each objective (model, architecture, board, validation, self-contained product) and what the device does end-to-end: presets from the IR remote, soft-start, closed loop, Grafana telemetry, fault latching.

4.2 Analytical



(a) Closed-loop poles vs k (0.05–2) for the 200 kHz (0.29 Ω), averaged (61 mΩ) and 400 kHz (1.7 Ω) plants with the deployed M; large dots $k = 1$.



(b) $+1$ V reference step on $V_{M2} - V_{M3}$, PV80 plant, $k = 0.1/0.3/1$.

Figure 17: Analytical $N = 3$ model with the deployed gain set: root locus and reference step.

▷ Findings: the voltage-loop poles stay real up to $k \approx 1$; the circulating loops go complex first; the same M on the 61 mΩ averaged plant crosses into the RHP near $k \approx 1.5$ (explains why the Simulink-bench gain set had to be rejected).

Table 4: DC gain matrix $G(N)$: analytical values for the two plants and the gains identified by each deployed gain set ($\mathrm{diag}(k_I)M^{-1}$). V and A per unit modal duty.

Source	v_{d1}/u_{c1}	v_{d2}/u_{c2}	i_{c1}/u_{d1}	i_{c2}/u_{d2}	cross v_{d1}/u_{d1}	cond(G)
analytical PV80, averaged R 61 mOhm (Simulink)	-40.4	-40.4	672.1	672.1	0.0	16.8
analytical PV80, switching R 0.29 ohm (200 kHz)	-38.2	-38.2	141.4	141.4	0.0	3.8
analytical PV80, switching R 1.72 ohm (400 kHz)	-29.2	-29.2	23.8	23.8	0.0	1.4
analytical BENCH QSPICE point (all CC), R 0.29 ohm	-46.6	-46.7	160.7	160.7	0.0	3.5
analytical BENCH ch1 CC / ch2,3 CV, R 0.29 ohm	-15.2	-7.2	155.2	172.4	0.0	26.3
gain set 20261015-1950-3mod-qspice200-bench	-46.3	-46.9	142.5	139.0	-24.7	3.2
gain set 20261011-1634-3mod-qspice	-37.4	-37.7	256.0	215.8	7.3	7.4
gain set 20261015-1754-3mod-simulink-bench	-46.0	-47.4	741.5	742.5	-0.6	16.1
gain set 20261016-1628-3mod-qspice300-bench	-45.3	-45.7	58.8	56.8	-17.0	1.6
gain set 20261016-1751-3mod-qspice400-bench	-41.6	-42.9	23.5	23.0	-14.4	2.3
gain set 20261016-1804-3mod-qspice400d1-bench	-42.7	-43.8	26.2	26.0	-10.3	1.9

- $G(N)$: analytical PV80 at 0.29 Ω gives −38 V / 141 A per unit; the identified 200 kHz set gives −46 V / 142 A on the bench point (table 4), i.e. the model and the identified plant agree on both blocks once R_{sw} is included; the voltage gain is $\approx -\bar{V}$ as predicted.
- Phase portraits (fig. 5): centre → focus → over-damped node as the plant resistance grows with f_{sw}; closed-loop trajectories approach first order along $i_m = -g_e v_d$.
- Design vs fitted: analytical τ for the +1 V step is 0.86–1.22× $1/(k\,2\pi\,150)$ over $k = 0.1 \ldots 1$ and both plants (table 8); no overshoot anywhere in the analytical model.
- Cross term not in the model: the identified sets show $v_{d,1}/u_{d,1} = -25 \ldots -10$ V per unit (table 4), a dead-time effect; the decoupling absorbs it at the deployed gains.

4.3 Simulink

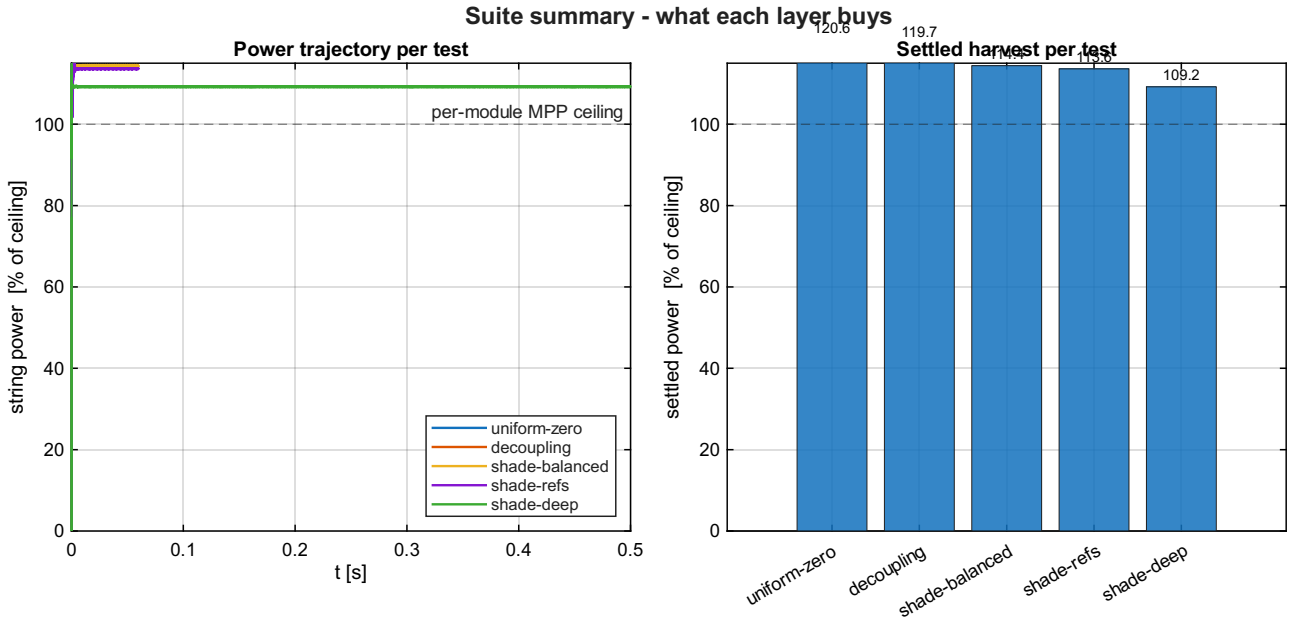
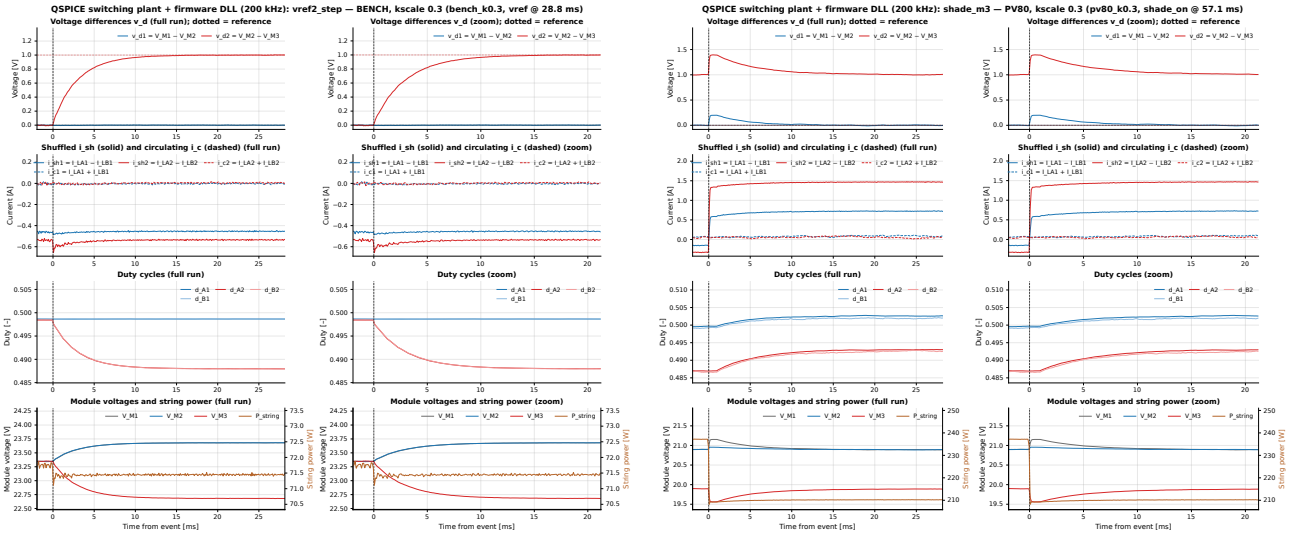


Figure 18: Carpark twin validate suite summary (tests 01–05: uniform, decoupling, half shade, shade with references, deep shade); every test passes its criteria. Per-test figures in appendix G.

- MIL is implied by the carpark twin working; keep the 7-test MIL summary (fig. 29) in the appendix.
- Important Simulink data: the twin reproduces the design bandwidth within 0.38–1.10× for τ (table 8), decouples the current loop during a voltage step (table 25), and shows the only overshoot at $k = 1$ (8.8 %, bench plant).
- Shade in the twin (S3, module 3 loses 1/3 of its current): $v_{d,2}$ is held and P_{str} recovers within 0.1 ms of the step at every k (table 26); the open-loop twin of the same event gives the benefit baseline.

▷ Robustness checks in the repository (`robustness_sweep.m`): ILB sensitivity ×4, V_{M1} source param vs Pi, sensor noise on.

4.4 QSPICE



(a) S4, bench plant, $k = 0.3$: +1 V reference step.

(b) S3, PV80 plant, $k = 0.3$: shade on module 3.

Figure 19: QSPICE integrated model with the firmware control DLL (100 μs bin means of the switching waveforms). All 25 runs in appendix H.

Table 5: Switching ripple of the circulating current: analytical law $\Delta i_c = \bar{V}/(2f_{sw}(L+M))$ with $L+M = 13.75\,\mu$H, the QSPICE device model, and what the 44 Hz telemetry shows instead (sensing noise, not ripple).

f_{sw} [kHz]	Source	Δi_c pk-pk [A]	per-leg pk-pk [A]	Δi_m pk-pk [A]	$\bar{V}$ [V]
200	QSPICE S10i (ud sweep, du = 0)	8.39	4.25	0.19	46.5
200	analytical Vbar/(2 f (L+M)), Vbar 46.6 V	8.47	4.24	0.00	46.6
200	hardware telemetry noise (44 Hz samples, 17 segments)	1.43	–	1.12	40.7
300	QSPICE S10i3.qraw raw, run 1 (du = 0), 2.5 ms	5.58	2.85	0.20	46.4
300	analytical Vbar/(2 f (L+M)), Vbar 46.6 V	5.65	2.82	0.00	46.6
300	hardware telemetry noise (44 Hz samples, 6 segments)	0.28	–	0.88	46.7
400	QSPICE S10i (ud sweep, du = 0)	4.09	2.18	0.39	46.0
400	QSPICE S10i4.qraw raw, run 1 (du = 0), 2.5 ms	4.13	2.19	0.41	46.0
400	analytical Vbar/(2 f (L+M)), Vbar 46.6 V	4.24	2.12	0.00	46.6
400	hardware telemetry noise (44 Hz samples, 5 segments)	0.45	–	0.46	46.7

▷ Raw QSPICE data: 25 runs (S1 closure, S3 shade, S4 reference step; bench and PV80; $k = 0.1/0.2/0.3/0.5/1$).
▷ Ripple: the device model reproduces $\Delta i_c = \bar{V}/(2f_{sw}(L+M))$ to 3 % (8.39 / 5.58 / 4.11 A pk-pk at 200 / 300 / 400 kHz, table 5); the wrong coupling sense would predict 25–47 A, so this also confirms $L+M$ is the circulating mode.
▷ Closed loop: τ is 0.52–1.17× design; first overshoot at $k = 0.5$ on PV80 (7.5 %); the 300/400 kHz gain sets were identified from S10i3/S10i4 and carry the larger R_{sw}.
▷ Limitation stated by the QSPICE verification: i_c and v_d at the mA/mV level depend on mΩ parasitics and ns edges and are not judged.

4.5 Bench Testing

Bench testing was a success. The setup was a 3-channel power supply (NGE103), an electronic load (DL3021), oscilloscope, laptop and the full system (fig. 14). Testing started at 200 kHz open loop to get baseline behaviour, then closed loop with gains of 0.1, 0.2 and 0.3 of the QSPICE values, then 300 kHz at 0.3 and 0.5, then 400 kHz at 0.3. 300 kHz seemed to be the most favourable: there is a trade-off

where the benefits reduce as the switching speed increases and the heat increases. An open-loop test at 100 kHz was also explored (13 Oct); the system sort of worked but with bizarre behaviour (appendix L). It is concluded that the power supplies cannot handle as much as the panels, or do not require as aggressive control.

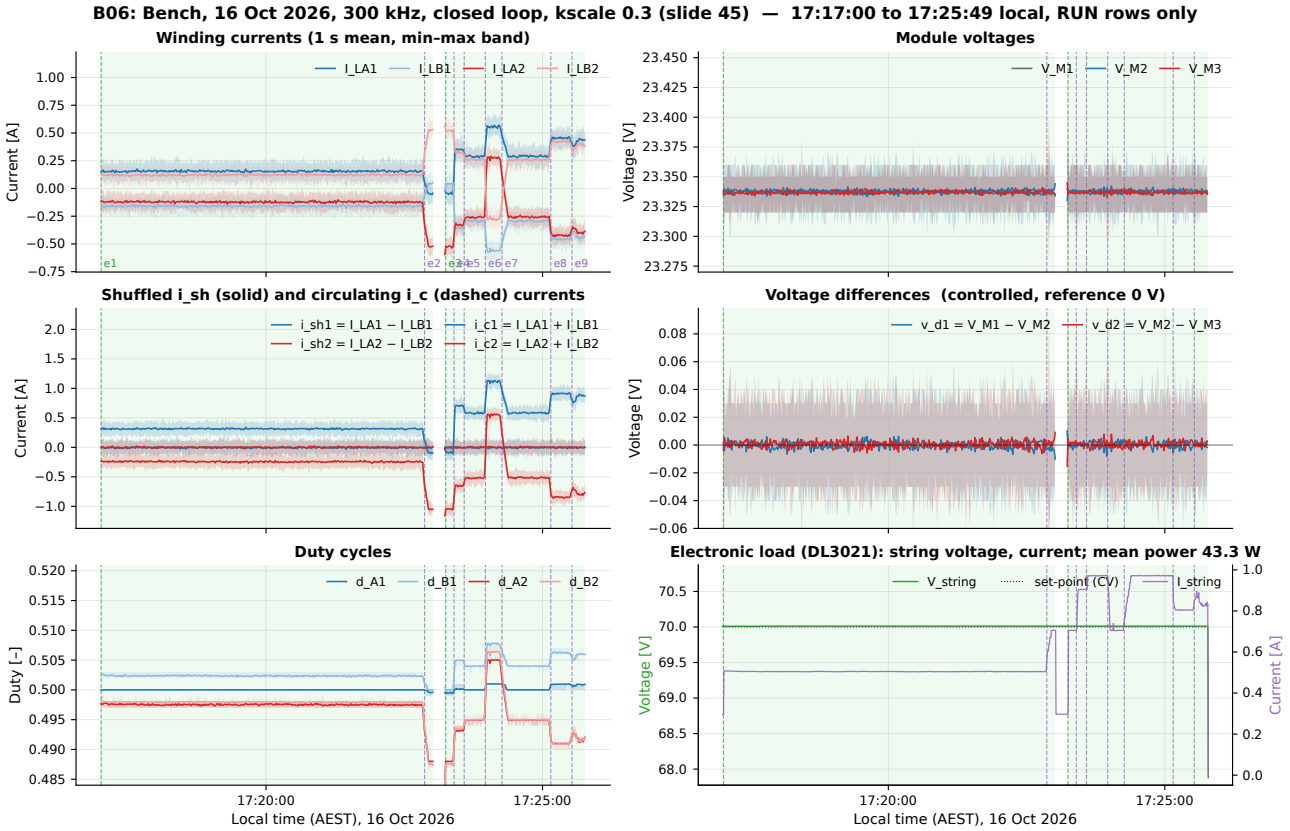


Figure 20: Bench, 300 kHz closed loop, $k = 0.3$ (slide p45 window, RUN rows only): winding currents and modal currents, module voltages and differences, duties, e-load. Numbered events are the PSU / e-load set-point steps (zooms in appendix I).

Table 6: Open-loop duty steps on the bench (B01s, 16 Oct, 200 kHz): measured modal responses. The averaged model predicts $\Delta v_d = -\bar{V}\Delta u_c$ with $\bar{V} = 46.6$ V.

Time	Command	Signal	y_0	y_∞	Δ	Rise 10–90	Settle 2 %	Noise
15:11:42	C,DUTYS,0.5,0.5,0.51,0.51	vd1	0.03	0.00	-0.03	-	0 ms	0.046
15:11:42	C,DUTYS,0.5,0.5,0.51,0.51	vd2	0.82	-1.05	-1.87	$\leq$ 25 ms	$\leq$ 3155 ms	0.030
15:11:42	C,DUTYS,0.5,0.5,0.51,0.51	ic1	0.42	0.46	0.04	-	0 ms	1.426
15:11:42	C,DUTYS,0.5,0.5,0.51,0.51	ic2	2.07	3.03	0.96	$\leq$ 48 ms	$\leq$ 50 ms	1.162
15:13:00	C,DUTYS,0.49,0.49,0.50,0.50	vd1	0.03	0.94	0.91	$\leq$ 38 ms	$\leq$ 67 ms	0.044
15:13:00	C,DUTYS,0.49,0.49,0.50,0.50	vd2	-1.06	-0.12	0.94	$\leq$ 37 ms	$\leq$ 5875 ms	0.015
15:13:00	C,DUTYS,0.49,0.49,0.50,0.50	ic1	0.54	0.82	0.28	-	0 ms	1.248
15:13:00	C,DUTYS,0.49,0.49,0.50,0.50	ic2	3.04	2.98	-0.06	-	0 ms	1.162
15:13:54	C,DUTYS,0.49,0.49,0.50,0.50	vd1	0.97	0.95	-0.02	-	3.41e+03 ms	0.029
15:13:54	C,DUTYS,0.49,0.49,0.50,0.50	vd2	-0.13	-0.12	0.01	-	0 ms	0.015
15:13:54	C,DUTYS,0.49,0.49,0.50,0.50	ic1	0.81	0.49	-0.33	$\leq$ 25 ms	$\leq$ 25 ms	1.333
15:13:54	C,DUTYS,0.49,0.49,0.50,0.50	ic2	2.58	3.14	0.57	$\leq$ 25 ms	$\leq$ 25 ms	1.026
15:15:15	C,DUTYS,0.51,0.51,0.50,0.50	vd1	0.97	-0.89	-1.86	$\leq$ 25 ms	$\leq$ 1731 ms	0.043
15:15:15	C,DUTYS,0.51,0.51,0.50,0.50	vd2	-0.13	-0.13	0.00	-	0 ms	0.030
15:15:15	C,DUTYS,0.51,0.51,0.50,0.50	ic1	0.43	0.40	-0.03	-	0 ms	1.214
15:15:15	C,DUTYS,0.51,0.51,0.50,0.50	ic2	2.82	2.76	-0.06	-	0 ms	1.257

Rise and settling times shorter than one telemetry frame (25 ms at 40 Hz) are bounds.

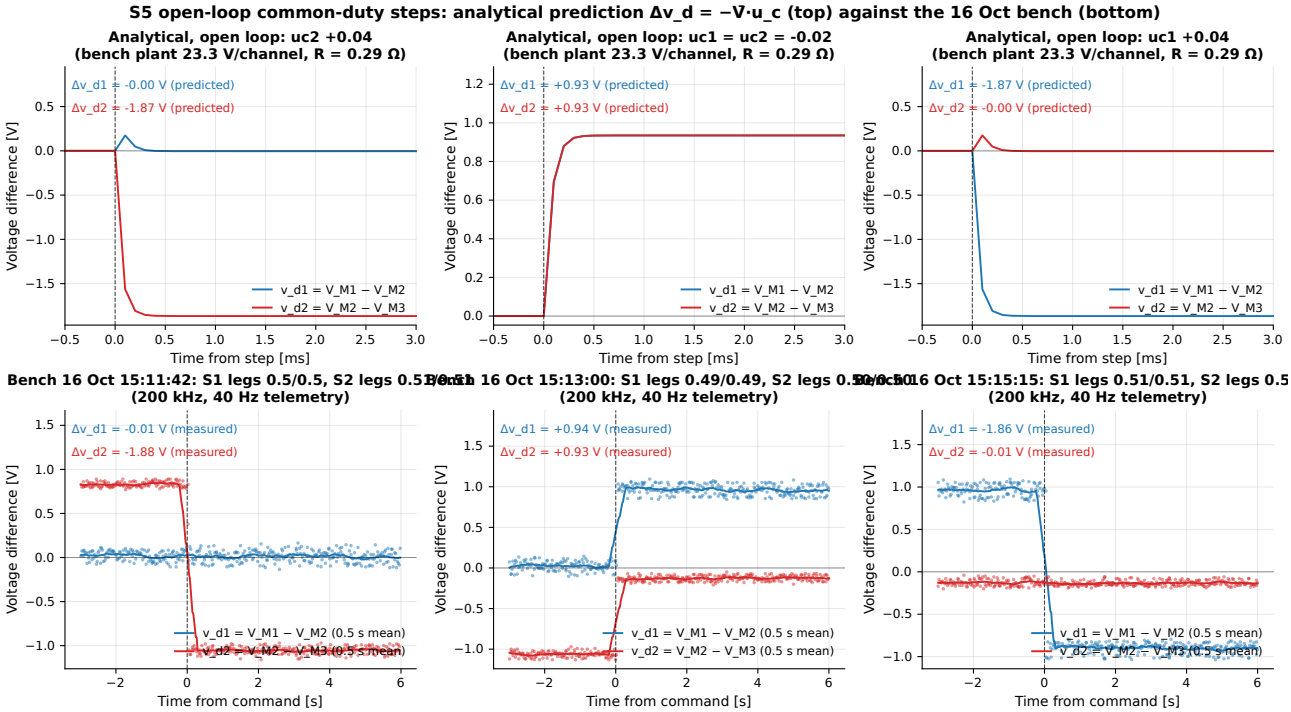


Figure 21: Open-loop duty steps (16 Oct 15:09–15:16, 200 kHz): hardware (orange) against the analytical model (grey). The measured Δv_d of $-1.87/+0.91/+0.94/-1.86$ V match $-\bar{V}\Delta u_c = \mp 1.86/+0.93$ V.

- ▷ Open loop confirms the model's DC gain on hardware: 1.2 % agreement on four duty steps (table 6, fig. 21); the open-loop residual $v_{d,2} \approx -0.17$ V at 200 kHz, -0.58 V at 300 kHz, -0.94 V at 400 kHz is $-R_m i_m$ growing with R_{sw} (table 18).
- ▷ Closed loop holds the references with no offset: over 22 settled segments $|v_{d,2}| \leq 0.010$ V and $|i_{c,2}| \leq 0.066$ A; the residual is sensing noise ($\sigma(v_{d,2})$ 0.013–0.17 V) (table 24). Compare to the draft's "±200 mV / ±200 mA": the measured bands are tighter.
- ▷ Loop closures (IDLE → SOFTSTART → RUN_CL): the ramp leaves 0–0.8 V on $v_{d,2}$; the loop removes it within one telemetry frame in 10 of 23 closures and within 3 s (noise-limited) in the rest (table 19); the loop's own 1–10 ms transient is below the 23 ms frame so rise/settle are bounds.
- ▷ PSU and e-load steps while closed (B06, fig. 20): $v_{d,2}$ stays inside ±0.1 V through 60 ↔ 125 V load steps and the PSU reconfiguration; what moves is i_m, as it should.
- ▷ Stability limit found: B07 at 300 kHz, $k = 0.6$ oscillated ($\sigma(v_{d,2}) = 1.41$ V, pk-pk 11.8 V, 6 s) before it was stopped; the 300 kHz set loses margin above $k \approx 0.5$, consistent with the root locus.
- ▷ Applying a gain change while running tripped the firmware three times (B04 $k \to 0.4$: module UV 0x10; B07 $k \to 0.6$ and $\to 0.3$: main-loop stall 0x2000): gains should be set before closing the loop; `C , APPLY` in RUN_CL stalls the loop. **[note]** A finding for the firmware future-work list.
- ▷ Telemetry rate: B02–B04 logged at 11–14 Hz (every time metric there is a ≥ 70 ms bound).
- ▷ Data-quality notes: slide "200 kHz kscale 20 %" ran at $k = 0.3$ (telemetry `kscale` 0.3, `pset` 0, boot default after the re-flash) and is reported as B03 $k = 0.3$; V_{M1} on the 200 kHz open-loop runs is reconstructed from the e-load.

Limitations. Can only do 2 shufflers. Boards cannot be stacked together, so system scalability cannot be commented on. The shufflers struggle to be driven independently as they are hard-wired together. The top-level INA239 does not function, so V_{M1} had to be inferred from the e-load value. The simulation has to be re-run every time the switching speed changes for best results.

4.6 Solar Test

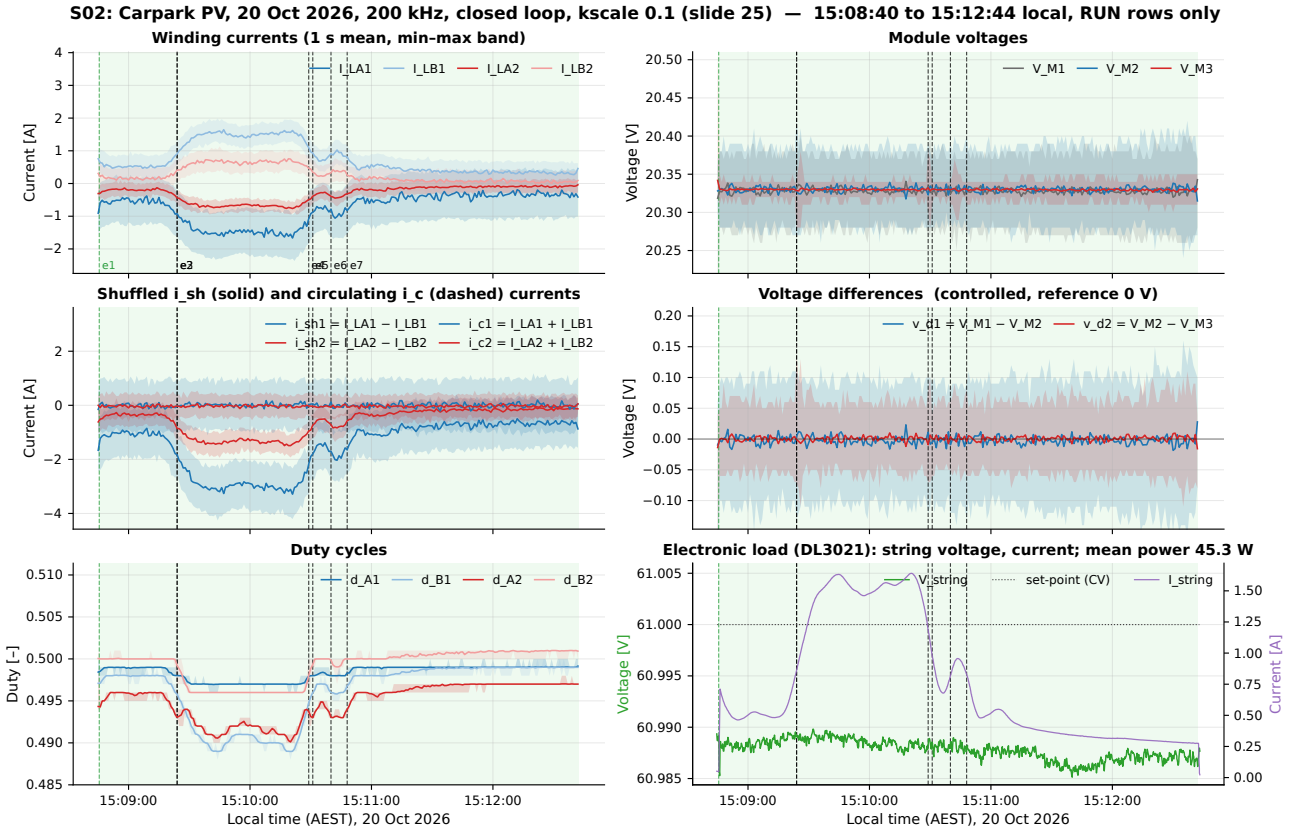
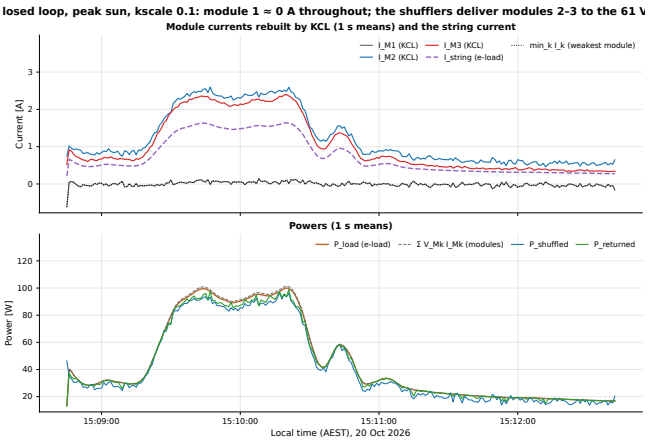
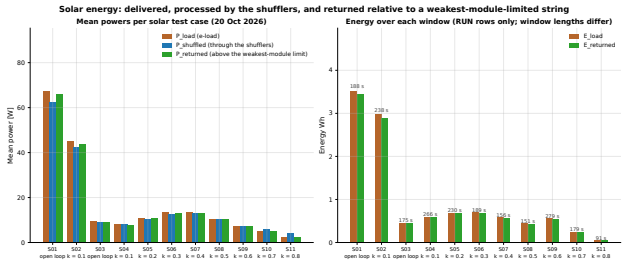


Figure 22: Carpark, peak sun, 200 kHz closed loop, $k = 0.1$ (slide p25 window): the shade event (15:09:20–15:10:40) moves i_m while v_d stays at zero. All eleven windows in appendix J.



(a) S02: module currents rebuilt by KCL, string current, and the powers delivered / shuffled / returned.



(b) Mean powers and energies per solar window.

Figure 23: Energy returned by the shufflers on 20 Oct 2026 (table 7).

Table 7: Solar test (20 Oct 2026, KAUSSP80 string into a 61 V CV load): powers in W, module currents in A (KCL reconstruction), energy delivered and energy returned above a weakest-module-limited string.

Case	Loop	k	T [s]	$\bar{P}_{load}$	$\bar{P}_{shuf}$	$\bar{P}_{ret}$	$P_{ret,max}$	I_{M1}	I_{M2}	I_{M3}	E_{load} [Wh]	E_{ret} [Wh]	η [%]
S01	OL	–	188	67.3	62.3	65.9	96.1	0.00	1.76	1.54	3.51	3.44	99.3
S02	CL	0.1	238	45.1	42.3	43.9	98.9	-0.00	1.21	1.03	2.98	2.90	99.2
S03	OL	–	175	9.4	9.3	9.3	11.1	-0.04	0.31	0.19	0.46	0.45	99.9
S04	CL	0.1	266	8.2	8.2	8.0	9.0	-0.04	0.28	0.16	0.60	0.59	99.9
S05	CL	0.2	230	10.9	10.6	10.7	12.4	-0.04	0.35	0.22	0.69	0.68	99.9
S06	CL	0.3	189	13.4	12.5	13.1	14.2	-0.03	0.42	0.27	0.70	0.69	99.9
S07	CL	0.4	156	13.5	13.0	13.3	14.2	-0.04	0.42	0.28	0.59	0.58	99.9
S08	CL	0.5	151	10.6	10.6	10.4	11.8	-0.04	0.34	0.22	0.44	0.44	99.9
S09	CL	0.6	279	7.3	7.3	7.2	8.5	-0.04	0.27	0.13	0.57	0.55	100.0
S10	CL	0.7	179	5.1	5.8	5.0	6.1	-0.04	0.21	0.09	0.26	0.25	100.1
S11	CL	0.8	91	2.3	4.4	2.3	3.2	-0.06	0.14	0.03	0.06	0.06	100.3

Module 1 carried ≈ 0 A in every window, so a plain series string would have delivered nothing to a 61 V load: $E_{ret} \approx E_{load}$.

- ▷ Solar testing was a success: 11 windows, 36 min of RUN time, closed loop from $k = 0.1$ to 0.8 without a fault.
- ▷ Raw peak-sun data (S01 open loop, S02 closed loop, fig. 22) and minimal-sun data (S03–S11): the open-loop residual $v_{d,2} = -0.32$ V (peak) / -0.07 V (minimal) is removed in closed loop (table 24).
- ▷ Module 1 carried ≈ 0 A in every window (by the firmware's own IM1r and by KCL) while modules 2 and 3 carried up to 2.5 A: the top panel was shaded or disconnected all afternoon (my shirt covered a panel during S02). The shufflers delivered the other two modules' power to the 61 V load that a plain series string could not have supplied at all: $E_{ret} = 10.6$ Wh $\approx E_{load}$ over 36 min, 99 W returned at the S02 peak (table 7). **[note]** Confirm what happened to module 1 and phrase the "massive energy return" accordingly.
- ▷ Shade events (table 22): on $|i_m|$ the mismatch steps are 0.5–1.5 A in 2–6 s (cloud / shirt speed), $v_{d,2}$ deviates < 0.3 V and recovers; the loop never reaches a duty limit (duties 0.45–0.55).
- ▷ Limitation: the weather. Minimal sun after 15:48 (P_{load} 2–13 W) gave the k sweep but not enough mismatch power to stress the loop; the closed-loop response to panel dynamics is therefore validated at low power and otherwise theoretical.
- ▷ Efficiency $E_{load}/E_{modules}$ reads 99.2–100 %: with V_{M1} from the e-load and module currents from KCL the measurement closes by construction; do not present it as a converter efficiency.

4.7 Synthesising Results

Table 8: +1 V reference step on v_{d2} (S4), 200 kHz gain set: fitted time constant τ, 10–90 % rise t_r and 2 % settling t_s in ms, analytical model | QSPICE switching model with the firmware DLL | Simulink carpark twin. $\tau_{design} = 1/(k\,2\pi\,150\,\text{Hz})$.

Setup	k	τ_{design}	τ	t_r	t_s	τ	t_r	t_s	OS %	τ	t_r	t_s	OS %
BENCH	0.1	10.61	10.54	23.2	41.3	9.72	19.8	30.7	0.0	10.15	22.9	41.4	0.0
BENCH	0.2	5.31	5.20	11.5	20.5	4.62	10.4	18.4	0.0	4.77	11.0	19.8	0.0
BENCH	0.3	3.54	3.42	7.6	13.6	2.84	6.5	11.9	0.0	2.97	6.9	12.4	0.0
BENCH	0.4	2.65	2.53	5.6	10.1	–	–	–	–	–	–	–	–
BENCH	0.5	2.12	1.99	4.5	8.0	1.47	3.5	6.2	0.0	1.51	3.7	6.8	0.0
BENCH	1.0	1.06	0.91	2.0	3.8	0.55	1.3	2.2	0.0	0.40	0.8	2.7	8.8
PV80	0.1	10.61	12.90	28.4	50.6	12.40	22.5	32.1	0.0	11.64	26.0	45.5	0.0
PV80	0.2	5.31	6.38	14.1	25.1	5.87	12.6	21.8	0.0	5.51	12.5	22.4	0.0
PV80	0.3	3.54	4.21	9.4	16.7	3.71	8.7	15.2	0.0	3.47	8.1	14.2	0.0
PV80	0.4	2.65	3.12	7.0	12.4	–	–	–	–	–	–	–	–
PV80	0.5	2.12	2.47	5.5	9.9	1.94	4.6	17.0	7.5	1.83	4.5	8.0	0.0
PV80	1.0	1.06	1.16	2.6	4.8	0.77	1.9	8.5	4.2	0.56	0.9	3.0	0.0

Hardware cannot resolve this transient (one telemetry frame = 23 ms); its steady-state error and noise are in Table 24.

- ▷ Comparison results in one table (table 8): τ relative to design is 0.86–1.22× analytical, 0.52–1.17× QSPICE, 0.38–1.10× Simulink; rise and settling follow $1/k$; overshoot appears only at $k \geq 0.5$ and stays below 8.8 %. Hardware agrees where it can measure: zero steady-state error, noise-limited residual, one-frame settling.

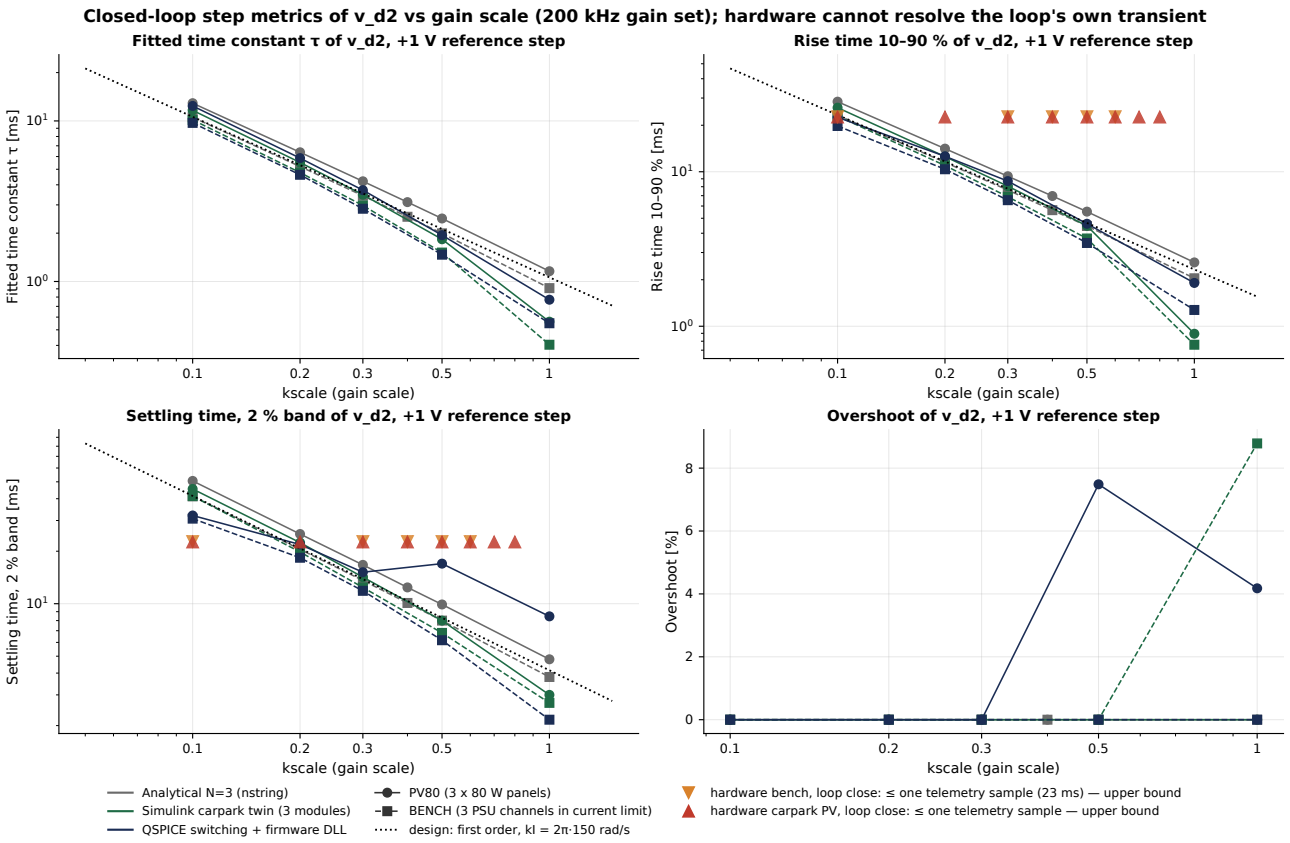
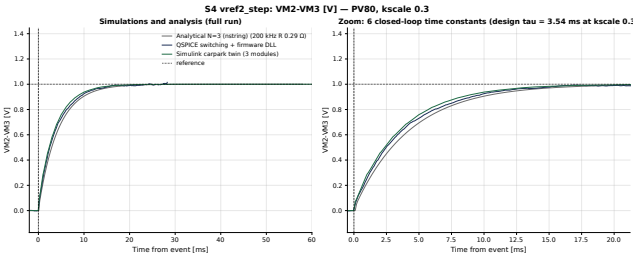
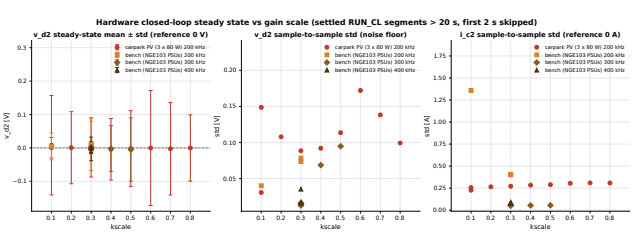


Figure 24: Closed-loop step metrics of $v_{d,2}$ vs gain scale (200 kHz gain set): fitted τ, 10–90 % rise, 2 % settling and overshoot for the analytical model, QSPICE and the Simulink twin; the dotted line is the first-order design $1/(k\,2\pi\,150)$; hardware closures are shown as one-sample upper bounds.



(a) S4, PV80, $k = 0.3$: $v_{d,2}$, three simulations overlaid.



(b) Hardware steady state vs k: $v_{d,2}$ error and noise, $i_{c,2}$.

Figure 25: Cross-phase comparison (all 81 overlays in appendix K).

- ▷ Agreement across the pipeline supports the gap claim: the same M decouples the voltage and current loops in an averaged model, a switching model with sensing quantisation and dead time, and on the board, at three switching frequencies.
- ▷ Where they disagree and why: the Simulink twin is faster than design at $k \geq 0.5$ (averaged plant, no R_{sw}); QSPICE at $k = 1$ shows 4–10 % overshoot (sampling delay + LPF); the 300 kHz set was identified from the bench point and loses margin earlier on hardware ($k = 0.6$ oscillation).
- ▷ Risk / ethics / sustainability outcomes in one paragraph: no injury, no component failure beyond U10, every fault latched by firmware, all data local, 10.6 Wh returned from a string with a dead module.
- ▷ Stakeholder feedback: supervisor's steer towards the averaged N-module model and the DC-gain identification; hardware teammate's QSPICE as the arbiter. **[note]** Add any direct feedback you received.

5 Conclusion

- ▷ Accuracy: the controller holds the module-voltage differences to within ± 0.010 V and the circulating currents to within ± 0.07 A on hardware (sensing-noise limited), and its bandwidth follows the design $\tau = 1/(k\,2\pi\,150)$ to within the spread of the three simulations (0.4–1.2×).
- ▷ Reliability: 21 test windows on two days, closed loop from $k = 0.1$ to 0.8 in the sun with no fault; the only instability was deliberately provoked ($k = 0.6$ at 300 kHz) and the only trips were live gain changes, both now understood.
- ▷ Significance: a complete, reproducible system (analytical model → Simulink → QSPICE → firmware → bench → sun, with one C core and one results pipeline) that can be used as a bouncing-off point for further investigation of MIMO control of DPP strings.
- ▷ Cycle back to the research question: the dynamics of multiple interleaved buck-boost converters are captured by the N-module averaged model with a diagonal-dominant DC gain; a decoupling integral controller on $[v_d, i_c]$ commands the currents so the string equalises; the MPP of a partially shaded string is reached when the module voltages are equal, which the loop enforces.
- ▷ Partial shading: with one panel covered (my shirt) the controller returned 99 W at peak and 10.6 Wh over the afternoon that a plain series string could not have delivered to the 61 V load.

Future work. Scalability: determine how the relationships change when more shufflers act in a string or how a shuffler handles a varying number of panels. Streamline the switching-speed gain calculation ($R_{sw}(f_{sw})$ is now a formula). Generate and compare control models (LQR, MPC, observer-based). Apply a machine-learning model / cost function to determine optimal gains. Fix the top-level voltage sensor. Consider effectiveness with different panel arrangements and dynamic configurations. Explore applications outside the solar industry (mining, computing, industrial, batteries); would this let you win a solar car race, and does the technology itself scale?

Recommendations. Explore other contexts. Beware of the electrical safety limits: they restrict the experiments that can actually be run. Go for hands-on development for understanding. The system likes high switching frequencies, so revisit GaN. Mechanical: downsize the heatsink, improve fasteners, reconfigure the PCB, redesign the enclosure. Review the GitHub repository, manufacturing notes and user guides to set up and continue the project. Consolidate the microcontroller and the microcomputer into one board.

- ▷ Blueprint for scaling up (presentation slide): mechanical (enclosure design, thermal analysis); electrical (decouple the two shufflers, provision for a chain of boards, centralised controller board, master–slave); software (many-N shuffler control, how further coupling affects dynamics, multiple strings, integrate MPPT); problem identification: re-evaluate the target DPP application, is solar where the most value is?
- ▷ Firmware items found in testing: make `C,APPLY` safe in RUN_CL (or refuse it), add the COMP/DAC hardware OC trip, rework U10's isolator.

Would like to see the theoretical model go toe to toe with the extended kalman filter.

In terms of immediate other work to do. Highly recommended to collect some more data and stress tets the system on a very sunny day. To properly get metrics for efficiency an mppt algorithm should be applied. My suggestion is to run it on the raspberry pi because that is what has remote access to the load through direct usb connection. If the same experimental setup is preserved will have to be aware

of the limitations of the load in relation to the panels.

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A Research Project Timeline

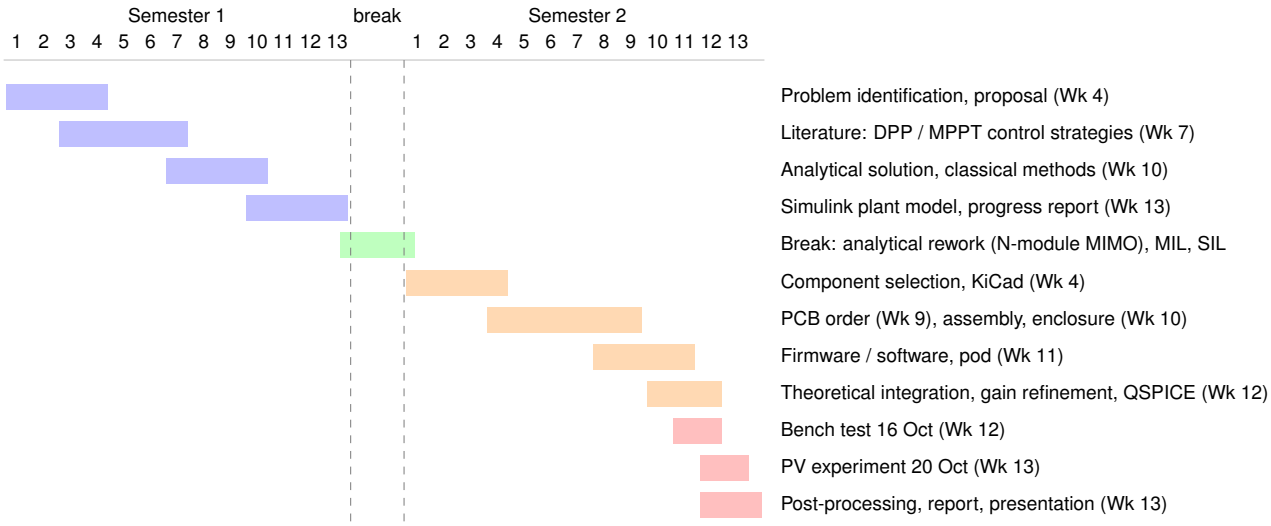


Figure 26: Project timeline across the two semesters (teaching weeks) with the milestone week of each phase.

- ▷ Changes from the proposal: PCB V1 deferred from Semester 1 Wk 10 to Semester 2 (progress report Appendix A); V2/V3 iterations collapsed into one board; MPPT integration deferred (the e-load / EcoFlow does string MPPT); had to compromise on PCB revisions and features.
- ▷ Interim milestones were reviewed weekly with the supervisor.

B Risk Register

Likelihood and consequence are rated Low (L), Medium (M), High (H) or Critical (C); the overall rating is the higher of the two. The register is the proposal's, with a *Status* column recording what happened.

Table 9: Milestones and progress achieved (proposal → progress report → delivery).

Semester	Week	Milestone	Progress / change
1	4	Problem identification, proposal	Research question, objectives, risk register, Gantt (34 weeks)
1	7	Existing DPP / MPPT control strategies	Literature review (Appendix A of the proposal), gap statement
1	10	Analytical solution using classical methods	Single-shuffler averaged model, equilibrium, linearisation, cascaded PI; marginal stability found
1	13	Simulink plant model	4-module PV string, P–V validation, inner current loop attempt (instability under shade)
Break	–	Analytical rework, MIL, SIL	MIMO reformulation (LQR + integral, then decoupling PI), N-module string model, MIL 7-test suite, SIL C core + golden replay
2	4	Component selection	STM32G474 Nucleo, TMCS1126, INA239, Raspberry Pi 4B; KiCad Version 0
2	9	PCB order	PCBWay board + stencil, components, reflow
2	10	Enclosure development	Enclosure, heatsink, pod wiring (IR, LCD, LED, DS18B20)
2	11	Firmware / software	Bare-metal C application, control core, telemetry contract, Pi stack (Docker InfluxDB + Grafana)
2	12	Theoretical integration / gain refinement	QSPICE integrated model with the control DLL; gain sets per f_{sw}; R_{sw} identified
2	12	Bench test	16 Oct 2026: 200/300/400 kHz, open and closed loop (table 23)
2	13	PV experiment	20 Oct 2026: 3 × KAUSSP80 in the carpark, $k = 0.1 \ldots 0.8$
2	13	Post-processing	Results pipeline (`make all`), REPORT.md, this report, presentation

Table 10: Project risk register with outcome.

#	Risk	Category	L	C	Rating	Mitigation / monitoring	Status at delivery
1	Scope creep	Project	H	H	H	Strict change control; new requirements deferred to the next revision; supervisor sign-off for additions	Materialised: one PCB revision only, MPPT deferred to the load, features compromised (slide: "had to compromise on PCB revisions and features")
2	No PV hardware available	Resource	M	C	H	Procure panels early; fallback QSPICE validation	Mitigated: 3 × KAUSSP80 80 W panels, DL3021 load; EcoFlow not used (V_{oc} 75 V > 60 V input)
3	Electronics failure	Hardware	M	H	H	Overload protection; spares; one-month buffer	Partly: U10 (top INA239) unusable (isolator wired reversed); worked around with V_{M1} from the e-load
4	Firmware failure	Control sys.	M	H	H	Simulation first (SIL, QSPICE DLL); unit tests; fault latching	Materialised in a contained way: 3 trips on live gain changes (UV, main-loop stall), all latched safely; E-stop and state telemetry worked
5	Monitoring software inaccuracy	Software	L	M	L	Sensor noise filtering; cross-validate against known points	Telemetry at 44 Hz is noise-limited; rise/settle reported as bounds; B02–B04 logged at 11–14 Hz
6	Data loss	Data	L	H	M	GitHub, OneDrive, local backups	No loss; every window re-exportable from InfluxDB
7	Unable to access data	Data	L	M	L	Shared cloud repositories, two admins	OK
8	Electrical hazard (50 V DC class)	Safety	M	C	C	QUT HSE protocols, PPE, isolation, formal risk assessment	No incident; 15 A fuse, OC/OV/UV envelopes, hardware + software E-stop
9	Sunburn (outdoor testing)	Safety	H	L	M	Sunscreen; sessions outside 10–14 h where practical	Carpark session 12:30–16:30 with shade for the operators

#	Risk	Category	L	C	Rating	Mitigation / monitoring	Status at delivery
10	Fire risk	Safety	L	C	H	Extinguisher; rated fuses; never unattended	No incident; heatsink and INA die temperatures logged (overtemperature trip at 85 °C, Pi E-stop at 80 °C)
11	Assignment deadline collision	Project	H	M	H	One-month buffer; weekly review	Buffer consumed by the PCB deferral; delivery on time
12	Skill gap between team members	Team	M	M	M	Division by domain; knowledge transfer at reviews	Worked: control vs power electronics split; QSPICE as the hand-over interface
13	Control–hardware integration failure	Control sys.	M	H	H	QSPICE validates firmware–hardware interaction first	Mitigated: the firmware control DLL in QSPICE caught the 5× current-gain error of the averaged model before hardware
14	MPPT fails to track global MPP	Control sys.	M	H	H	Validate across all shading scenarios in simulation	Descoped: string MPPT delegated to the load; equalisation (the enabling condition) validated
15	Intellectual property dispute	Legal	L	H	M	Timestamped GitHub; monitor IP Australia	No issue
16	Poor commercial adoption	Strategic	L	L	L	Outside scope	–
17	Data breach / cybersecurity	Data	L	M	L	No personal data; private repos; credentials excluded	Pi services local-only; params page has no authentication (note for future work)
18	Team member illness	Team	M	M	M	Buffer; control work proceeds in isolation	No impact
19	Safe outdoor testing (rain, wind)	Safety	M	M	M	Weather check; enclosure; abort criteria	Dry day; minimal sun after 15:48 limited the closed-loop stress test
20	Inaccurate model	Control sys.	M	H	H	Identify the plant at each f_{sw}; compare three models	Materialised and resolved: R_{sw} found (0.29 / 0.80 / 1.70 Ω), gain sets re-identified per f_{sw}

#	Risk	Category	L	C	Rating	Mitigation / monitoring	Status at delivery
21	Sensor faults	Hardware	M	M	M	Zero-cal at boot; plausibility checks; redundancy via KCL	U10 unusable; ADC phase-B fix on 16 Oct; KCL reconstruction cross-checks the module currents

C Analytical Solution

C.1 Single shuffler averaged model (lead-up, Semester 1)

(a) Topology and sign conventions.

(b) Coupled inductor model: $L_k = 0.5L_3$, $L_m = L_2 - 0.5L_3$; ERUC23-4R1K: R_{DC} 1.85 mΩ, L_k 2.05 µH, L_m 5.85 µH [39].

Figure 27: Single-shuffler model elements.

State, inputs. $x = [I_{L1}, I_{L2}, V_{C1}, V_{C2}]^T$, $u_d = [d_A, d_B]^T$, $u_e = [I_{PV1}, I_{PV2}, I_{LOAD}]^T$. $I_{Lk} > 0$ flows $V_{mid} \to L_k \to$ switch node; $V_{C1} = V_{top} - V_{mid}$, $V_{C2} = V_{mid}$.

Switching states. $2^4 = 16$ combinations, 7 forbidden (shoot-through), 9 valid; the 4 dual-switch states form the averaged basis with dwell fractions $d_A d_B$, $d_A(1-d_B)$, $(1-d_A)d_B$, $(1-d_A)(1-d_B)$; the single-switch and all-off states are dead-time transients.

Averaged model (with $\mathbf{L} = \begin{bmatrix} L & M \\ M & L \end{bmatrix}$):

$$\mathbf{L}\dot{\mathbf{I}} = \begin{bmatrix} -d_A V_{C1} + (1-d_A)V_{C2} \\ -d_B V_{C1} + (1-d_B)V_{C2} \end{bmatrix} - R\mathbf{I} \tag{4}$$

$$C_1 \dot{V}_{C1} = d_A I_{L1} + d_B I_{L2} + I_{PV1} - I_{LOAD} \tag{5}$$

$$C_2 \dot{V}_{C2} = I_{PV2} - (1-d_A)I_{L1} - (1-d_B)I_{L2} \tag{6}$$

Rule: C_1 receives current from each phase whose high-side switch conducts; C_2 supplies current to each phase whose low-side switch conducts.

Equilibrium. Volt-second balance forces $D_A = D_B \equiv D$ and

$$\frac{\bar{V}_{C2}}{\bar{V}_{C1}} = \frac{D}{1-D}, \qquad D = \frac{I_{LOAD} - I_{PV1}}{I_{LOAD} - I_{PV1} + I_{PV2}}, \qquad \bar{I} = \frac{I_{PV2}}{2(1-D)}. \tag{7}$$

The differential duty $d_A - d_B$ is a transient input only; a finite load is required for a non-trivial equilibrium.

Linearisation and modes. $\tilde{A}$ has identical rows 1,2 and columns 1,2 ($L_1 = L_2$). Common-mode duty $\tilde{u}_c = (\tilde{d}_A + \tilde{d}_B)/2$ drives $(I_\Sigma = I_{L1} + I_{L2}, V_{C1}, V_{C2})$; differential duty $\tilde{u}_\delta = \tilde{d}_A - \tilde{d}_B$ drives $I_\Delta = I_{L1} - I_{L2}$ through a pure integrator $L\dot{\tilde{I}}_\Delta = -V_\Sigma \tilde{u}_\delta$.

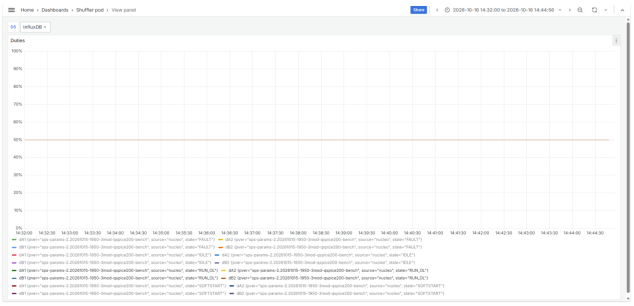
Stability. $\sigma(\tilde{A}) = \{0, 0, \pm j\omega_0\}$, $\omega_0^2 = 2D^2/(LC_1) + 2(1-D)^2/(LC_2)$; with $L = 8.2\,\mu$H, $C = 100\,\mu$F, $D = 0.5$: $\omega_0 \approx 34.9$ krad/s (5.56 kHz), $Z_0 = 0.286\,\Omega$. With the coupled inductor and $R_w = 1.85$ mΩ: $-67.3 \pm j26968$, -902, 0 (f_0 4.29 kHz, $\zeta = 0.0025$). The open-loop plant is marginally stable; the controller must provide all damping.

Controllability / observability. Controllable if $\bar{I} \neq 0$ (no DCM), $V_\Sigma \neq 0$ and $D \in (0,1)$. Observability: $\{V_{C1}, V_{C2}\}$ gives rank 3 of 4 (differential current hidden); adding I_{L1} gives rank 4. Per-phase current sensing is therefore mandatory (sensor set of section 3.5).

Discretisation. ZOH at $T_s = 5\,\mu$s: poles $0.9848 \pm 0.1737j$, 1, 1 on the unit circle.

Cascaded PI (Semester 1 design, superseded). Crossovers 10 kHz / 1 kHz / 5 kHz, PM 84°: $K_{p,i} = 0.00678$, $K_{i,i} = 42.6$; $K_{p,v} = 0.628$, $K_{i,v} = 395$; $K_{p,\delta} = 0.00339$. Three SISO loops treated the coupled states as independent; the fixed-current-reference inner loop drove the shaded module to open circuit (progress report Fig. 8). Figure 28 shows the pole map and phase portraits that motivated the MIMO reformulation.

(a) Open-loop eigenvalues: idealised model on the $j\omega$ axis; winding resistance and coupling pull the resonant pair into the LHP.



(b) Full 4-state MIMO plant: LQR + integral pulls the resonant pair off the $j\omega$ axis into a well-damped region.

(c) Phase portraits: (a) ideal, centre; (b) passive damping, $\zeta \approx 0.003$; (c) inner current loop active, stable focus.

Figure 28: Single-shuffler stability pictures (Semester 1 and break).

Re-interpretation after the hardware convention was fixed. The sensed $I_{LA} + I_{LB}$ *is* the circulating current; relabelled, the Appendix-B model gives -77 V per unit against -76.5 measured and -200 A per unit with the $0.386\,\Omega$ battery-path resistance, so the Semester 1 derivation was right in kind (DESIGN_LOG 6.1).

C.2 N-module string model (deployed)

Variables (fig. 4). Node potentials ϕ_k, $k = 1..N+1$ (node $N+1$ is ground); module voltage $v_k = \phi_k - \phi_{k+1}$. Shuffler j spans modules j, $j+1$: rails ϕ_j, ϕ_{j+2}, midpoint ϕ_{j+1}, $\bar{V}_j = v_j + v_{j+1}$. Modal duties $u_c = d_A + d_B - 1$, $u_d = d_A - d_B$ (inverse $d_A = \frac{1}{2}(1 + u_c + u_d)$, $d_B = \frac{1}{2}(1 + u_c - u_d)$). Modal currents $i_m = i_A + i_B$ (shuffled; telemetry $I_{LA} - I_{LB}$ because the B sensor is reversed) and $i_c = i_A - i_B$ (circulating; telemetry $I_{LA} + I_{LB}$). Outputs $v_{d,j} = v_j - v_{j+1} = \phi_j - 2\phi_{j+1} + \phi_{j+2}$ and $i_{c,j}$.

State equations (eqs. (1) to (3)): S injects $+1$ at the midpoint and $-\frac{1}{2}$ at each rail of shuffler j; G_n is the nodal conductance of the module slopes g_k with G_{load} added at node 1; w the module short-circuit current injections. Rail currents are $\frac{1}{2} i_m (1 \pm u_c) \pm \frac{1}{2} i_c u_d$.

Inductances. As built (ERUC23 windings 1:−1) the circulating mode sees $L + M = 13.75\,\mu$H and the shuffled mode $L - M = 2.05\,\mu$H ($L_{self} = 7.9\,\mu$H, $M = 5.85\,\mu$H). The ripple law $\Delta i_c = \bar{V}/(2 f_{sw}(L+M))$ confirms the assignment (table 5).

Equilibrium. $D_j\bar{V}_j = v_{j+1} + \frac{1}{2}R_m I_{m,j} \Rightarrow D_j/(1-D_j) = v_{j+1}/v_j$. Charge balance on the ladder: $T(N)\,I_m = \delta I$, $T = \mathrm{tridiag}(-\frac{1}{2}, 1, -\frac{1}{2})$, $(\delta I)_j = I_j - I_{j+1}$, with the closed form $T^{-1}_{jk} = \frac{2}{N}\min(j,k)(N - \max(j,k))$; string current $I_{str} = I_1 - D_1 I_{m,1} = \frac{1}{N}\sum_k I_k$. For $N = 3$: $i_{m1} = (4I_1 - 2I_2 - 2I_3)/3$. Open loop at $D = 0.5$: $v_{d,j} = -R_m i_{m,j}$. Check against QSPICE S10i ($R_m = 0.29\,\Omega$): $i_m = -0.466/-0.534$ A (model) vs $-0.464/-0.539$ A (QSPICE); I_{str} 1.032 vs 1.024 A; $R_m i_m = -0.135$ V vs $v_{d,1} = -0.135$ V.

DC gain.

$$G(N) = \begin{bmatrix} G_v & 0 \\ 0 & \mathrm{diag}(\bar{V}_j/R_c) \end{bmatrix}, \qquad G_v = -(R_m I + Z_e)^{-1} Z_e\,\mathrm{diag}(\bar{V}_j), \qquad Z_e = -D_v G_n^{-1} S, \quad (8)$$

where D_v maps $\phi \mapsto v_d$. $G(2)$: $-\bar{V}/(R_m g + 1)$. $G(3)$ voltage block: diagonal $-\bar{V}(4R_m g + 3)/(4R_m^2 g^2 + 8R_m g + 3)$, off-diagonal $2R_m \bar{V} g/(\cdot)$: diagonal $\approx -\bar{V}$ with $O(R_m g)$ neighbour terms. Numbers in table 4.

Controller. $M = G^{-1}\mathrm{diag}(k_I)$ with $k_{I,v} = 2\pi \cdot 150\,\mathrm{s}^{-1}$, $k_{I,i} = 2\pi \cdot 400\,\mathrm{s}^{-1}$; $u = k\,(M_p e + M\xi)$, $M_p = M\,\mathrm{diag}(k_{p,v}, k_{p,v}, k_{p,i}, k_{p,i})$, $k_{p,v} = 1.58 \times 10^{-4}$ s, $k_{p,i} = 1.08 \times 10^{-4}$ s, $\xi \mathrel{+}= T_s e$, $T_s = 100\,\mu$s, first-order LPF on y with $f_c = 2$ kHz ($a = 2\pi f_c T_s/(1 + 2\pi f_c T_s) = 0.557$), one tick of delay; back-calculation anti-windup $\xi = M^{-1}(u_{act} - M_p e)$. Firmware layout $u = [u_{c,1}, u_{c,2}, u_{d,1}, u_{d,2}]$, $e = [e_{vd1}, e_{vd2}, e_{ic1}, e_{ic2}]$. Design: $\tau_v = 1/(k k_{I,v})$, $\tau_i = 1/(k k_{I,i})$, rise $= 2.197\tau$, 2 % settle $= 3.91\tau$.

Table 11: Design time constants vs gain scale and the closed-loop pole frequencies at $k = 1$.

k	τ_v [ms]	τ_i [ms]	rise $2.197\tau_v$ [ms]	closed-loop poles at $k = 1$
0.1	10.61	3.98	23.3	model M: 160 / 164 Hz (v_d), 593 Hz (i_c)
0.3	3.54	1.33	7.8	deployed M: 127 / 135 Hz (v_d), 582 / 604 Hz (i_c)
1.0	1.06	0.40	2.3	

Plants used in the analysis. PV80 ($N = 3$): $V = 20.5$ V, $g = 0.19$ S per module, $C = [130, 260, 130]\,\mu$F, $G_{load} = 20$ S (DL3021 CV), $R_w = 0.061\,\Omega$ (averaged) or R_{sw}. BENCH-QSPICE: $V = 23.3$ V, $g = 0$, $I = [0.8, 1.0, 1.3]$ A (all NGE103 channels in current limit, the 16 Oct point), same C. Reduced voltage mode: $f_0 = 1/(2\pi\sqrt{L_m C_e})$ with $C_e = 173\,\mu$F gives 8.4 kHz; $\zeta = \frac{1}{2}(R_m\sqrt{C_e/L_m} + g_e\sqrt{L_m/C_e}) = 0.29$ at 61 mΩ, 1.34 at 0.29 Ω, 7.8 at 1.7 Ω.

PBH ranks (PV80). $n = 7$ states; controllable from u (7), from u_c alone (5), from u_d alone (2); observable from y (7), from v_d alone (5): circulating modes are reachable only from u_d and seen only by the current sensors; the string-sum mode (176 Hz with a current sink) is unreachable/unobservable with equal capacitors and belongs to the load / MPPT. $\mathrm{cond}(G(N))$ for PV80 grows only from 16.6 ($N = 2$) to 18.9 ($N = 8$), i.e. the decoupling scales.

C.3 Switching resistance R_{sw}

The averaged 61 mΩ plant over-predicts the current-loop gain 5×. Open-loop u_d sweeps on the QSPICE device model (S10i / S10i3 / S10i4) with and without reverse recovery and device capacitances isolate the mechanism: pole-capacitance commutation during the dead time costs volt-seconds proportional to the current,

$$R_{sw} = \frac{4\bar{V}^2 C_p f_{sw}}{\Delta i^2} = 64\,C_p\,f_{sw}^3\,(L+M)^2, \quad (9)$$

which with $C_p = 2.2$ nF gives 0.21 / 0.72 / 1.70 Ω at 200 / 300 / 400 kHz; the identified gains (142 / 58 / 23 A per unit) imply 0.27 / 0.79 / 1.66 Ω; the analysis uses 0.29 / 0.80 / 1.70 Ω. The switching twin (`nswitch.py`, exact per-configuration integration with $C_p = 2C_{oss}$) reproduces QSPICE point for point (170/165/148/111/74/66 vs 172/163/148/112/71/66 A per unit).

C.4 Open items

Unexplained 6.34 kHz / ζ 0.53 resonance seen in June 2026; C_{oss} is voltage dependent so R_{sw} should be re-measured per operating point; the $u_d \to v_d$ cross term ($-10 \ldots -25$ V per unit) is not in the model; at 400 kHz the dead time sits at the ZVS boundary and the linear model stops near $u_d \approx 0.02$.

D Controller, Supervisor and Firmware Pseudocode

```
function ctrl_step(VM[1..N], IL_A[1..NS], IL_B[1..NS], rv[1..NS], Vref[1..NS]):
    # outputs: module-voltage differences and circulating currents
    y = [VM[1]-VM[2], ..., VM[N-1]-VM[N], IL_A[1]+IL_B[1], ..., IL_A[NS]+IL_B[NS]]
    y_f += a * (y - y_f)                    # LPF, a = 2*pi*fc*Ts/(1+2*pi*fc*Ts), fc = 2 kHz
    r   = [rv + Vref, 0 ... 0]              # supervisor relaxation + manual offsets; ic refs are zero
    e   = r - y_f
    xi_try = xi + Ts * e                    # integrator candidate
    u   = kscale * (Mp @ e + M @ xi_try)    # u = [uc1..ucNS, ud1..udNS]; M = inv(G) diag(kI)
    for each shuffler j:
        dA[j] = clamp(0.5 + 0.5*(u[j] + u[NS+j]), dmin, dmax)
        dB[j] = clamp(0.5 + 0.5*(u[j] - u[NS+j]), dmin, dmax)
    u_act = [dA + dB - 1, dA - dB]          # what was actually applied
    xi = Minv @ (u_act / kscale - Mp @ e)   # back-calculation anti-windup (every tick)
    return dA, dB
```

Listing 1: MIMO decoupling PI, one 100 μs tick (`shuffler_ctrl.c`).

```
function supervisor_step(IM[1..N], IL_A, IL_B, d[1..NL], en):
    if not en: beta = 0; rv = 0; return rv
    id   = IL_A - IL_B                      # shuffled current per shuffler
    imf += alpha * (IM - imf); idf += alpha * (id - idf) # alpha = Ts3/TF = 0.2
    dl   = max(imf) - imf                   # deficit of each module vs the strongest
    stress = max( max|idf| / I_RATED, max|d - 0.5| / 0.45 ) # I_RATED = 10 A
    if   stress > S_HI: beta += Ts3 * KB * (stress - S_HI) # S_HI 0.9, KB 60
    elif stress < S_LO: beta -= Ts3 * KREL * (S_LO - stress) # S_LO 0.7, KREL 6
    beta = clamp(beta, 0, BMAX)                          # BMAX 4
    for each shuffler k: rv[k] = clamp(beta * (dl[k+1] - dl[k]), -RVMAX, RVMAX) # 30 V
    return rv                               # relaxes equalisation when a shuffler is stressed
```

Listing 2: Feasibility governor / supervisor, 1 ms (`shuffler_supervisor.c`).

```
function recon(IL_A, IL_B, dA, dB):
    for each shuffler k:
        ia = IL_A[k]; ib = -IL_B[k]         # B sensor is reversed on the board
        itop[k] = dA[k]*ia + dB[k]*ib       # averaged current drawn at the top rail
        ic[k]   = ia + ib                   # injected at the midpoint
        ibot[k] = ic[k] - itop[k]           # returned at the bottom rail
    IM[N] = 0
    for j = N-1 down to 1:
        IM[j] = IM[j+1] + ic[j] - (j < NS ? itop[j+1] : 0) - (j >= 2 ? ibot[j-1] : 0)
    return IM                               # relative module currents (offset-free deficits)
```

Listing 3: Module-current reconstruction by KCL (`shuffler_recon.c`); the string current is unobservable so $I_N = 0$ is the reference.

```
function mppt_step(Vstr, Istr, beta, en):
    if not en: Iref = IREF0; dir = +1; boot = 1; return Iref # IREF0 8 A
```

```
    acc += Vstr * Istr; n += 1
    if n == NWIN:                                          # 20 ticks = 20 ms
        P = acc / NWIN; acc = 0; n = 0
        if |beta - beta0| > 1e-3: boot = 1      # governor moved: hold the reference
        elif boot: Pprev = P; Iref += dir * S0; boot = 0      # S0 0.25 A
        else:
            dP = P - Pprev
            step = clamp(KA * |dP|, SMIN, SMAX)               # 0.01, 0.05 .. 1 A
            if   dP > DB: Iref += dir * step                  # DB 1 W
            elif dP < -DB: dir = -dir; Iref += dir * step
            Pprev = P
        beta0 = beta; Iref = clamp(Iref, IMIN, IMAX)          # 2 .. 8.5 A
    return Iref
```

Listing 4: Adaptive-step perturb-and-observe MPPT, 1 ms (`shuffler_mppt.c`); compiled in but disabled on hardware because the load performs the string MPPT.

```
on HRTIM_master_repetition:                  # every 20 / 30 / 40 PWM periods
    Sensors_Tick()                           # 80-sample means per channel; OC / rail / DMA-stall latch
    IL = Sensors_Current_A()                 # (code - zero) * 3300/4096 mV / (100 mV/A)
    if Sensors_FaultLatched(): trip(FLT_OC_LATCH)
    prt_isr_service()                        # pending C,APPLY: swap gains, xi = Minv * u_act(d)
    VM = snapshot of the 1 kHz INA task (VM1 from vstr_pi - VM2 - VM3 when vm1_src = 2)
    if state == SOFTSTART:
        d = ramp(d_start, target, n_tick / 1500)      # 150 ms linear ramp
    elif state == RUN and mode is CL:
        out = stack_step({VM, IL, Vstr = sum VM, Vref, en_sup, en_mppt = 0})
        d = clamp(out.d, d_prev - slew, d_prev + slew) # slew 0.01 per tick
    elif state == RUN: d = open-loop duties (C,DUTY / C,DUTYS)
    else:              d[i] = Vbot / (Vtop + Vbot)    # pre-bias while idle (0.5 if unknown)

    for each leg: hw_set_duty(leg, d)      # clamp to the f_sw limits, write CMP1 (preloaded)
```

Listing 5: Control ISR, 10 kHz (HRTIM master repetition interrupt, `app.c`).

```
function stack_step(in):
    if tick % STACK_DECIM == 0:              # 10 -> 1 kHz
        IM   = recon(in.IL_A, in.IL_B, dA, dB)
        rv   = supervisor_step(IM, in.IL_A, in.IL_B, d, in.en_sup)
        Iref = mppt_step(in.Vstr, in.Istr, beta, in.en_mppt)
    dA, dB = ctrl_step(in.VM, in.IL_A, in.IL_B, rv_delayed, in.Vref) # rv through a unit delay
    tick += 1
    return {dA, dB, rv, beta, stress, IM, Iref}
```

Listing 6: Stack scheduler (`stack.c`): controller every tick, supervisory layers every 10th.

```
every 10 ms:
    kick IWDG                                  # 0.3 s window
    if state in {RUN, SOFTSTART}:
        if E-stop pin low:                     trip(FLT_ESTOP)
        OV / UV / string-OV counters: 3 consecutive samples -> trip(code) # 30 V, 3 V (CL), 85 V
        if CL and now - last_rx > wdg_s:       trip(FLT_COMMS_WDG)       # 5 s without a C line
        if CL and vm1_src == 2 and VM1 stale: trip(FLT_VSTR_STALE)
        if any INA die > 85 C:                 trip(FLT_OVERTEMP)
```

```
        if CL and a needed VM unreadable:   trip(FLT_INA)
    switch state:
        IDLE:     if enable and mode != IDLE: pins -> HRTIM; arm synchronised enable; state
             = SOFTSTART
        SOFTSTART: if not enable: state = STOP; elif ramp done: stack_init(); state = RUN
        RUN:      if not enable or mode == IDLE: state = STOP
        STOP:     legs off; enable = 0; state = IDLE
        FAULT:    hw_outputs_off()          # every 10 ms until C,CLEAR
    send S line on every state change; send F line once per latched fault

function trip(code):
    hw_outputs_off(); gates low             # first, from any context
    faults |= code; enable = 0; state = FAULT
```

Listing 7: Safety task and state machine, 100 Hz (`app.c`); `trip()` may run from any context.

```
while IDLE: every leg pair is held at d0 = Vbot / (Vtop + Vbot) # matches the natural
    midpoint
on enable: outputs switch on at master CMP1 = PER/4 (centred first pulse, 16-try fallback
    window)
SS_TICKS = 150 ms * 10 kHz = 1500
tick 0: d_start[i] = d_applied[i]
each tick n: frac = min(n / SS_TICKS, 1)
    target = 0.5 if CL else OL duty[i]
    d[i] = d_start[i] + (target - d_start[i]) * frac
n == SS_TICKS: ss_done -> safety task moves to RUN; stack_init() so the controller starts
    from zero integrators
```

Listing 8: Soft start (`app.c` + `hw_legs_enable_sync`).

```
pod.py:
    RX thread: read line from /dev/ttyACM0 @ 921600 -> telemetry_parser.parse_line(line)
               -> InfluxDB points (telemetry / status / fault / note / cmd), batched 200
                   pts / 500 ms
    TX thread: one byte every 2 ms (firmware RX is one IRQ per byte); C,PING every 1 s (
        comms watchdog)
    main loop (0.2 s): DS18B20 heatsink temperature -> Influx (C,ESTOP if > 80 C); LCD
        state/values; RGB LED by state
    control socket 127.0.0.1:5017: podctl / params_web / remote.py send commands; IR keys
        -> presets (actions.py)
eload.py:
    5 Hz:  :MEAS:VOLT? :MEAS:CURR? :MEAS:POW? -> Influx (source = eload)
    1 Hz:  mode, levels, ranges, questionable status
    2 Hz:  C,SET,vstr_pi,<V> -> Nucleo (VM1 source when U10 is unreadable)
```

Listing 9: Pi pod main loops (`pod.py`, `eload.py`).

E Firmware, Peripherals and Telemetry Contract

E.1 Scheduling

Clock: HSI 16 MHz → PLL → SYSCLK 170 MHz (AHB/APB 170 MHz, flash latency 4, boost scale); the 400 kHz build feeds the ADCs from PLL-P at 56.7 MHz. IWDG 0.3 s. ISR cost is measured with the DWT cycle counter against the 100 μs budget (Release `-O2`; `-O0` reached 75 μs idle and $>$ 100 μs in CL at 300 kHz). Toolchain: CMake $\geq$ 3.22, Ninja, arm-none-eabi-gcc (Cortex-M4F, hard float, nano specs), presets Debug/Release; `build_wsl.sh` uses the Windows STM32Cube bundle from WSL.

Table 12: Execution contexts of the firmware (bare-metal superloop plus interrupts; no RTOS).

Context	Trigger / priority	Rate	Work
`app_ctrl_isr`	HRTIM master repetition IRQ, NVIC prio 1	10 kHz ($T_s = 100\,\mu$s)	hall averages + OC latch, gain swap, soft-start / `stack_step` / OL duties / pre-bias, slew limit, write CMP1
`app_tick_1ms`	SysTick, prio 0	1 kHz	ms counter; main-loop stall check (> 20 ms while switching $\Rightarrow$ fault 0x2000)
`app_task_sense_v`	main loop	1 kHz	INA239 $\times 3$ over SPI3 $\rightarrow V_{M1..3}$, die temperatures
`app_task_safety`	main loop, every 10th pass	100 Hz	IWDG kick, OV/UV/string-OV, watchdogs, state machine
`app_task_telemetry`	every 20th pass	50 Hz	`T` frame; `S` line at 1 Hz and on state change
`app_task_cmd`	every pass	1 kHz	parse `C,...` lines, answer `A,...`
E-stop	EXTI0 on PC0, prio 0	event	`trip(FLT_ESTOP)`
LPUART1 RX	prio 6, 1 byte per IRQ	event	512-byte ring buffer

E.2 Peripherals

E.3 Fault bits

E.4 Telemetry and command contract

Telemetry frame (50 Hz, 27 fields):

```
T,ms,state,fault(hex),VM1,VM2,VM3,ILA1,ILB1,ILA2,ILB2,dA1,dB1,dA2,dB2,rv1,rv2,beta,stress,
    IM1r,IM2r,T1,T2,T3,kscale,pset,pver
S,ms,state,fault,mode,fw,pver     F,ms,code,channel,value,raw     A,<cmd>,OK|ERR,<why>
    P,name,val[,min,max,class]
```

States: `BOOT`, `SELFTEST`, `IDLE`, `SOFTSTART`, `RUN_OL`, `RUN_CL`, `RUN_OL1`, `RUN_CL1`, `RUN_OL2`, `RUN_CL2`, `STOP`, `FAULT`. Outside RUN_CL the fields `rv`, `beta`, `stress`, `IM1r`, `IM2r` repeat their last value (the "stale values" seen on the Grafana screenshots); the analysis keeps RUN rows only.

Commands (Pi $\rightarrow$ Nucleo): `C,MODE,<IDLE|OL|CL|OL1|CL1|OL2|CL2>`; `C,TEST,0..3`; `C,EN,0|1`; `C,DUTY,leg,d`; `C,DUTYS,dA1,dB1,dA2,dB2`; `C,VREF,rv1,rv2`; `C,SUP,0|1`; `C,SET,name,value`; `C,SETM,m00..m33`; `C,GET,name`; `C,LIST`; `C,APPLY`; `C,REVERT`; `C,CAL,ch,mV/A,zero`; `C,ESTOP`; `C,CLEAR`; `C,PING`. Parameter classes: STAGED (take effect on `C,APPLY`: `kscale`, `kp_v`, `kp_i`, `fc_lpf`, `filt_ma`, `dmin`, `dmax`, `m00..m33`); LIVE (`sup_*`, `vref1/2`, `hall_mva*`, `oc_a`, `vm_ov`, `vm_uv`, `vstr_ov`, `vstr_param`, `vstr_pi`, `vm1_prot`, `slew`, `wdg_s`); IDLE-only (`hall_zero*`, `vm1_src`). `pset` increments on every swap; a swap rebases the integrators ($\xi = M^{-1}u_{act}$) so the duties do not jump.

E.5 Firmware images

E.6 Raspberry Pi stack

Docker: `influxdb:2.7` on :8086 (org `boost`, bucket `shuffler`, infinite retention) and `grafana/grafana-oss:11.2.0` on :3000 (provisioned Flux datasource, dashboard "Shuffler pod": mode, active, init check, temperatures, state, module voltages, inductor currents, duties, supervisor, reconstructed module currents, string voltage INA-sum vs e-load, e-load status/V/I, string power, command log, run-time parameters). Services (systemd): `pod.service` (`pod.py`, 921600 baud), `eload.service` (Rigol DL3021 over USB-TMC, 5 Hz measurements, 2 Hz V_{str} forward, limits CC 15 A / CV 150 V / CP 200 W), `params_web.service` (stdlib HTTP on :8080, no authentication, allow-listed commands). IR: Elegoo NEC remote via `gpio-ir` on GPIO17 (POWER = E-stop, FUNC =

Table 13: Peripheral configuration (`shuffler_hw.c/h`, `sensors.c`, `ina2x9.c`, `comms.c`).

HRTIM1	$f_{HRTIM} = 170\,\text{MHz} \times 32 = 5.44\,\text{GHz}$ (184 ps). Master PER 27200 (200 kHz) / 18133 (300) / 13600 (400). Legs A–D: A legs reset on master PER, B legs on master CMP2 = PER/2 (180° interleave). Duty = leg CMP1, preloaded, update on reset. Output 1 set on the master event, reset on CMP1; output 2 is the complement from the dead-time generator. Dead time `HW_DEADTIME_NS` 100 (136 steps × 0.735 ns, both edges; was 200 ns until 16 Oct). Pins: leg 0 S1A PA8/PA9, leg 1 S1B PA10/PA11, leg 2 S2A PB12/PB13, leg 3 S2B PB14/PB15 (AF13); a constructor drives them low before `main()`. Duty limits 0.05–0.95 (200 kHz), 0.08–0.95 (300), 0.12–0.88 (400). Synchronised enable at master CMP1 = PER/4. Hardware fault inputs not used (COMP/DAC OC trip reserved, bit 0x002).
ADC1 / ADC2	12-bit, regular group of 2 ranks, trigger HRTIM ADC trigger 1 = TA CMP3/CMP4 + TB CMP3/CMP4 (4 per period at PER/8 + kPER/4). Sample time 6.5 cycles, clock PCLK/4 = 42.5 MHz, no oversampling, offset calibration at start. ADC1 IN1 = PA0 (I_{LA1}), IN2 = PA1 (I_{LA2}); ADC2 IN17 = PA4 (I_{LB1}), IN7 = PC1 (I_{LB2}). DMA1 ch1/ch2 circular half-word, 160 samples = 4 triggers × 20 periods × 2 ranks = one control tick; the ISR polls the TC flags.
Hall conversion	$I = (\overline{\text{code}} - \text{zero}) \cdot (3300/4096\,\text{mV})/S$, $S = 100$ mV/A on all four channels (B phases were 25 before 16 Oct; `params.yaml` still carries 25 for channels 1/3); nominal zero 2048 (1.65 V); boot zero-cal averages 64 ticks, falls back to 2048 and sets `FLT_ZEROCAL` if off by $> \pm 124$ codes. B-phase sensors are physically reversed. Run-time `hall_mva0..3` (± 400, $\lvert x \rvert \geq 10$), `hall_zero0..3` (1800–2300). Box average of 80 samples per channel per tick (nulls at f_{sw}, $2f_{sw}$, … and every 10 kHz multiple). OC latch: $\lvert I \rvert >$ `oc_a` (12 A) for 20 ticks, code < 40 or > 4055 for 3 ticks, or DMA not wrapping for 3 ticks.
SPI3	Master, mode 1, 8-bit MSB-first, software NSS, prescaler 128 ⇒ 1.33 MHz; PB3 SCK, PB4 MISO, PB5 MOSI; CS1 PA7 (U12, PV3+ to GND), CS2 PB6 (U11, PV2+ to GND), CS3 PC7 (U10 behind ISO7762 U13, PV1+ to PV3+). INA239 ADC_CONFIG 0xF248 (continuous, 84 µs, AVG 1); VBUS 3.125 mV/LSB, die temperature 125 m°C/LSB; INA229 auto-detected (195.3 µV/LSB); absent devices re-probed at 1 Hz. $V_{M3} = U12$, $V_{M2} = U11 - U12$, $V_{M1} = -U10 - V_{M2}$ (U10 wired reversed) or, by default (`vm1_src` = 2), $V_{M1} = v_{str,pi} - U11$ (stale after 2 s).
LPUART1	ST-LINK VCP PA2/PA3, 921600 8N1, blocking TX, protocol version 2, firmware `shuffler-fw-0.4.1`.
GPIO	E-stop PC0 (pull-up, falling-edge EXTI, prio 0); B1 toggles enable; LD2 PA5.
Controller filter	IIR $y_f \mathrel{+}= a(y - y_f)$, $a = 0.5569$ at $f_c = 2$ kHz; optional moving average of 1–8 ticks (`filt_ma`). Supervisor filter $\alpha = 0.2$.

Table 14: Fault bitmask (`app.h`); OV/UV/string-OV need 3 consecutive 10 ms samples; V_{M1} is excluded unless `vm1_prot` = 1.

Bit	Meaning	Bit	Meaning
0x001	OC latch / ADC rail / DMA stall	0x080	over-temperature (INA die $>$ 85 °C)
0x002	hardware OC (COMP, not implemented)	0x100	watchdog reset
0x004	E-stop	0x200	software E-stop (`C,ESTOP`)
0x008	module OV $>$ 30 V	0x400	string OV $>$ 85 V
0x010	module UV $<$ 3 V (closed loop only)	0x800	comms watchdog (CL, 5 s without a command)
0x020	zero-calibration	0x1000	$v_{str,pi}$ stale
0x040	INA bus / VM unreadable	0x2000	main loop stalled $>$ 20 ms while switching

Table 15: Firmware variants used on 16 and 20 Oct (`7_Pi/firmware/VERSION.txt`).

Image	f_{sw} [kHz]	PER	periods / tick	duty limits	gain set (`pver`)
`shuffler.bin`	200	27200	20	0.05–0.95	`20261015-1950-3mod-qspice200-bench`
`shuffler_300k.bin`	300	18133	30	0.08–0.95	`20261016-1628-3mod-qspice300-bench`
`shuffler_400k.bin`	400	13600	40	0.12–0.88	`20261016-1751-3mod-qspice400-bench`

disable, digits = presets, PLAY = run, VOL± = kscale ±0.1, EQ = status pages). Presets (`actions.py`): 0 init check, 1 open loop, 2 closed loop, 3/4 load on/off, 5/6 load remote/local, 7/8 kscale 0.3/1.0, 9 clear fault. Flashing: OpenOCD (`interface/stlink.cfg`, `target/stm32g4x.cfg`) with the pod stopped around the flash; `stage_firmware.sh` builds Release and copies the `.bin`; `sync.sh` rsyncs `7_Pi` to the Pi; `setup.sh` installs packages, udev rules, overlays (I^2C LCD 0x27/0x3F, 1-wire DS18B20 GPIO4, IR GPIO17), the venv and the services.

▷ Inconsistencies found while documenting (fix in the repo or mention): README says IWDG 2 s and dead time 200 ns, code says 0.3 s and 100 ns; telemetry.md says `%.4g`, code prints fixed decimals; `params.yaml` still has 25 mV/A for the B-phase halls; no script builds the 300k/400k images (built by hand with `HW_PWM_KHZ`); no MISRA statement anywhere (the proposal promised MISRA-C).

F Gain Sets

Every gain set is a JSON file in `6_RTOS/ctrl_core/params/` (`version`, `n_mod`, `ctrl`, `sup`, `mppt`); `gen_params.py` renders it to `params.h` (M, M_p, M^{-1} as float32 literals). All 3-module sets use $T_s = 100\,\mu$s, $f_c = 2$kHz, $d_{min/max} = 0.08/0.92$, $k_{p,v} = 1.58 \times 10^{-4}$, $k_{p,i} = 1.08 \times 10^{-4}$. Rows are $[u_{c,1}, u_{c,2}, u_{d,1}, u_{d,2}]$, columns $[e_{vd1}, e_{vd2}, e_{ic1}, e_{ic2}]$.

Table 16: Base gain matrices M of the deployed and candidate gain sets.

Set (`version`, sps-params-2…)	row	e_{vd1}	e_{vd2}	e_{ic1}	e_{ic2}
`20261015-1754-3mod-simulink-bench` (rejected)	$u_{c,1}$	−20.477	0.010	−0.046	0.092
	$u_{c,2}$	−0.008	−19.903	−0.089	0.045
	$u_{d,1}$	0	0	3.390	0
	$u_{d,2}$	0	0	0	3.385
`20261015-1950-3mod-qspice200-bench` (200 kHz, deployed)	$u_{c,1}$	−20.284	0.122	−9.384	0.105
	$u_{c,2}$	0.207	−20.010	0.055	−10.521
	$u_{d,1}$	−0.146	−0.044	17.569	−0.202
	$u_{d,2}$	−0.195	−0.131	−0.102	18.019
`20261016-1628-3mod-qspice300-bench` (300 kHz, deployed)	$u_{c,1}$	−20.739	0.400	−15.980	0.196
	$u_{c,2}$	0.127	−20.602	0.581	−18.513
	$u_{d,1}$	−0.125	−0.167	42.703	−1.068
	$u_{d,2}$	−0.052	−0.066	−1.532	44.231
`20261016-1751-3mod-qspice400-bench` (400 kHz, deployed)	$u_{c,1}$	−22.683	0.684	−36.942	−0.076
	$u_{c,2}$	−0.306	−22.038	−0.106	−40.278
	$u_{d,1}$	0.152	0.197	107.293	−3.903
	$u_{d,2}$	0.246	0.191	−3.838	109.866
`20261016-1804-3mod-qspice400d1-bench` (400 kHz, 100 ns dead time, not deployed)	$u_{c,1}$	−22.048	0.847	−23.059	−0.907
	$u_{c,2}$	0.004	−21.495	−1.370	−25.126
	$u_{d,1}$	−0.080	−0.083	95.867	−0.201
	$u_{d,2}$	−0.117	−0.072	−0.892	96.629
`20261011-1634-3mod-qspice` (S9i, 100 kHz, 150 W panels)	$u_{c,1}$	−25.225	−0.538	1.947	0.248
	$u_{c,2}$	−0.273	−24.997	−0.471	−2.092
	$u_{d,1}$	0.069	−0.018	9.999	1.180
	$u_{d,2}$	−0.030	0.101	1.867	11.874

▷ Simulink vs QSPICE: the voltage rows agree ($u_c/e_{vd} \approx -20$, i.e. $k_{I,v}/\bar{V}$), the current rows differ 5× (3.39 vs 17.6) because the switching plant's current-loop gain is $\bar{V}/R_{sw}$, not $\bar{V}/R_{phase}$; the QSPICE sets also carry the u_c/e_{ic} cross terms (−9 to −37) that cancel the dead-time coupling v_d/u_d.

▷ Deployment record (telemetry `pver`): 16 Oct bench qspice200 12:59–16:29, qspice300 from 16:29 (brief returns to 200 at 16:56–17:03 and 17:13–17:14), qspice400 from 17:52; 20 Oct solar qspice200 throughout, k from 0.1 to 0.8. The MIL 4-module set (`sps-params-1.0.0-mil4`, 6×6) is only used by the MIL model at $k = 1$.

▷ The ident script refuses to write a set whose $u_d \to i_c$ diagonal is ≤ 0 and warns if $\mathrm{cond}(G) > 100$; the previous JSON is archived under its version name.

G Simulink Validate Suites

Figures exported by `shuffler_string_validate.m` (vector PDF, light theme, zoom tiles) for the MIL model (4 modules, 3 shufflers, $k = 1$, cold start) and the carpark twin (3 modules, 2 shufflers, sensing model). Each test has a *control* figure (v_d, i_c, duties, with a 0–10 ms zoom), a *string* figure (module voltages, currents, powers) and a *supervisory* figure (stress, β, r_v). Pass criteria: settled ($|\Delta v_d| < 0.05$ V, $|\Delta i_c| < 0.5$ A between two 10 ms windows), track ($|v_d - r_v| < 0.25$ V, $|i_c| < 1$ A), MPPT ($\sum P_{mod} > 98.5\,\%$ of the ceiling), protect ($|i_c| < 1$ A, $|i_{diff}| \leq 10$ A).

Table 17: Validate test matrix.

#	Test	irradiance on the shaded module	r_v	T_{stop}	mode	result (MIL / carpark)
01	uniform-zero	1000 W/m^2	0	0.06 s	track	PASS / PASS
02	decoupling	1000	[0 2 0] (carpark [0 2])	0.06	track	PASS / PASS
03	shade-balanced	500	0	0.06	track	PASS / PASS
04	shade-refs	500	[0 2 −2] (carpark [2 −2])	0.06	track	PASS / PASS
05	shade-deep	200	0	0.50	protect	PASS / PASS
06	mppt-uniform	1000	0	0.60	mppt	PASS / (CC load only)
07	mppt-shade	500	0	0.60	mppt	PASS / (CC load only)

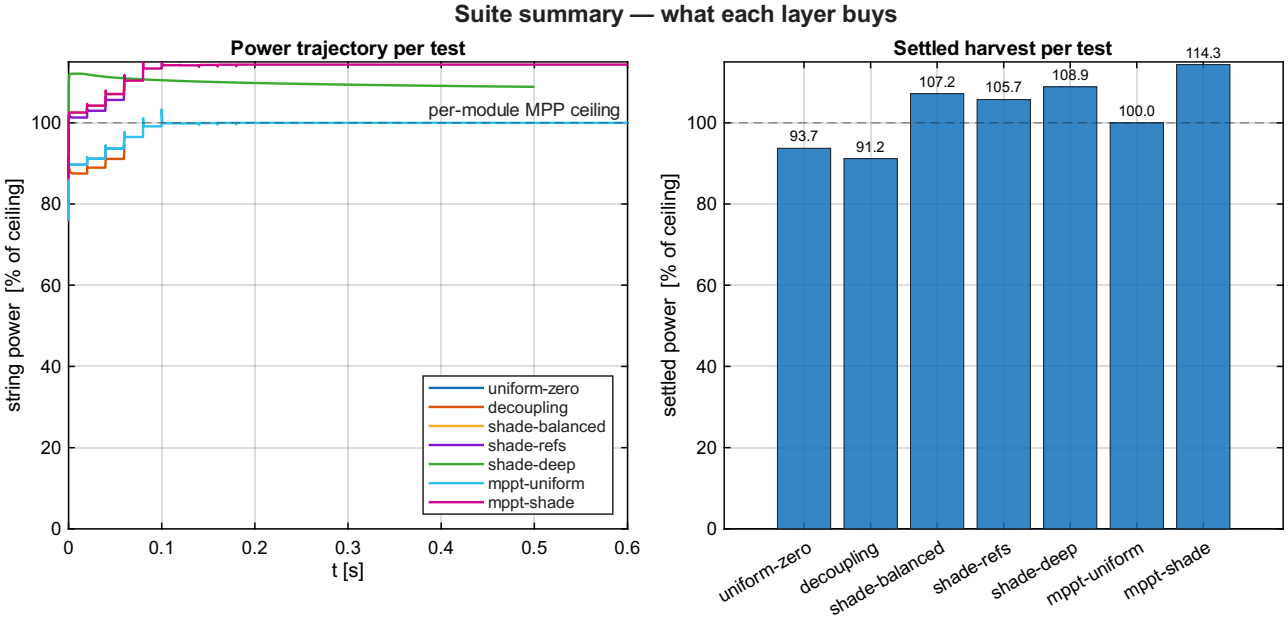


Figure 29: MIL validate suite summary (7 tests).

G.1 MIL (4 modules, 3 shufflers)

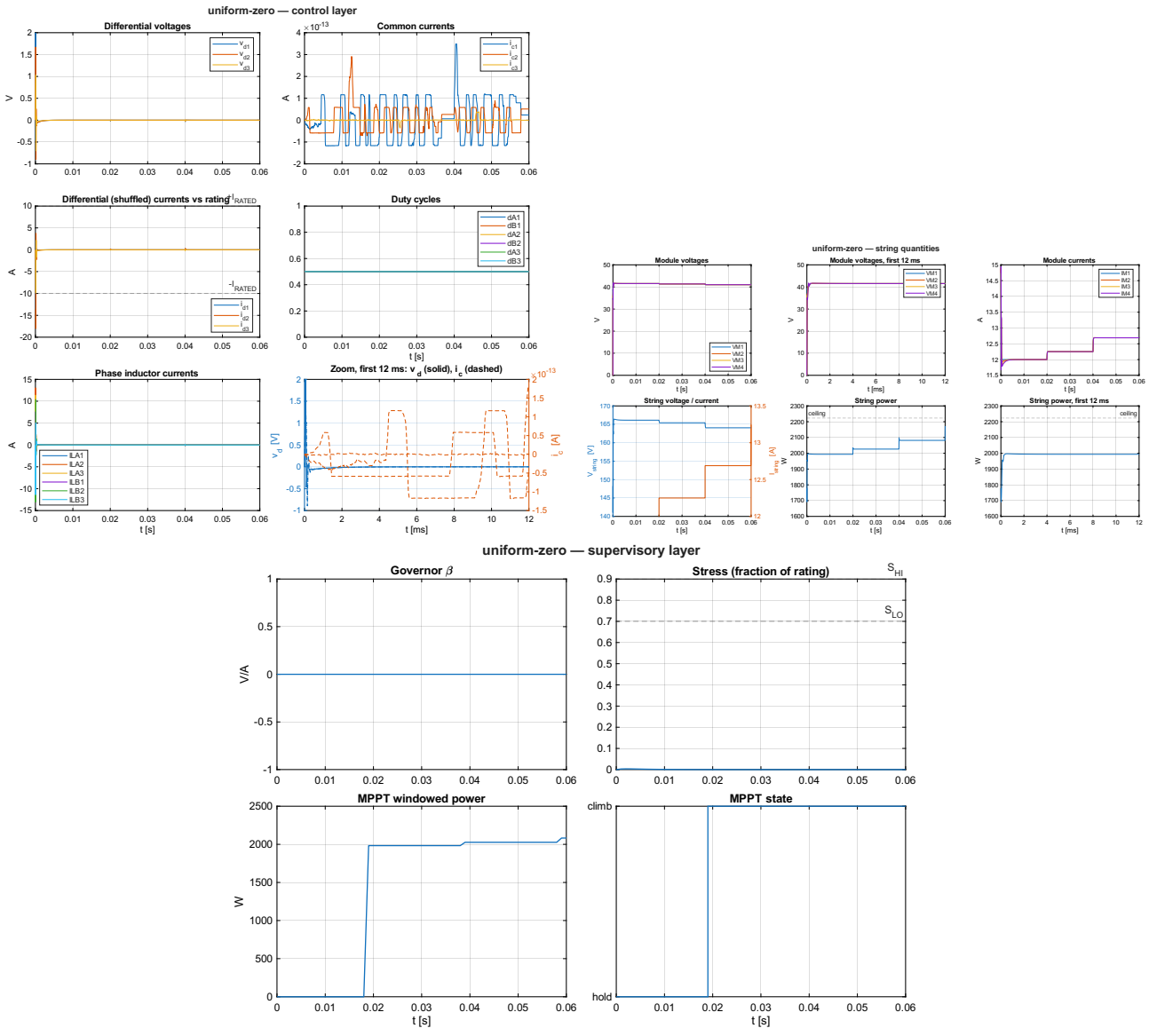


Figure 30: MIL test 01 (uniform-zero): control, string and supervisory views.

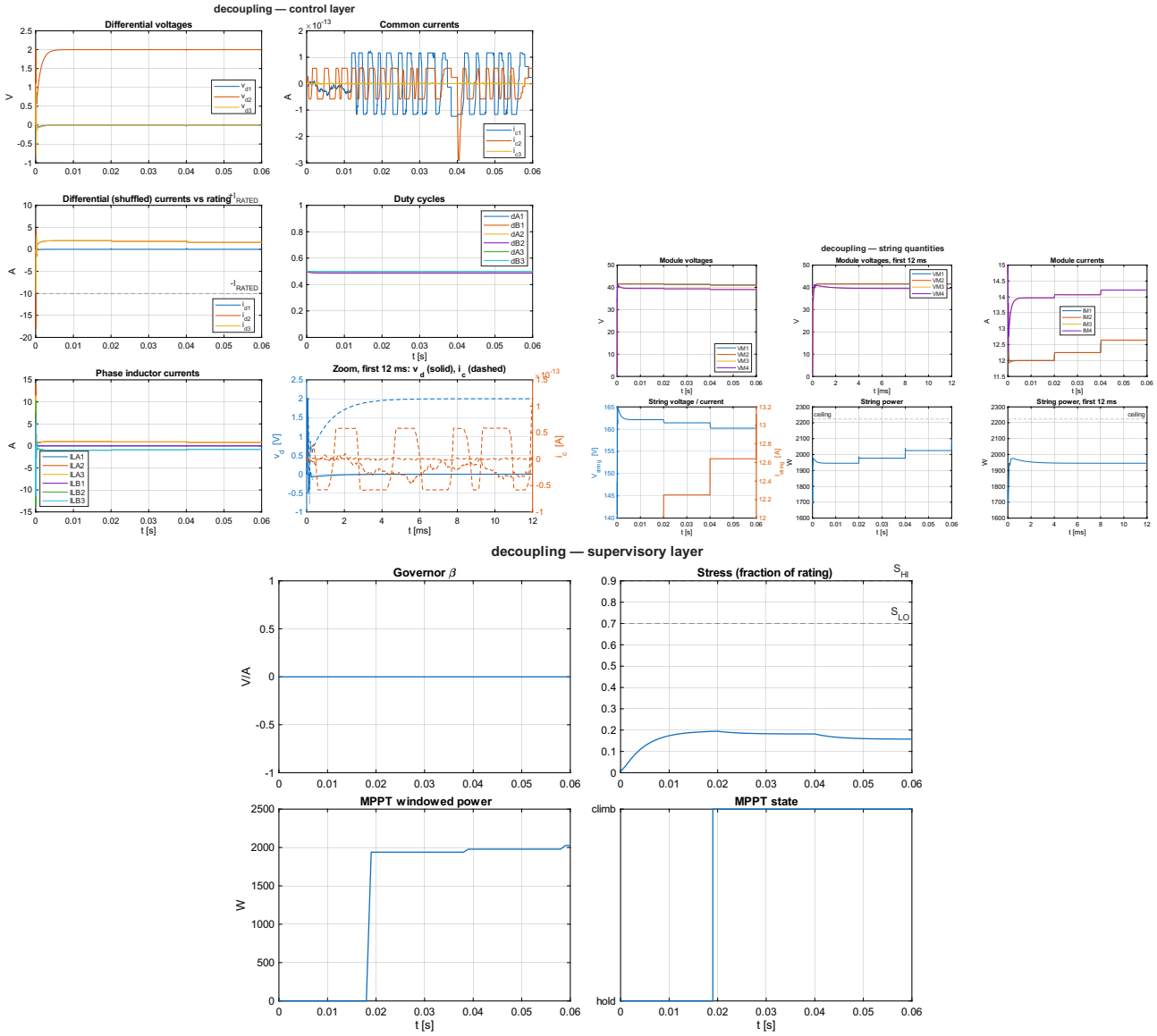


Figure 31: MIL test 02 (decoupling): control, string and supervisory views.

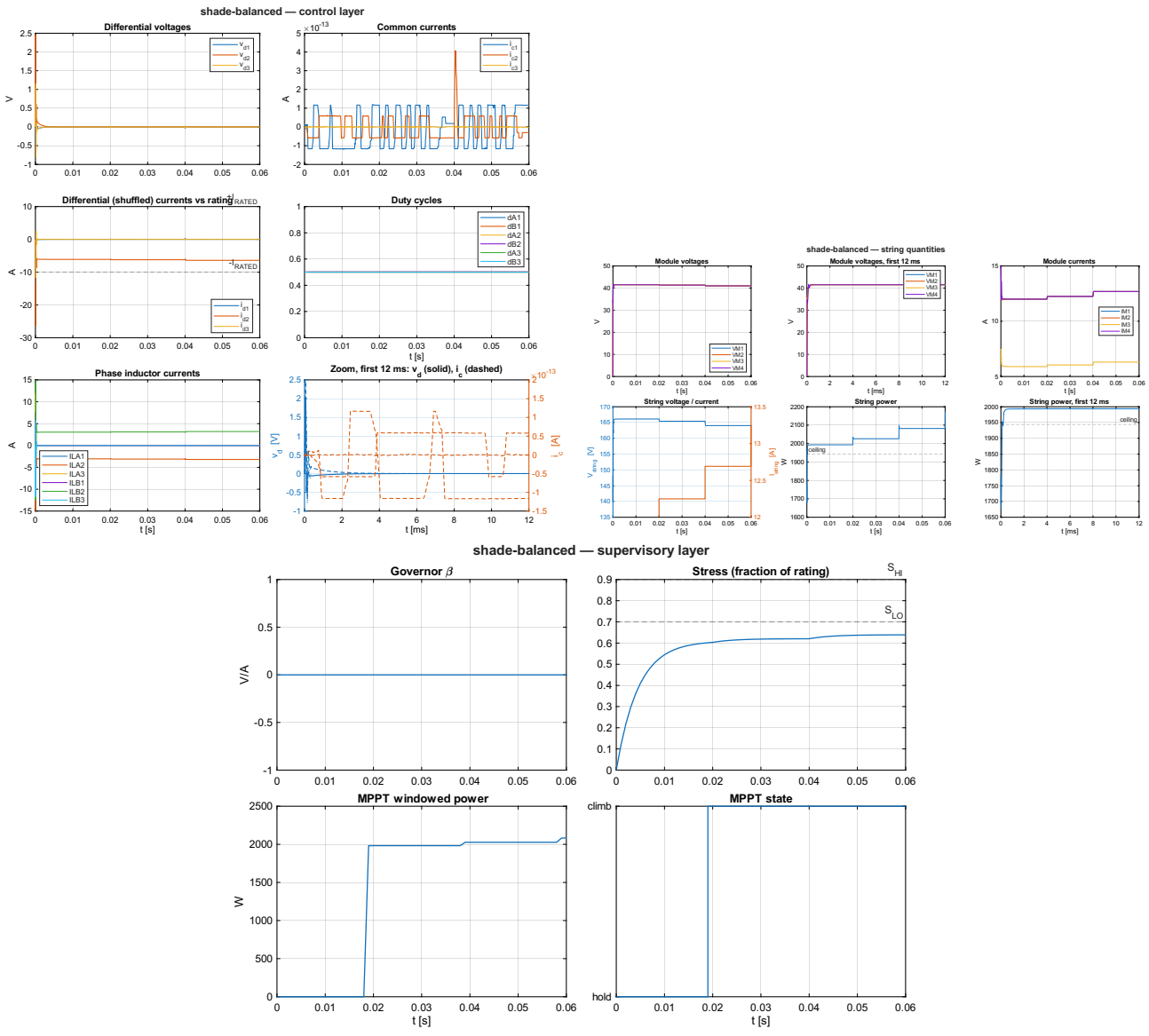


Figure 32: MIL test 03 (shade-balanced): control, string and supervisory views.

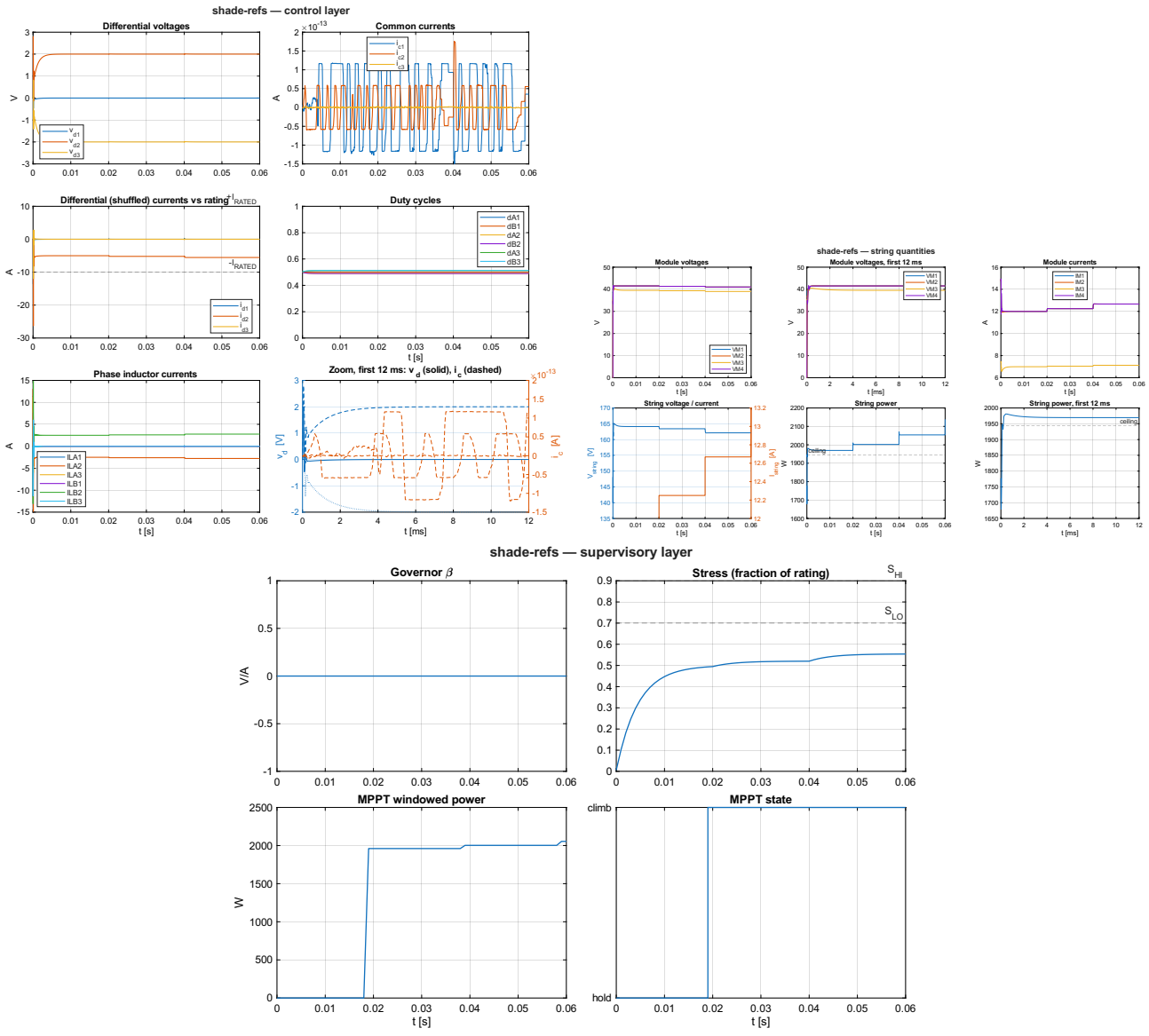


Figure 33: MIL test 04 (shade-refs): control, string and supervisory views.

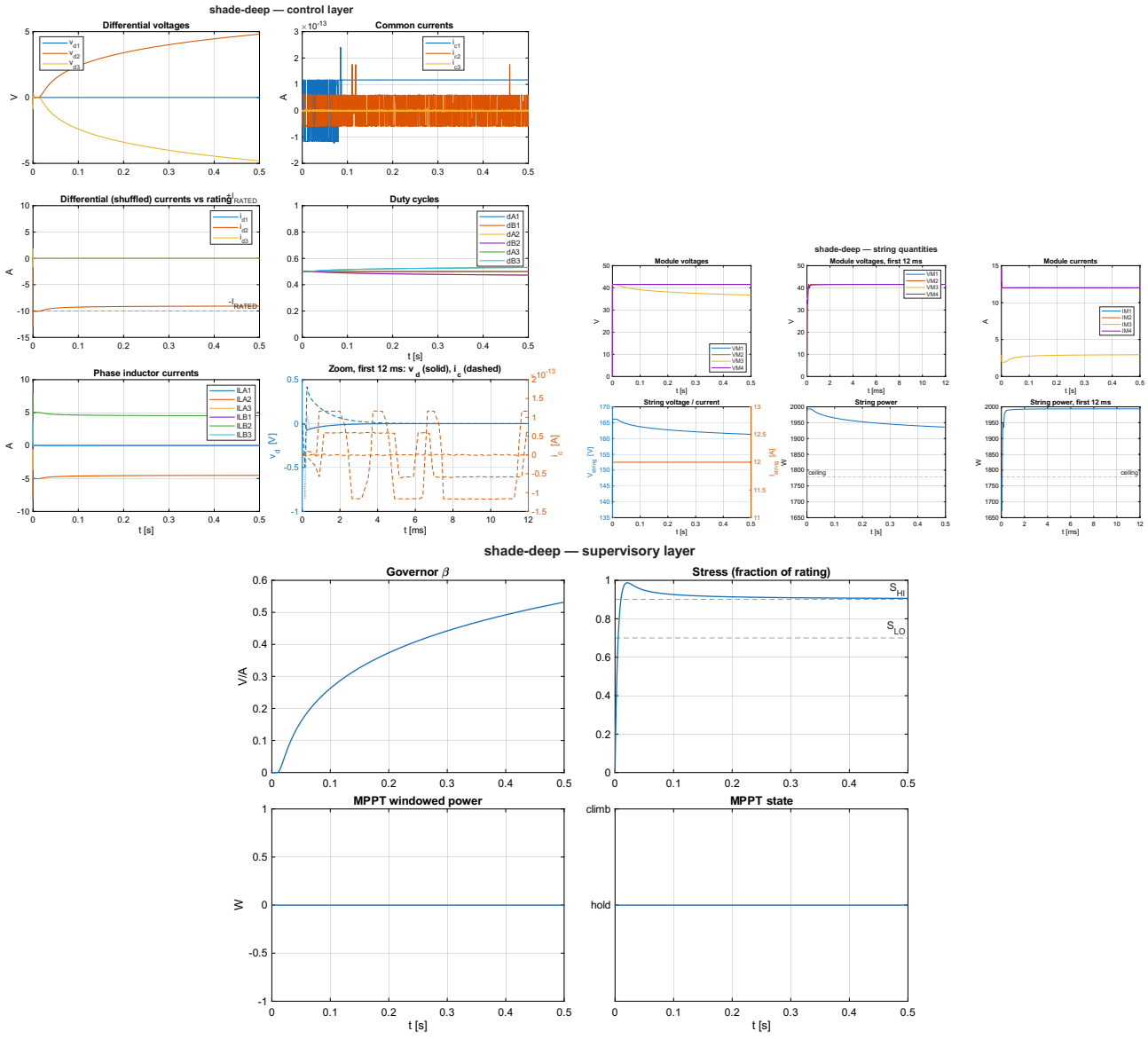


Figure 34: MIL test 05 (shade-deep): control, string and supervisory views.

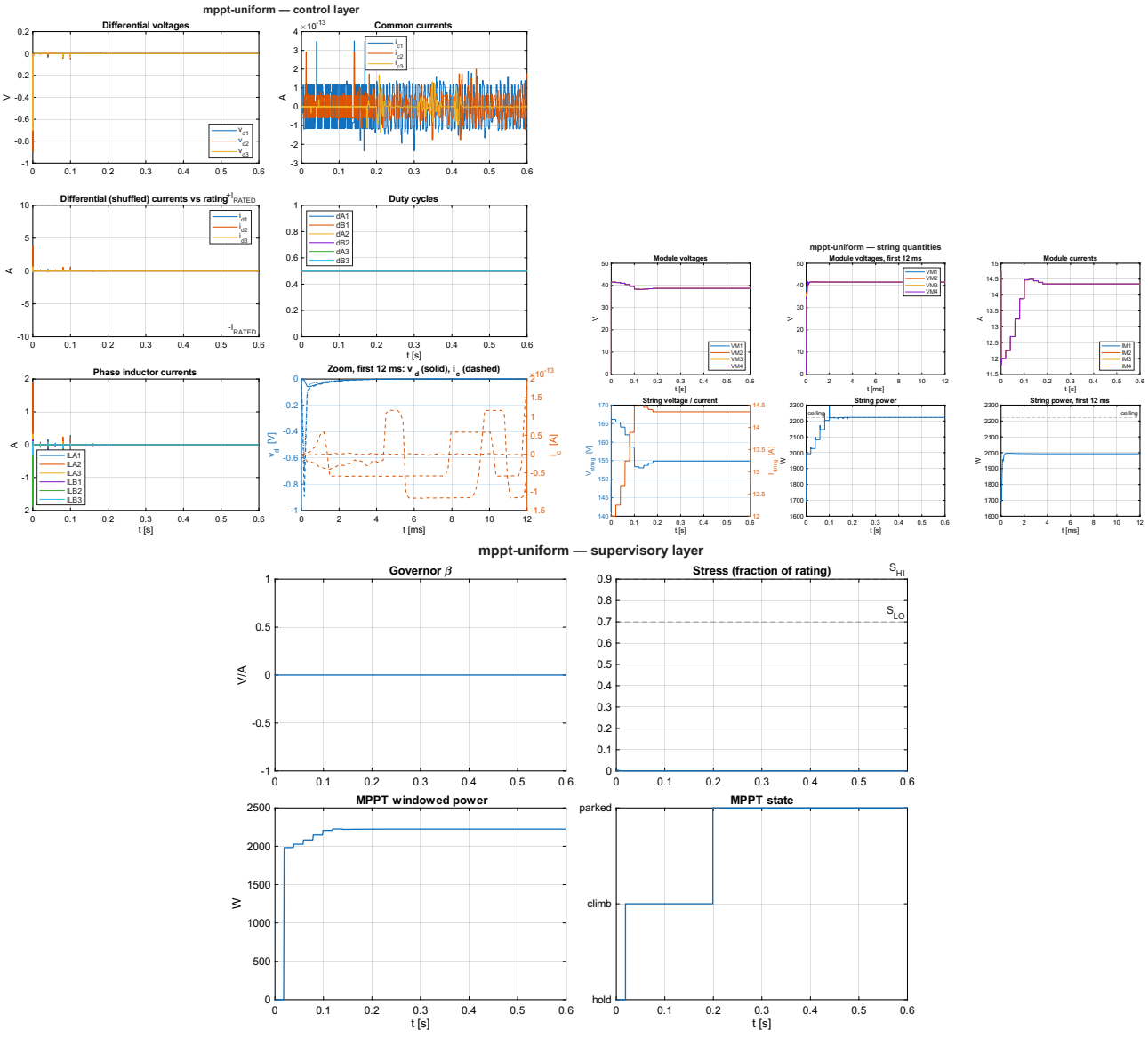


Figure 35: MIL test 06 (mppt-uniform): control, string and supervisory views.

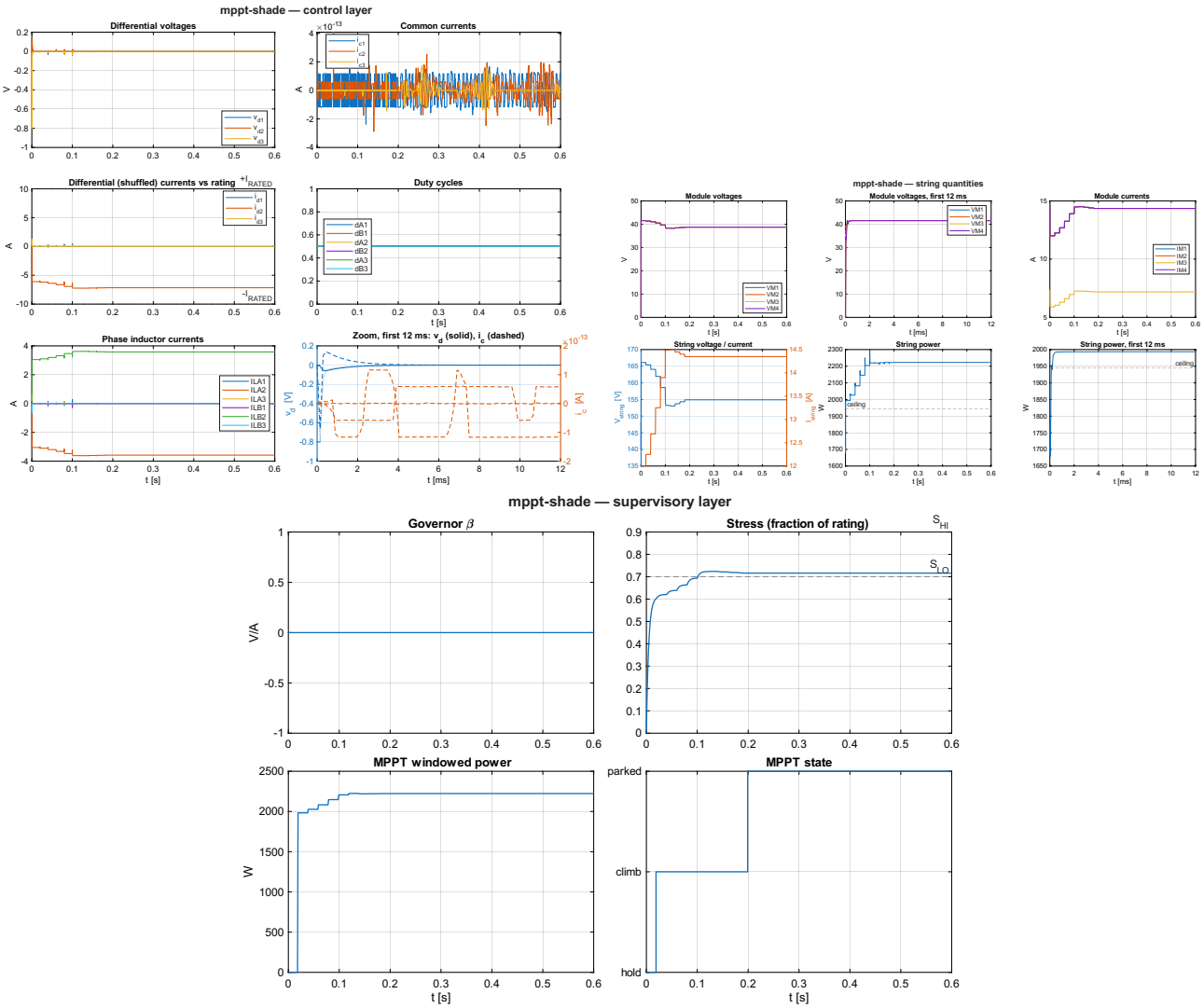


Figure 36: MIL test 07 (mppt-shade): control, string and supervisory views.

G.2 Carpark twin (3 modules, 2 shufflers)

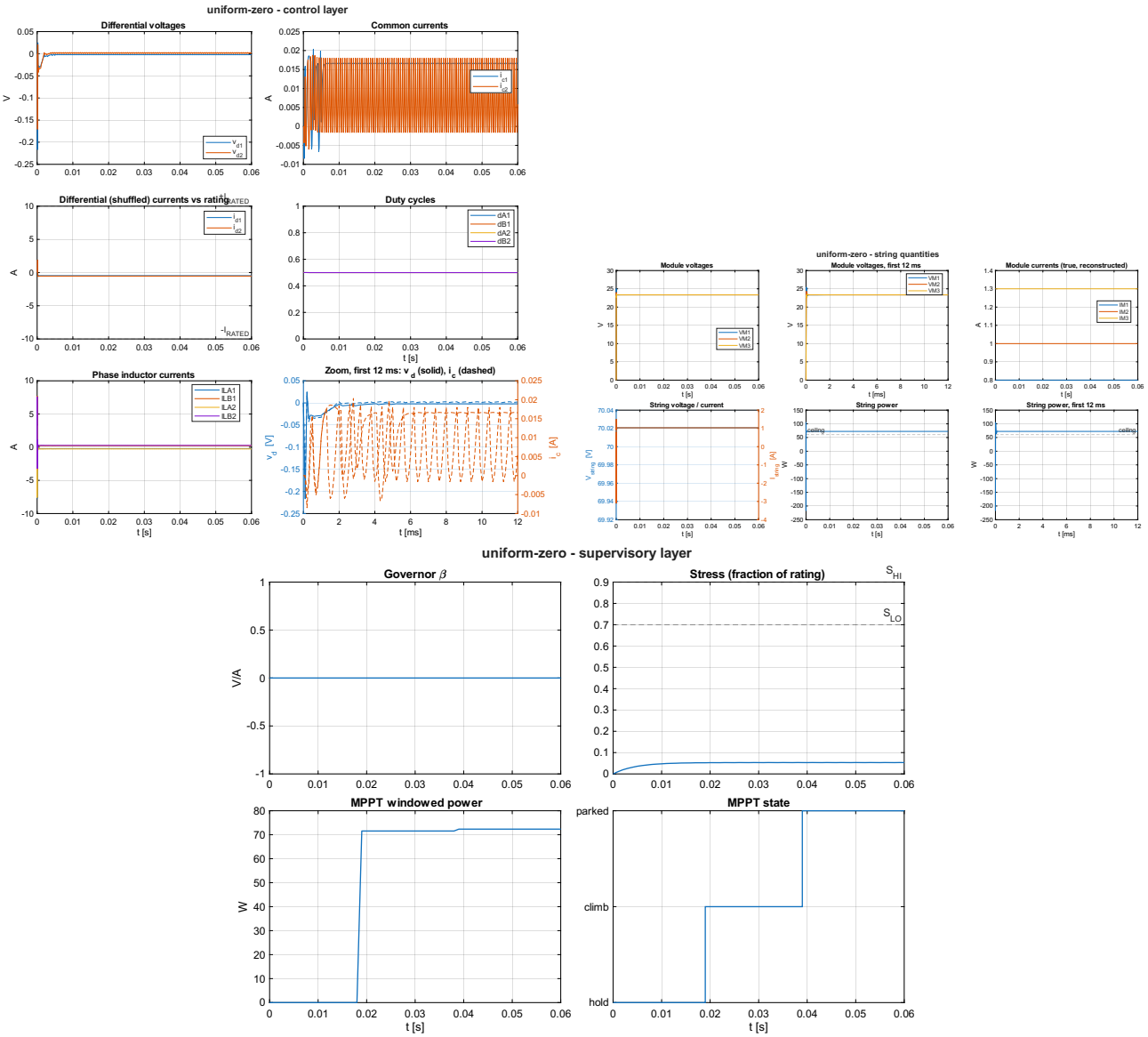


Figure 37: Carpark twin test 01 (uniform-zero): control, string and supervisory views.

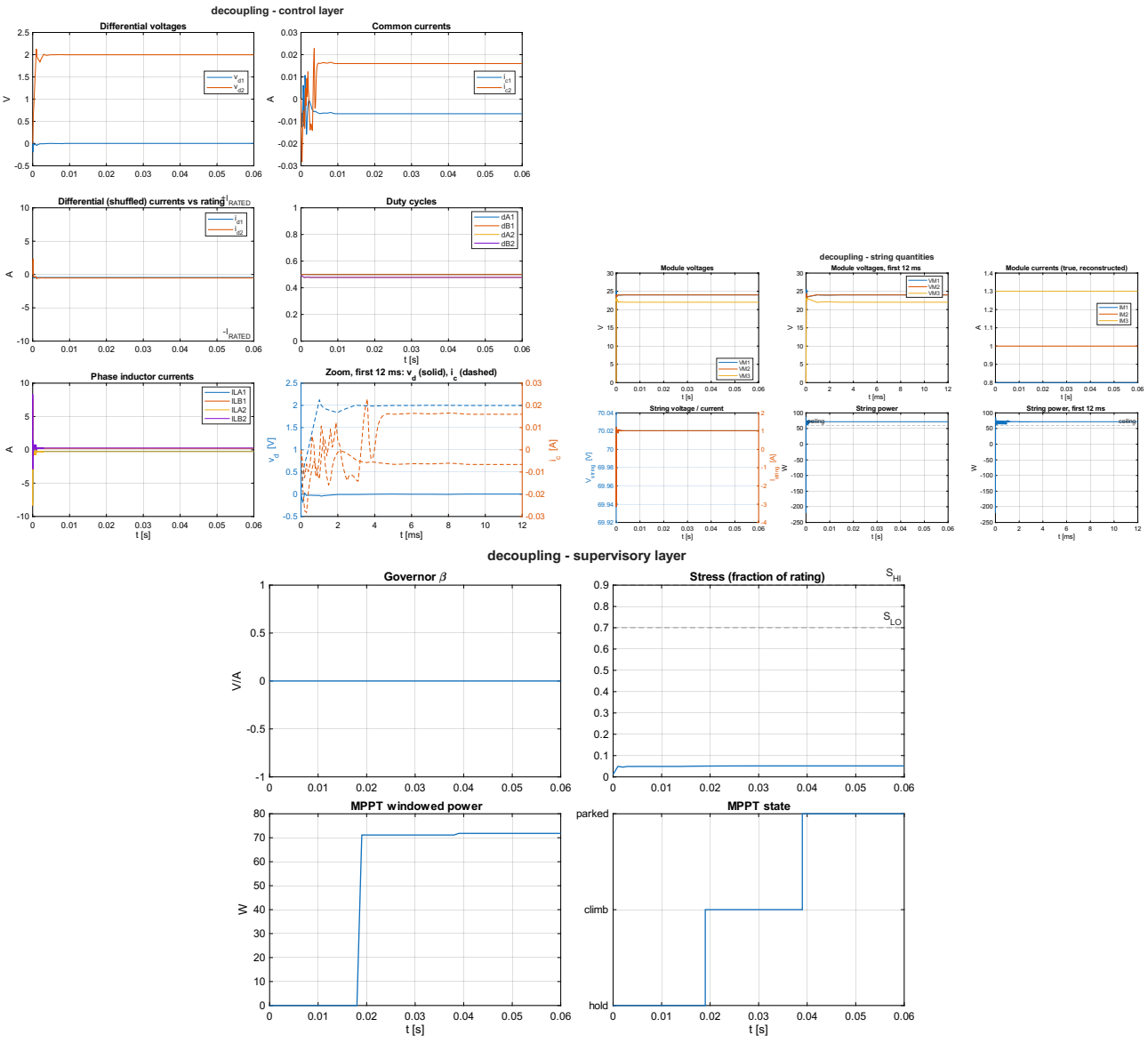


Figure 38: Carpark twin test 02 (decoupling): control, string and supervisory views.

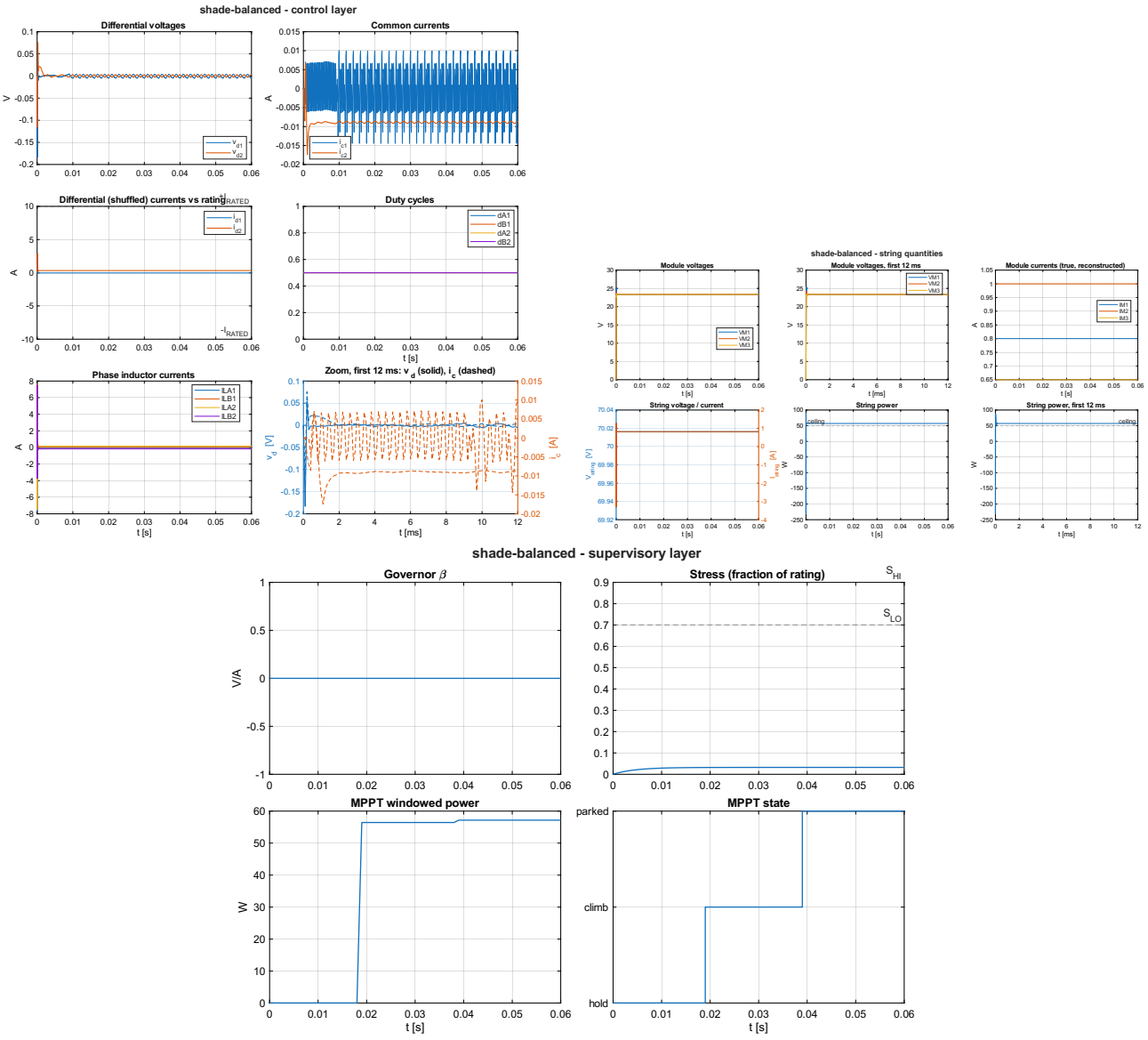


Figure 39: Carpark twin test 03 (shade-balanced): control, string and supervisory views.

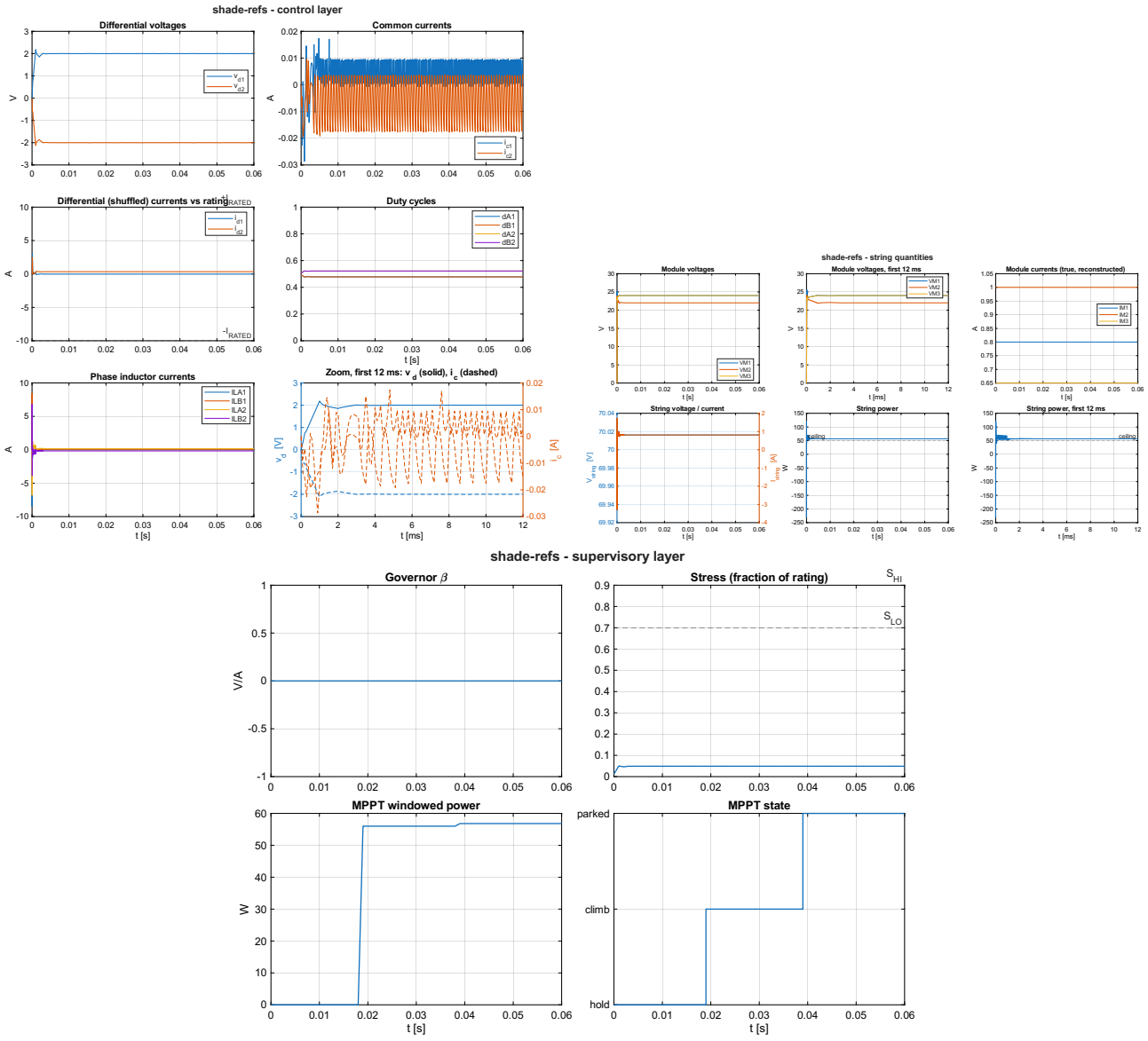


Figure 40: Carpark twin test 04 (shade-refs): control, string and supervisory views.

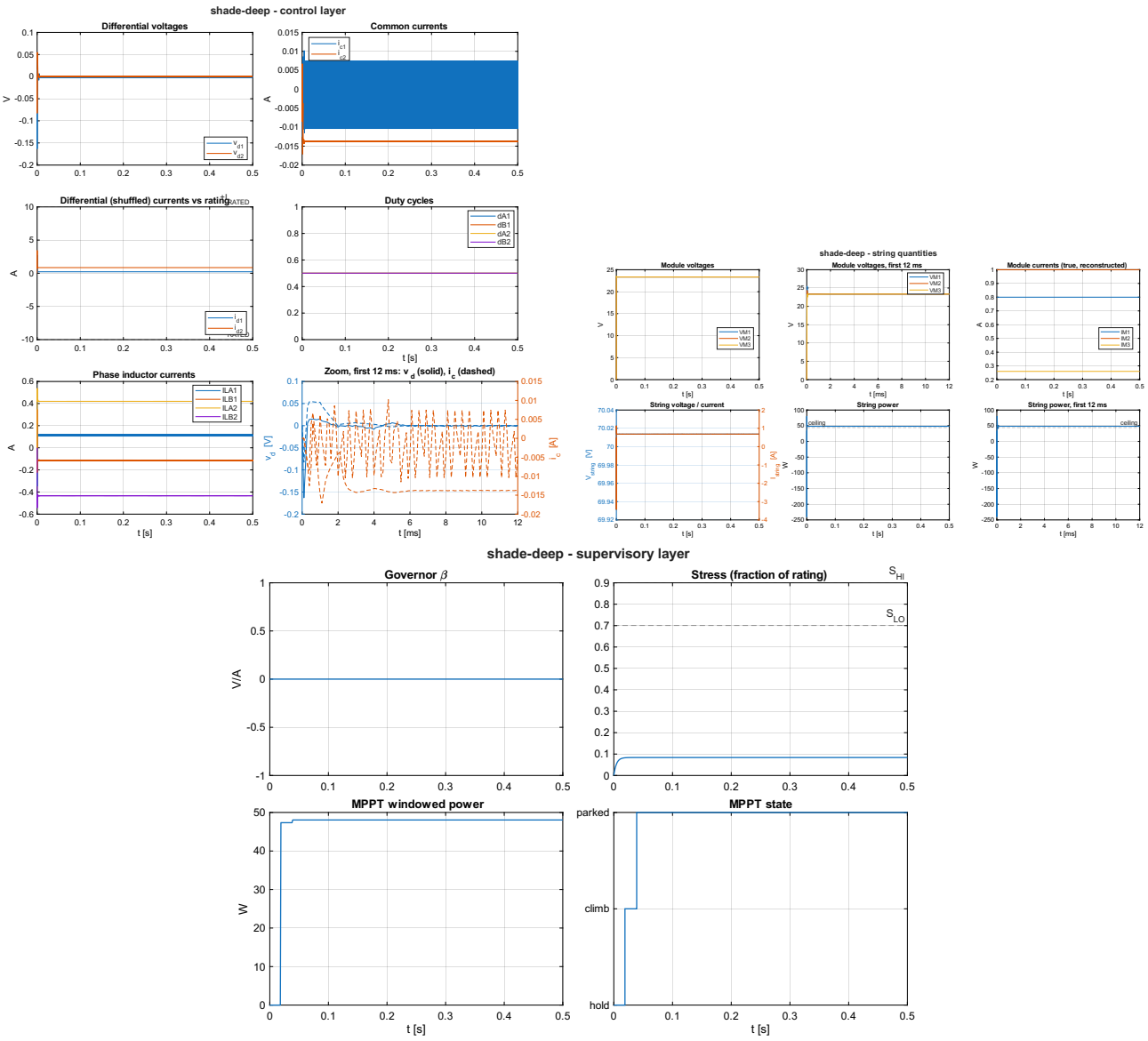


Figure 41: Carpark twin test 05 (shade-deep): control, string and supervisory views.

G.3 Re-plotted in the common style

The Python pipeline re-plots every validate test on the common axes used for the hardware (`out/figures/mil_validate` and `carpark_validate`); two examples:

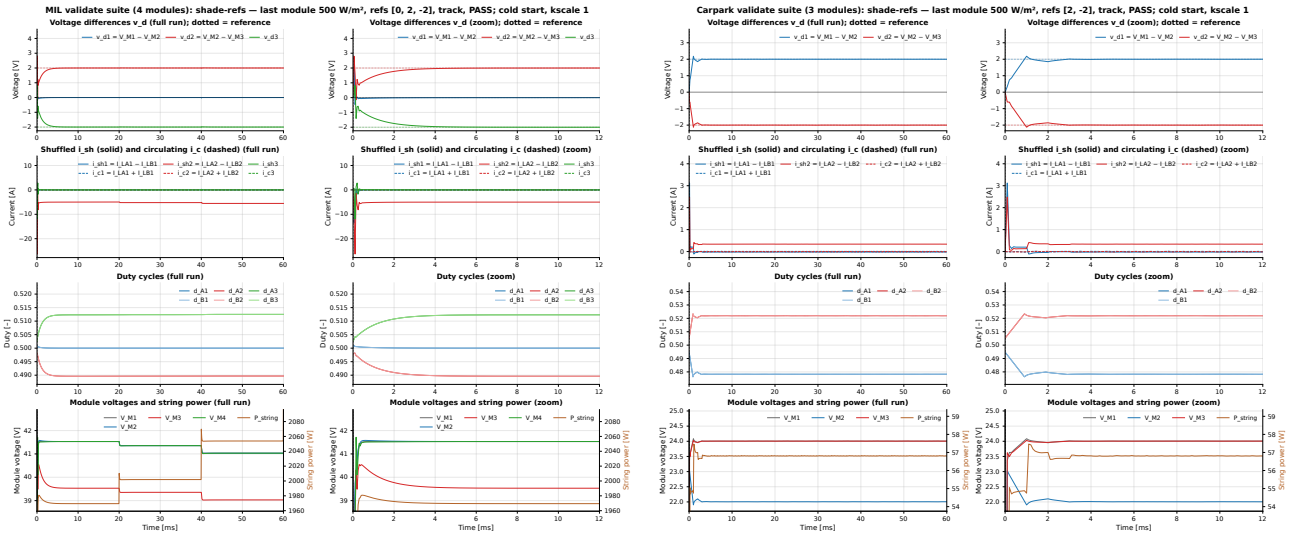


Figure 42: Test 04 (shade with references) re-plotted: MIL (left) and carpark twin (right).

H Analytical, QSPICE and Simulink Scenario Runs

Scenarios: S1 loop closure on the open-loop residual, S3 module 3 loses 1/3 of its current (PV80), S4 $+1$ V reference step on $V_{M2} - V_{M3}$, S5 open-loop duty steps as sent on the bench. Every run is plotted full-length and zoomed; the metrics are in appendix K. Only the $k = 0.1/0.3/1$ runs are reproduced here; all 28 analytical, 25 QSPICE and 25 carpark-twin figures are in `8_Results/analysis/out/figures/`.

H.1 Analytical $N = 3$ model (deployed gain set per f_{sw})

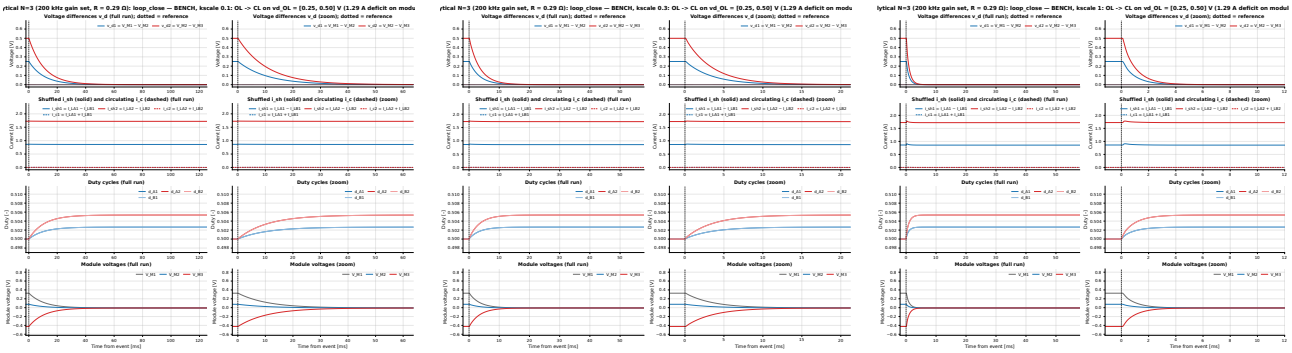


Figure 43: Analytical S1, BENCH plant, 200 kHz: $k = 0.1, 0.3, 1$.

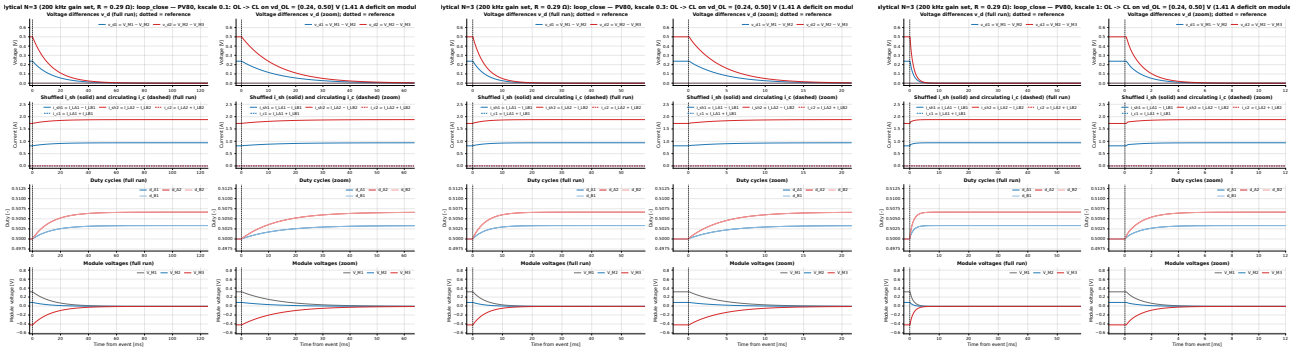


Figure 44: Analytical S1, PV80 plant, 200 kHz: $k = 0.1, 0.3, 1$.

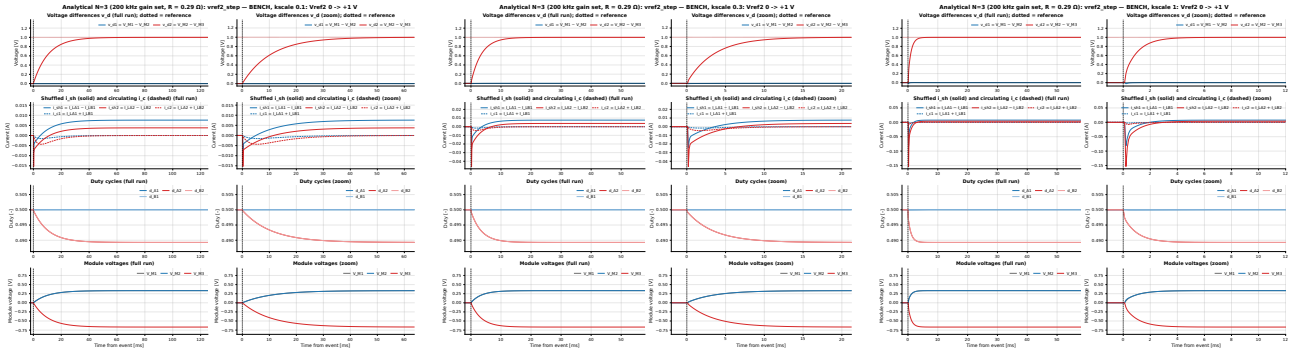


Figure 45: Analytical S4, BENCH plant, 200 kHz: $k = 0.1, 0.3, 1$.

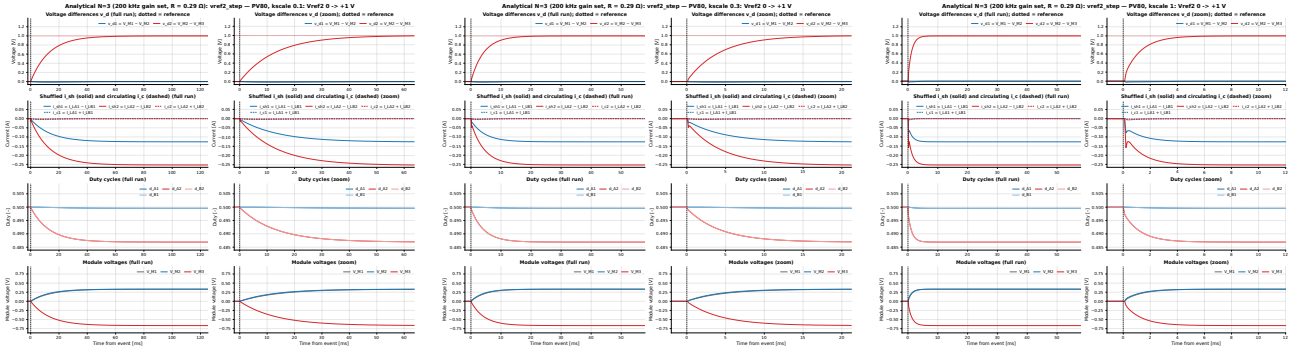


Figure 46: Analytical S4, PV80 plant, 200 kHz: $k = 0.1, 0.3, 1$.

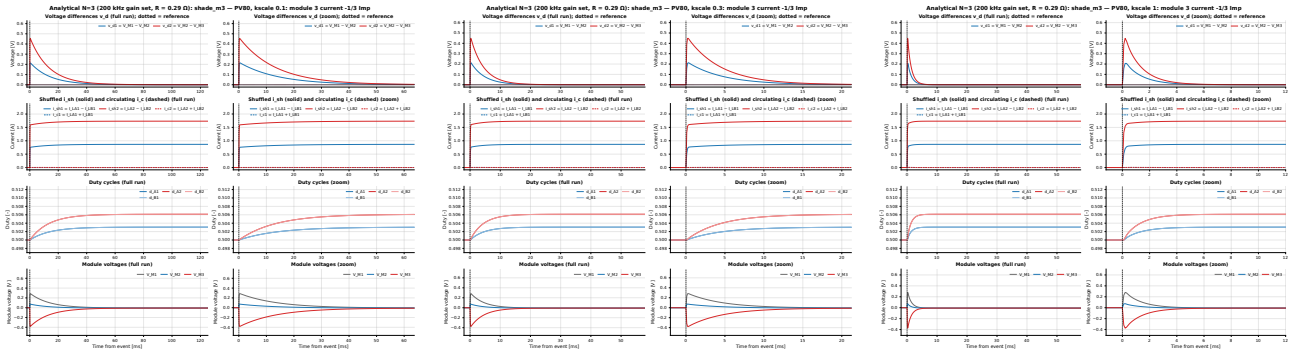


Figure 47: Analytical S3 (shade on module 3), PV80, 200 kHz: $k = 0.1, 0.3, 1$.

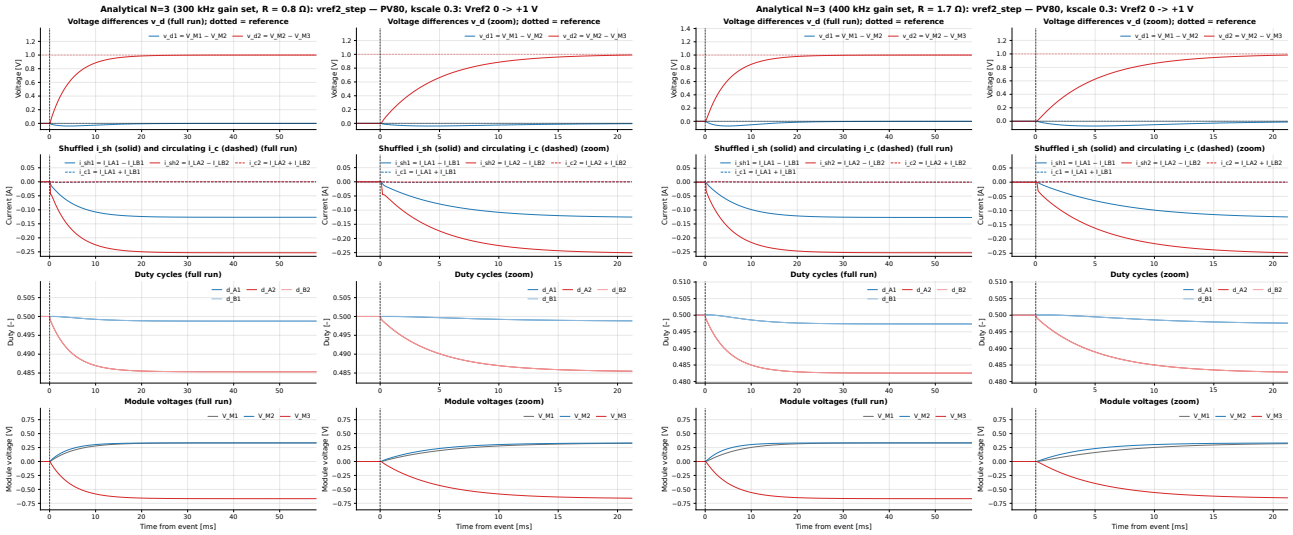


Figure 48: Analytical S4 with the 300 kHz (left) and 400 kHz (right) gain sets and plant resistances, $k = 0.3$.

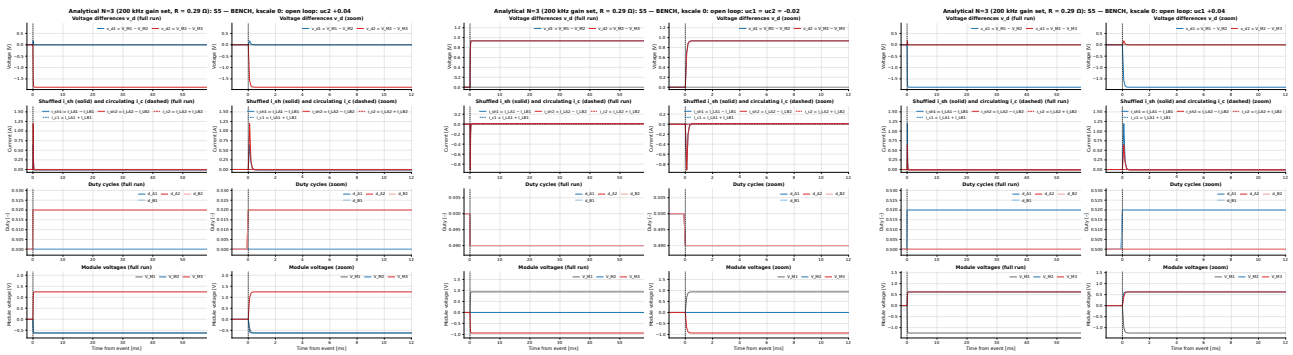


Figure 49: Analytical S5, open-loop duty steps as commanded on 16 Oct ($u_{c,2}$ +0.04; $u_{c,1} = u_{c,2} = -0.02$; $u_{c,1}$ +0.04).

H.2 QSPICE integrated model with the firmware DLL (200 kHz)

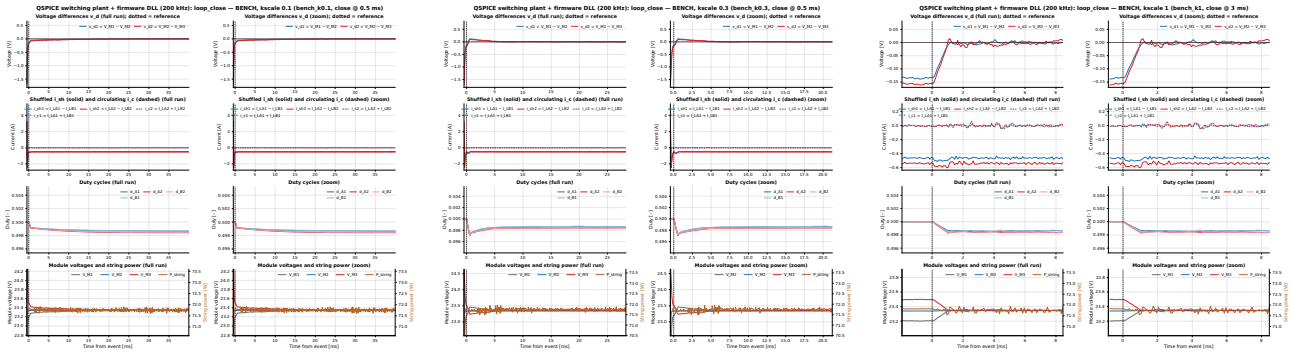


Figure 50: QSPICE S1, BENCH plant: $k = 0.1, 0.3, 1$ (100 μs bin means).

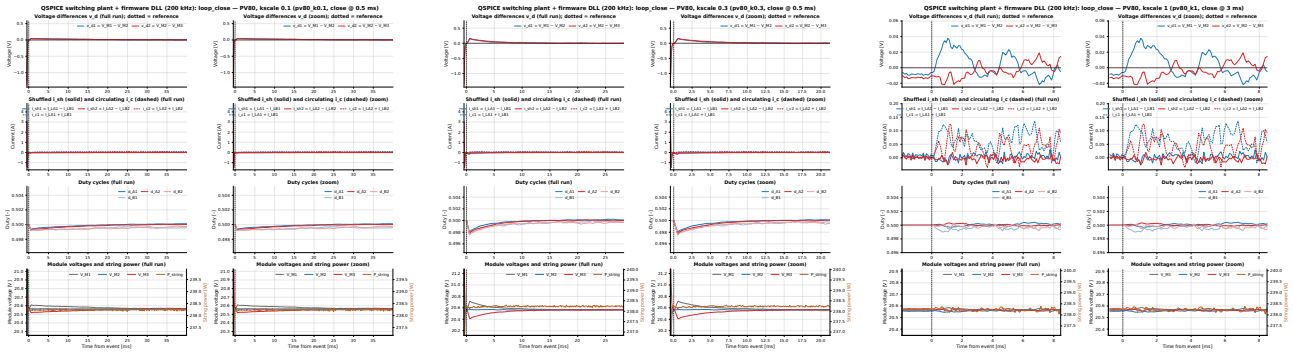


Figure 51: QSPICE S1, PV80 plant: $k = 0.1, 0.3, 1$ (100 μs bin means).

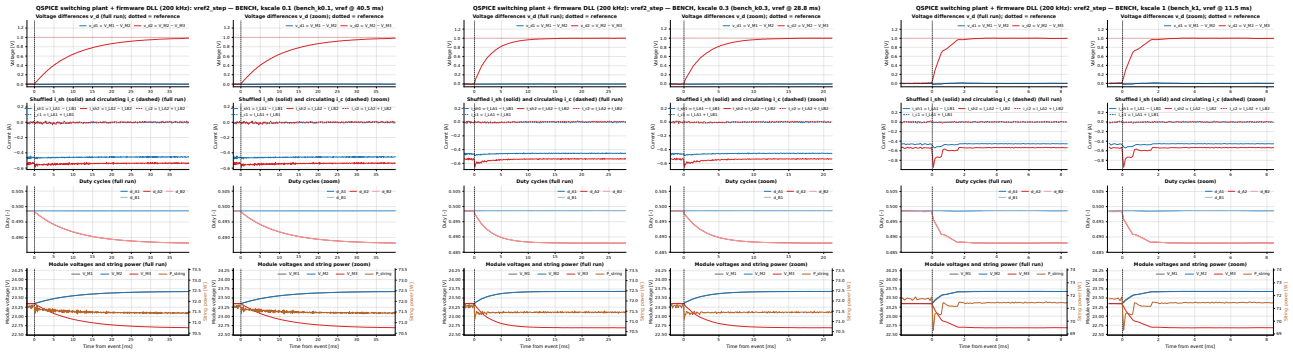


Figure 52: QSPICE S4, BENCH plant: $k = 0.1, 0.3, 1$ (100 μs bin means).

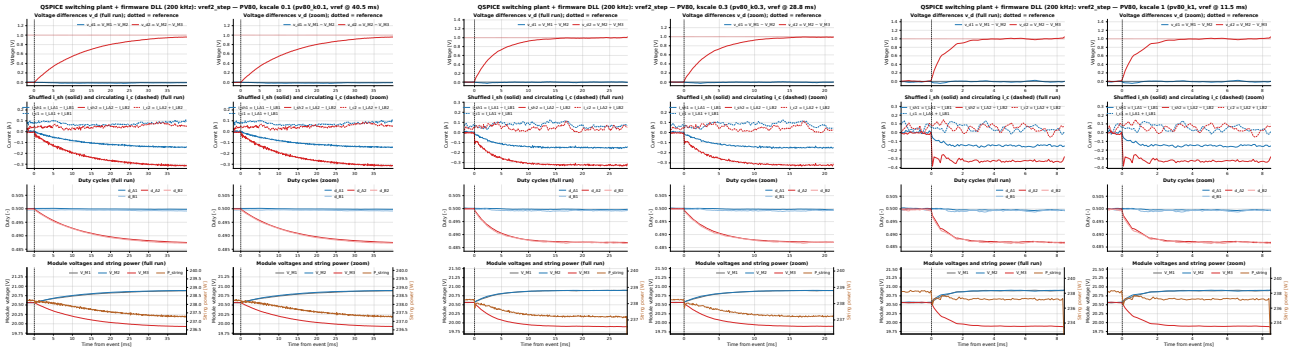


Figure 53: QSPICE S4, PV80 plant: $k = 0.1, 0.3, 1$ (100 μs bin means).

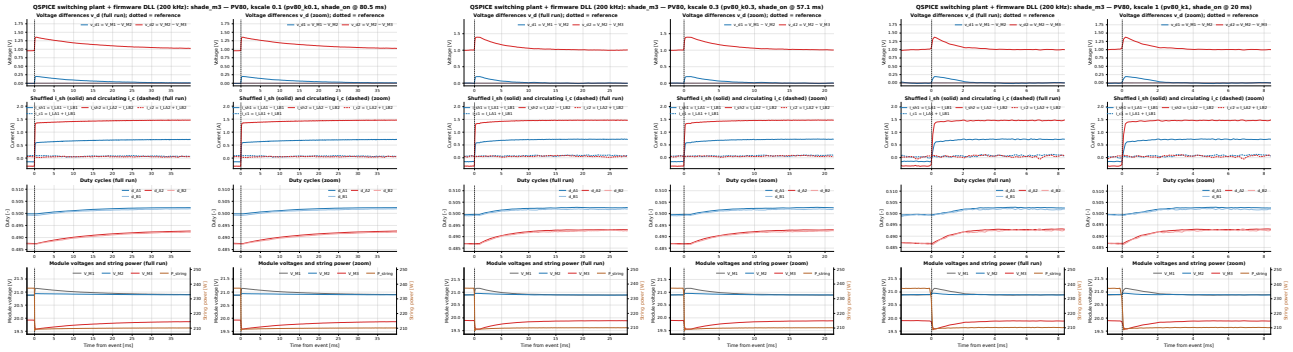


Figure 54: QSPICE S3 (shade on module 3), PV80: $k = 0.1, 0.3, 1$.

H.3 Simulink carpark twin (averaged switches, sensing model)

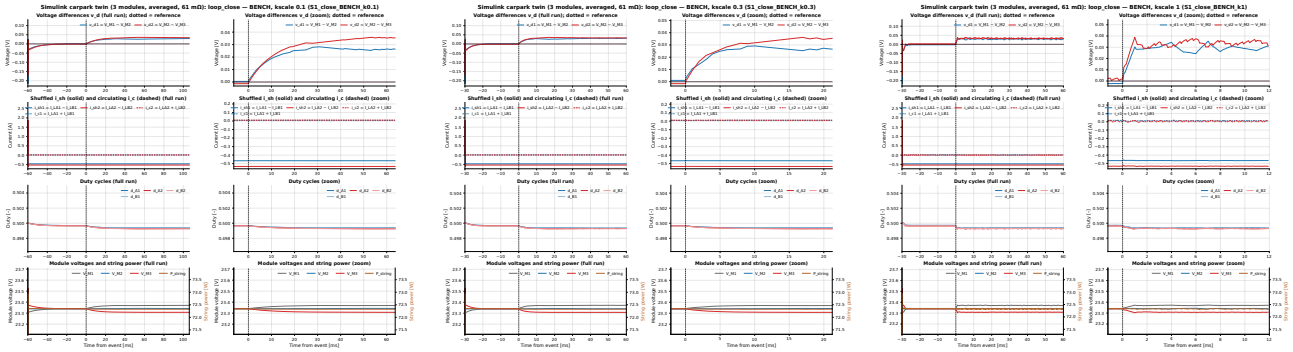


Figure 55: Simulink carpark twin S1, BENCH plant: $k = 0.1, 0.3, 1$.

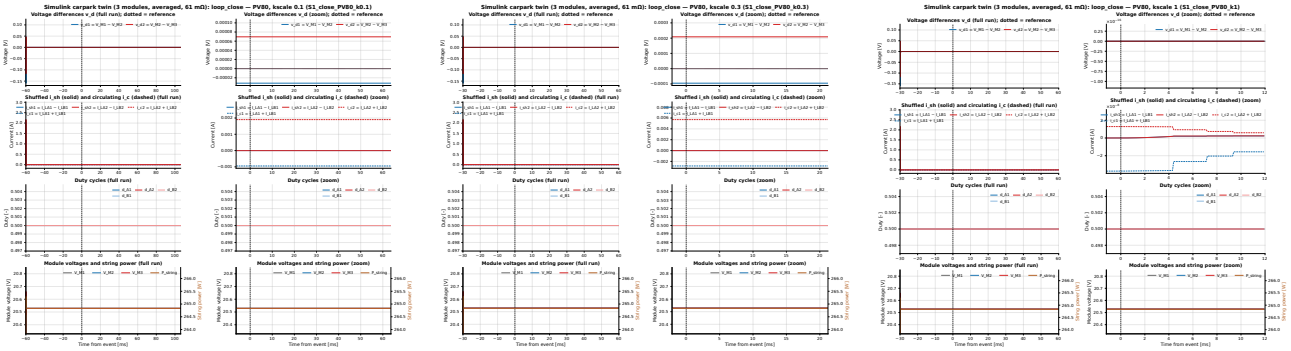


Figure 56: Simulink carpark twin S1, PV80 plant: $k = 0.1, 0.3, 1$.

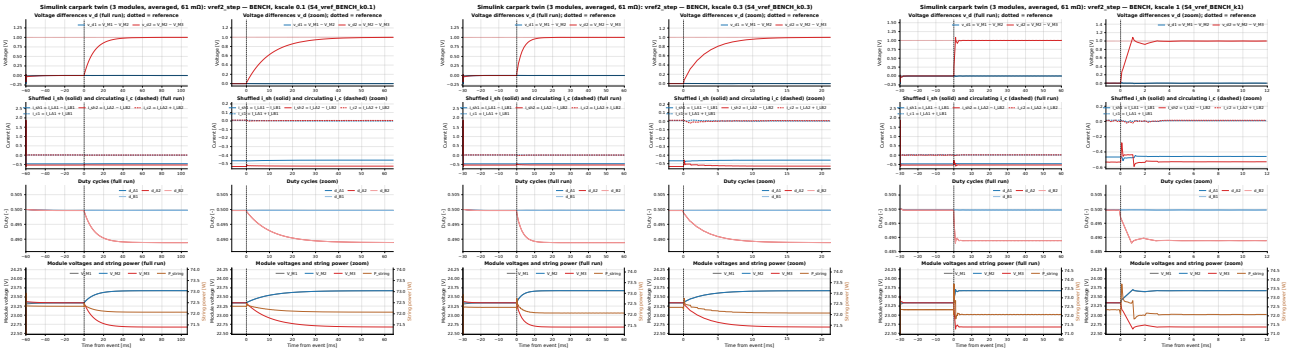


Figure 57: Simulink carpark twin S4, BENCH plant: $k = 0.1, 0.3, 1$.

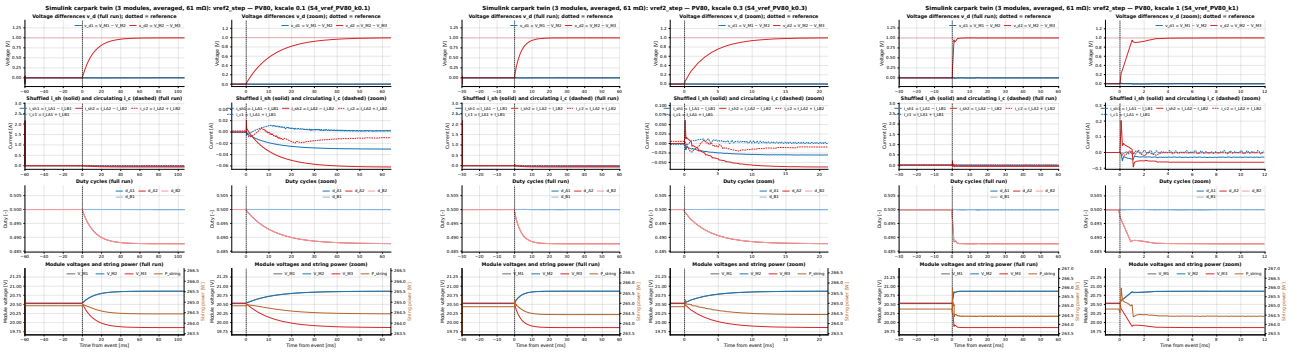


Figure 58: Simulink carpark twin S4, PV80 plant: $k = 0.1, 0.3, 1$.

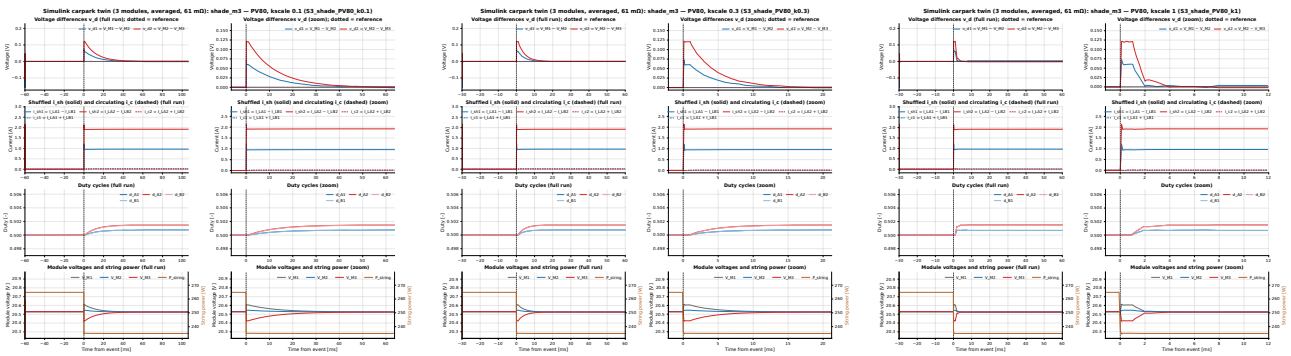


Figure 59: Simulink carpark twin S3 (shade on module 3), PV80: $k = 0.1, 0.3, 1$.

I Bench Test Raw Data (16 Oct 2026)

One overview figure per presentation slide window (RUN_OL / RUN_CL rows only, re-exported from InfluxDB; stale values outside RUN removed), followed by the event zooms the metrics are taken from. Panels: winding currents and modal currents (i_m, i_c); module voltages and v_d; duties; e-load voltage / current / power with set-points. The telemetry rate is printed on each figure; 200 kHz closed-loop runs logged at 11–14 Hz.

I.1 200 kHz

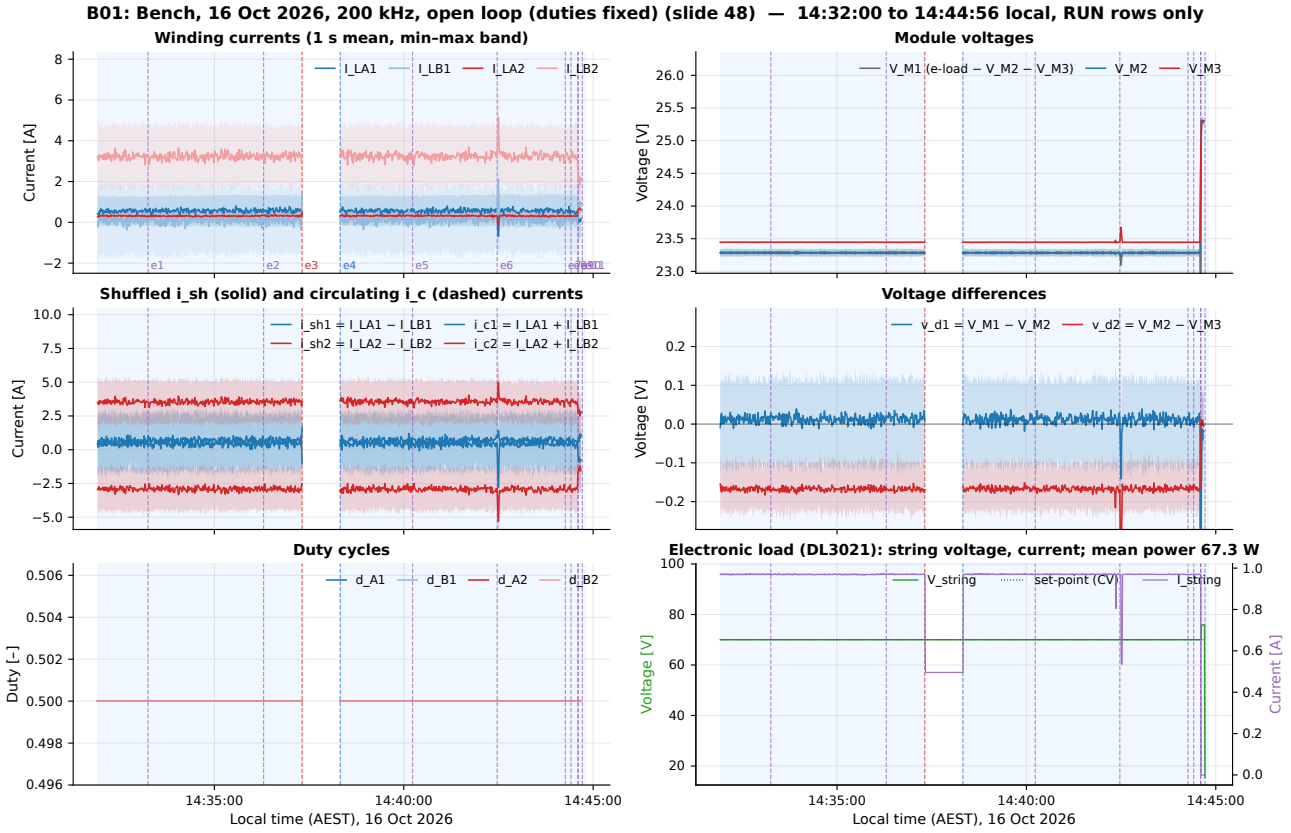


Figure 60: B01: 200 kHz open loop, duties 0.5, e-load CV steps 125 ↔ 60 V; V_{M1} from the e-load (slide p48).

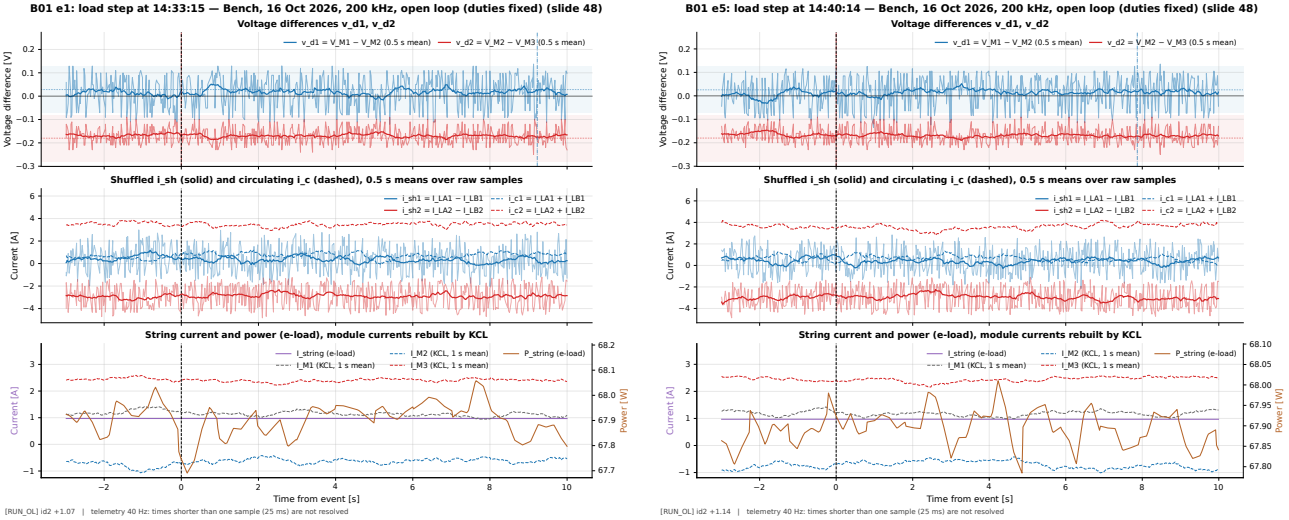


Figure 61: Event zooms for the window above (left: `B01_e1_load_step`, right: `B01_e5_load_step`).

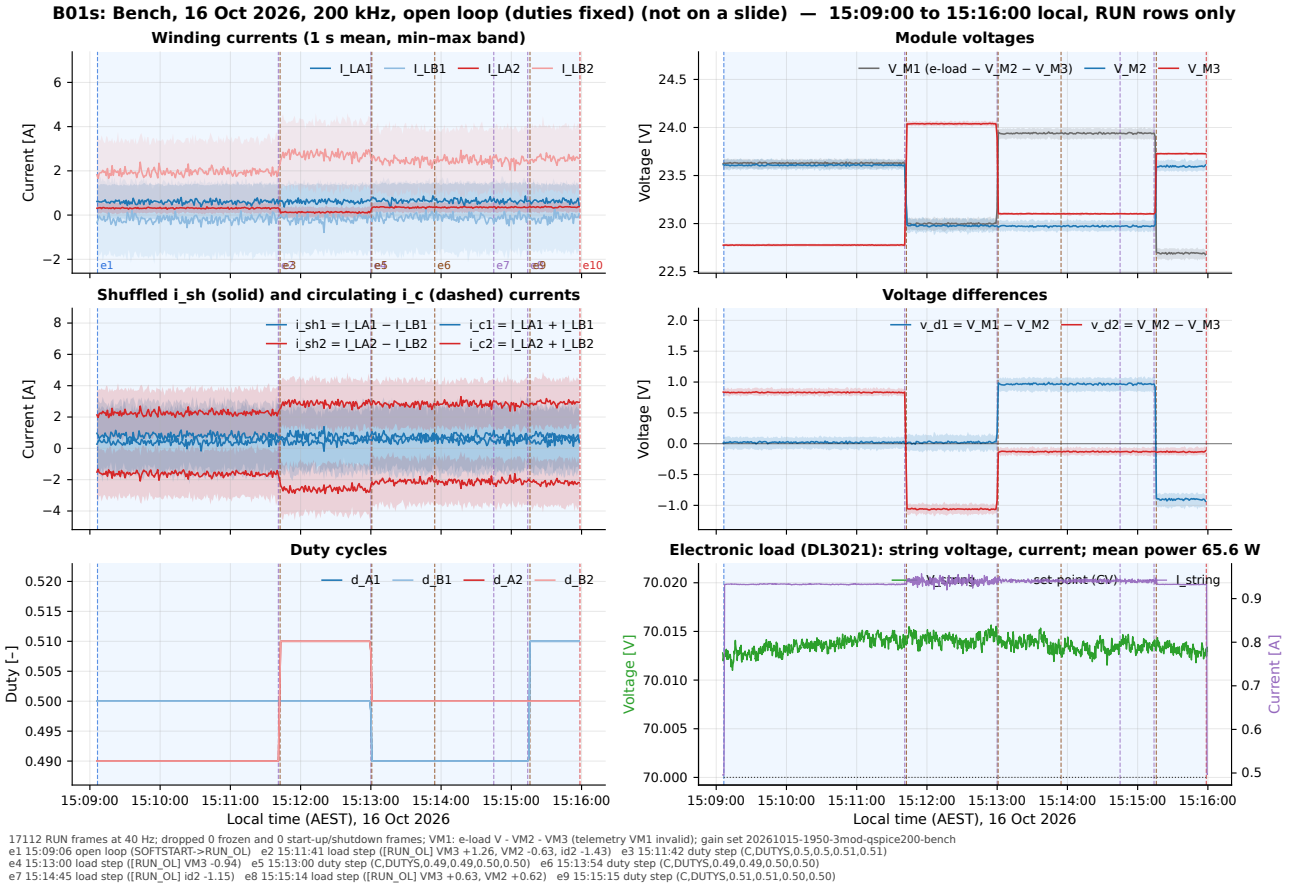


Figure 62: B01s: 200 kHz open loop with the duty steps (15:11:42 S2 +0.02/leg, 15:13:00 S1 & S2 −0.01/leg, 15:15:15 S1 +0.02/leg); ends in FAULT (slide not on a slide).

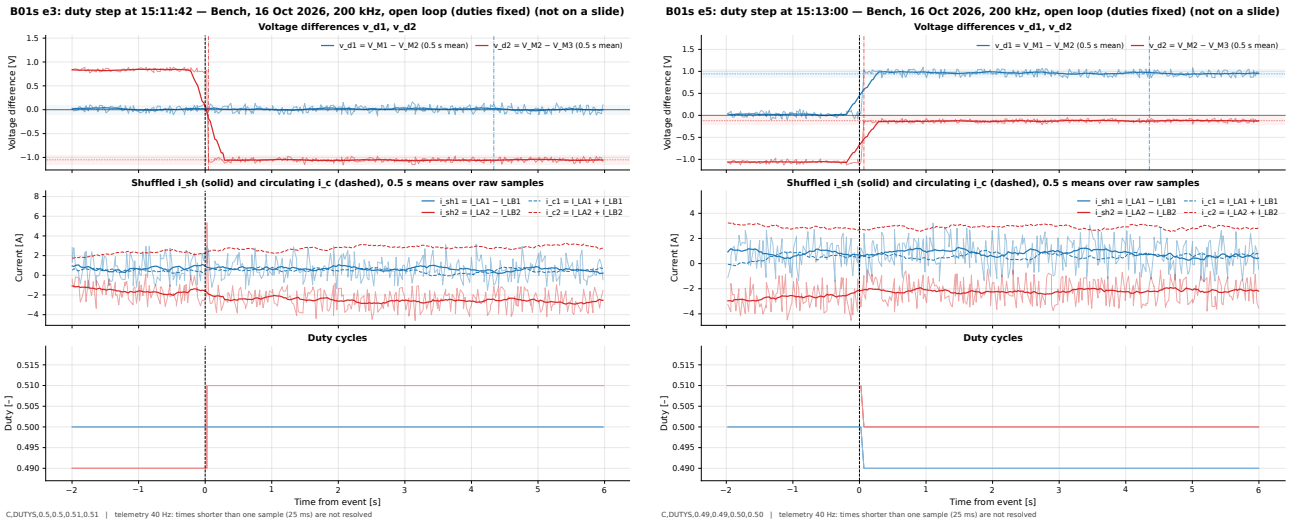


Figure 63: Event zooms for the window above (left: `B01s_e3_duty_step`, right: `B01s_e5_duty_step`).

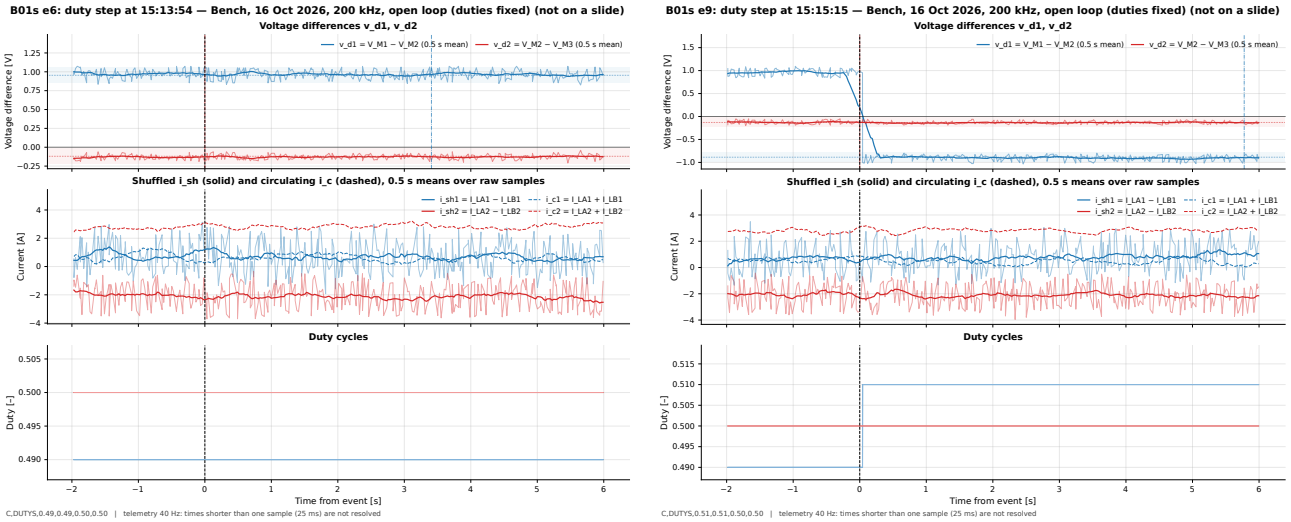


Figure 64: Event zooms for the window above (left: `B01s_e6_duty_step`, right: `B01s_e9_duty_step`).

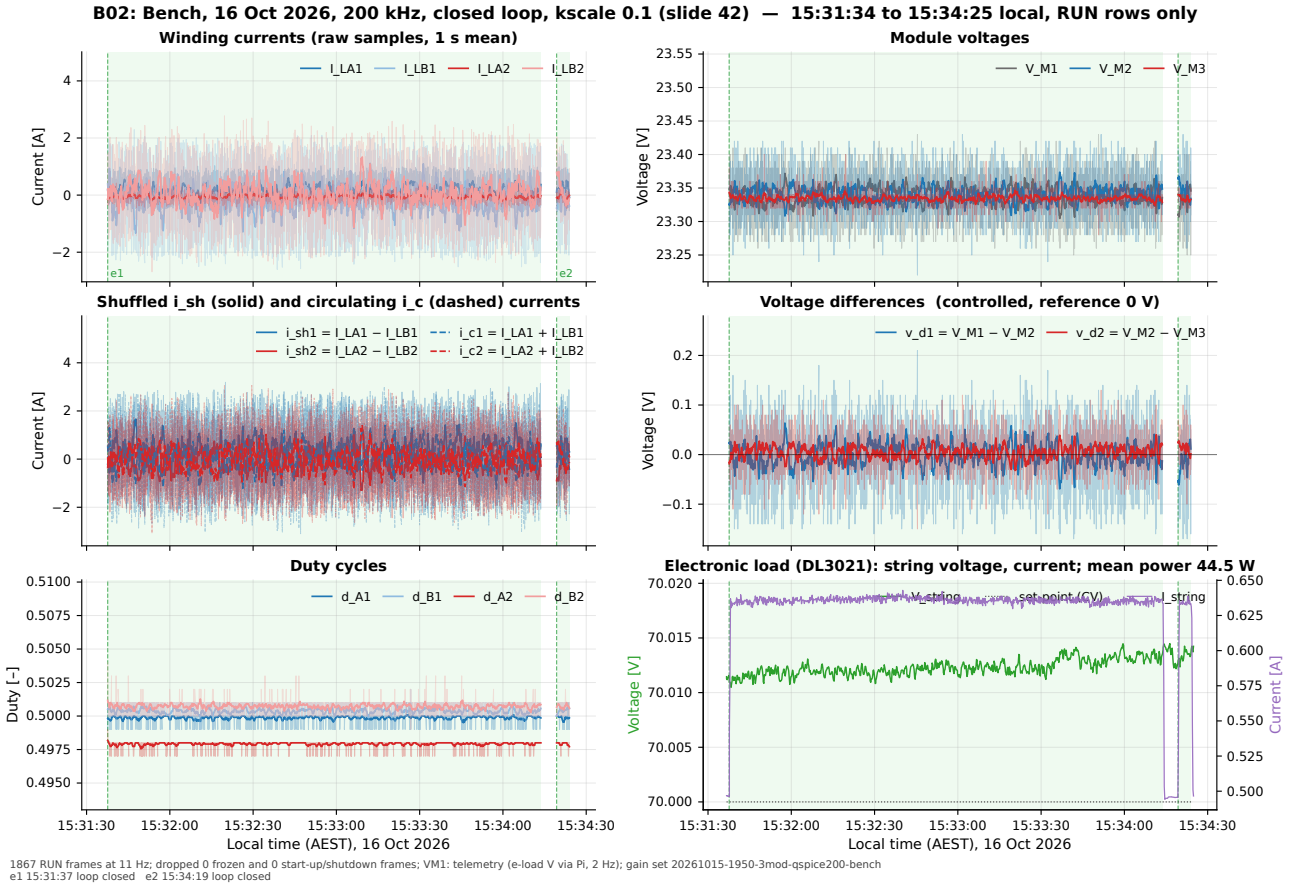


Figure 65: B02: 200 kHz closed loop, $k = 0.1$, before the ADC phase-B fix, PSU unstable (slide p42).

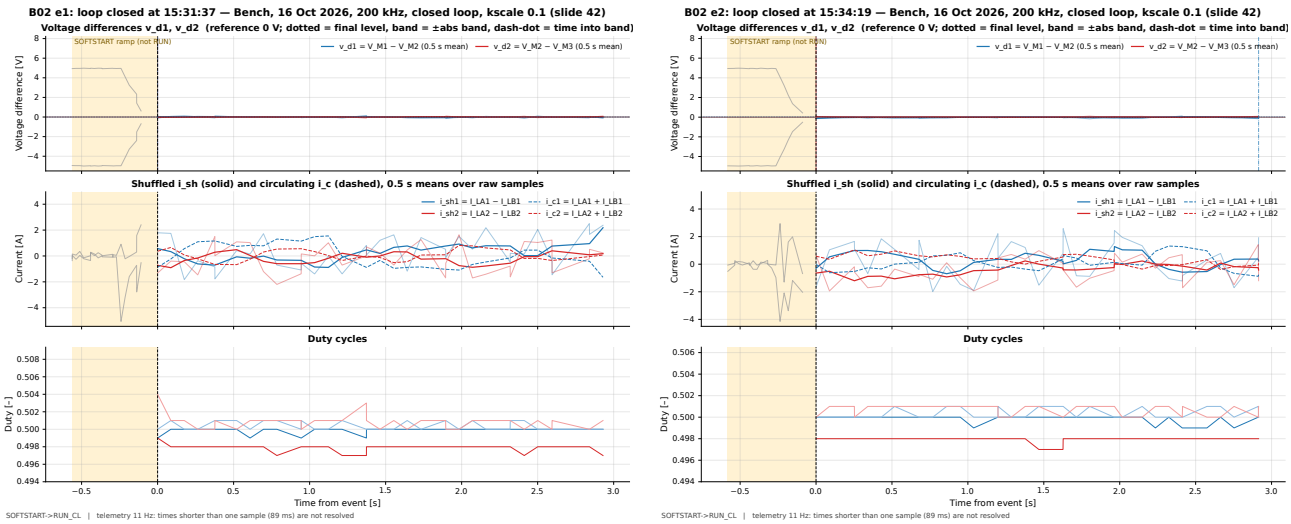


Figure 66: Event zooms for the window above (left: `B02_e1_close`, right: `B02_e2_close`).

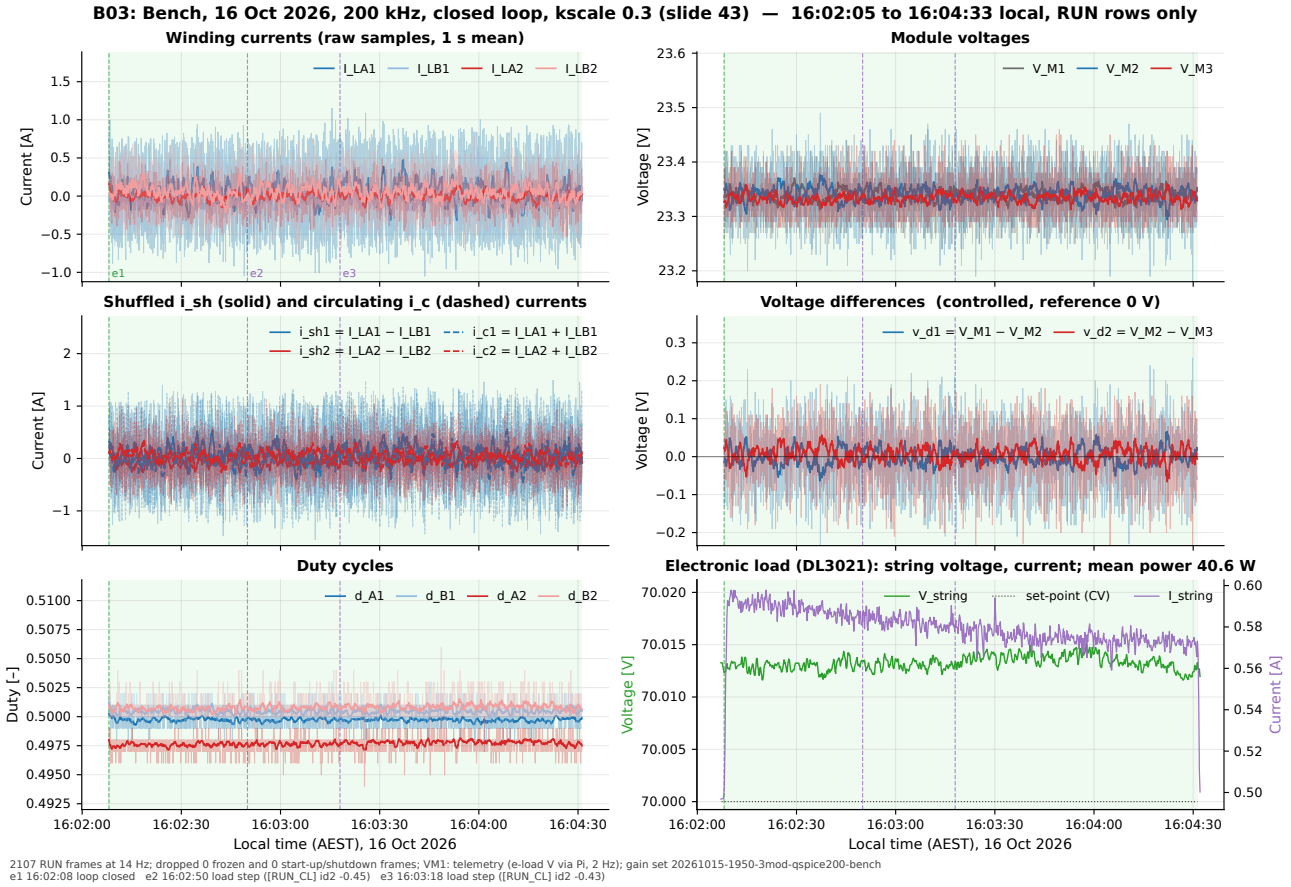


Figure 67: B03: 200 kHz closed loop after the ADC fix; slide says 20 %, telemetry says $k = 0.3$ (pset 0, boot default) (slide p43).

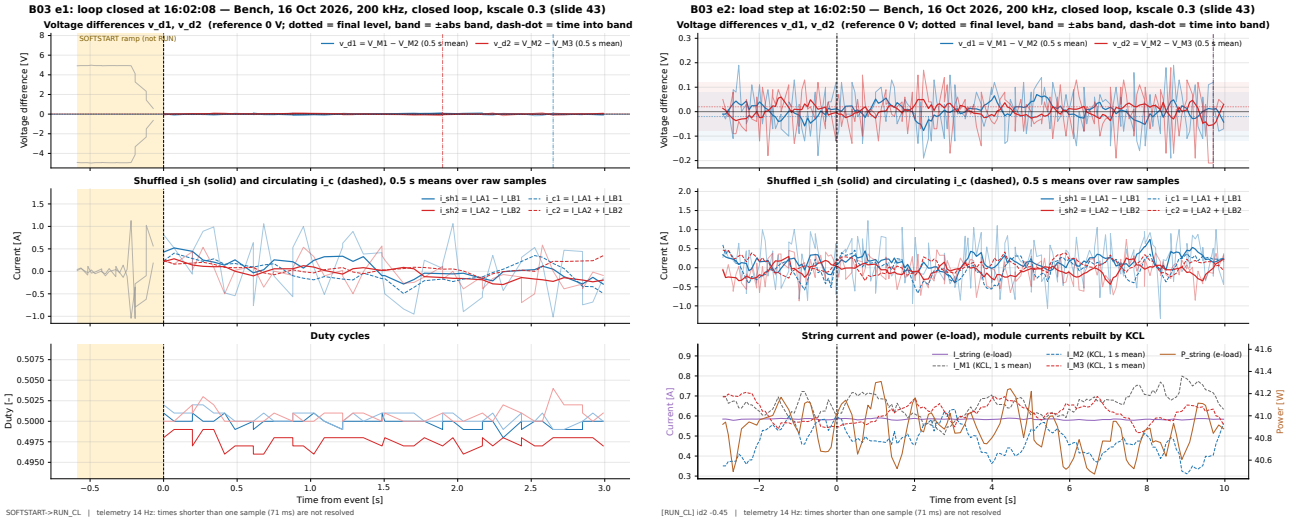


Figure 68: Event zooms for the window above (left: `B03_e1_close`, right: `B03_e2_load_step`).

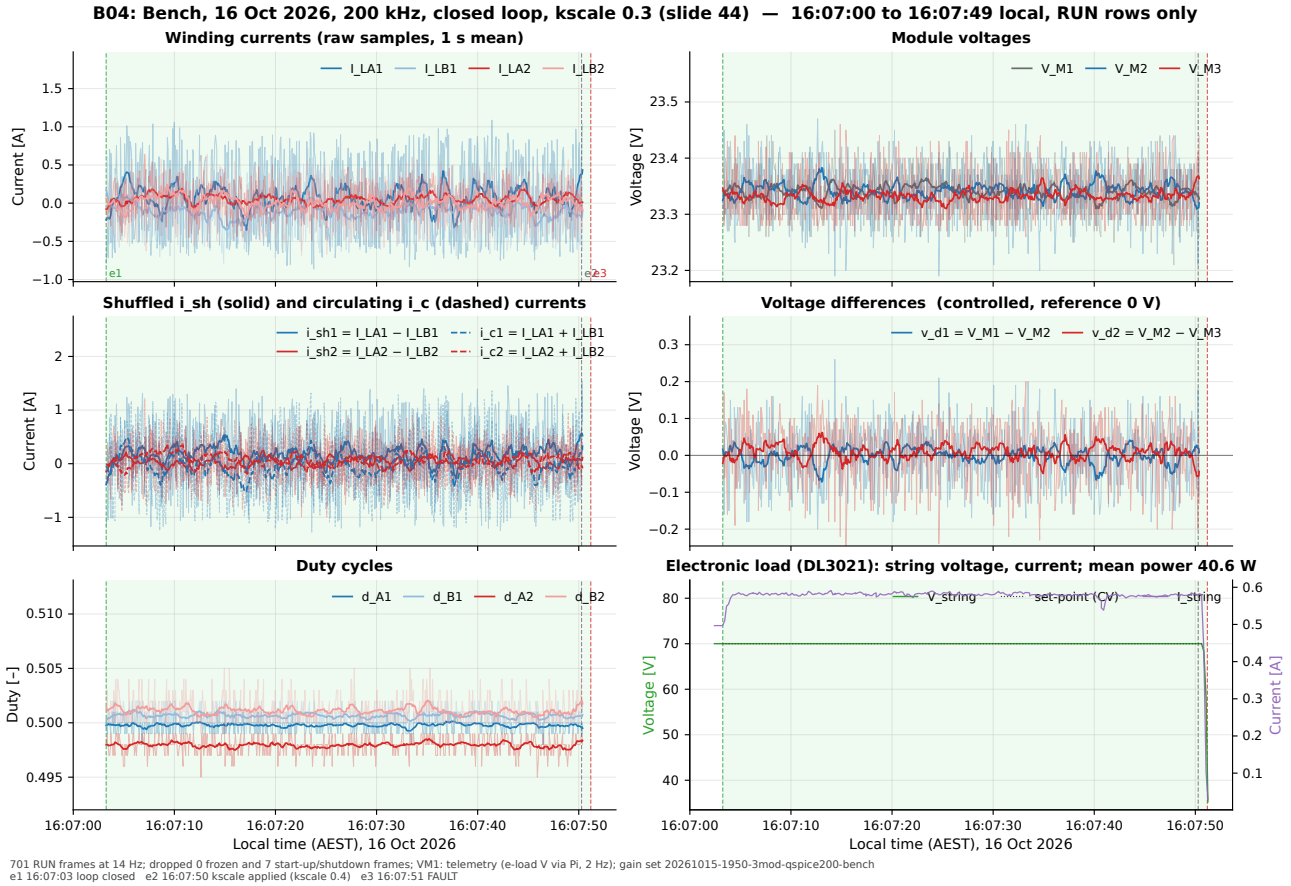


Figure 69: B04: 200 kHz closed loop, $k = 0.3$; $k \to 0.4$ applied live at 16:07:50 $\Rightarrow$ module-UV fault within 1 s (slide p44).

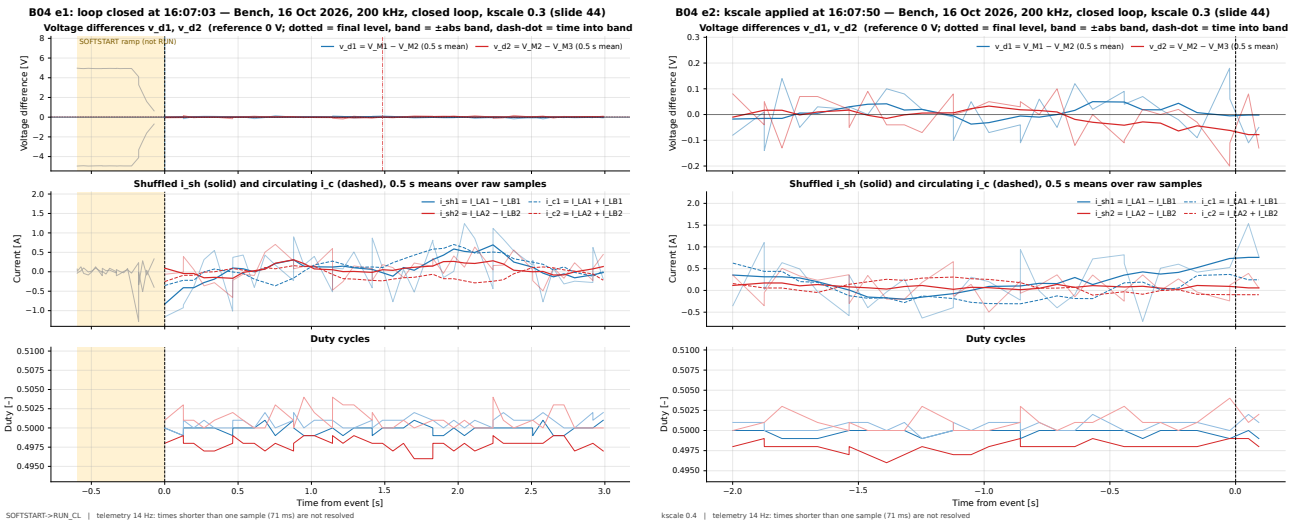


Figure 70: Event zooms for the window above (left: `B04_e1_close`, right: `B04_e2_kscale_apply`).

I.2 300 kHz

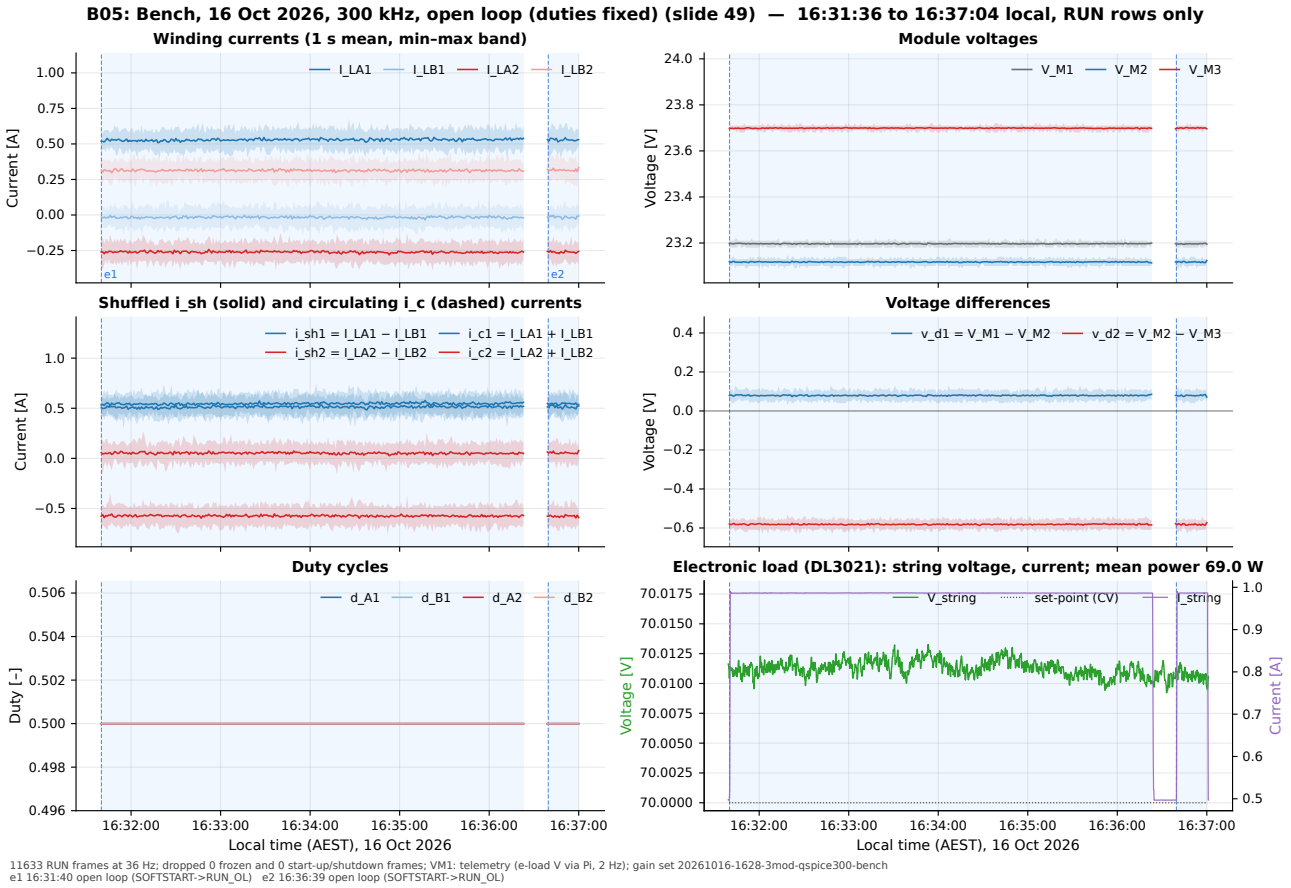


Figure 71: B05: 300 kHz open loop (slide p49).

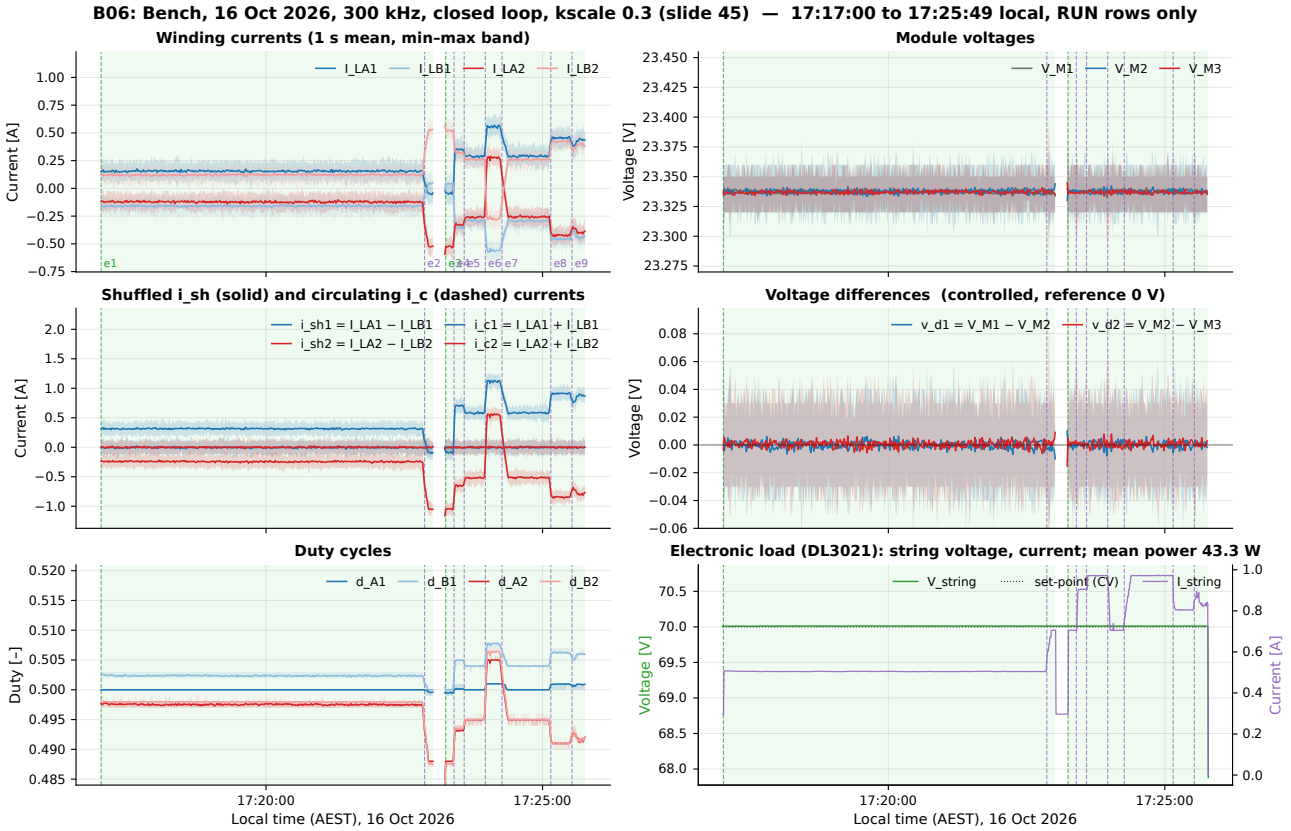


Figure 72: B06: 300 kHz closed loop, $k = 0.3$, PSU / e-load set-point steps 17:22:30–17:25:30 (slide p45).

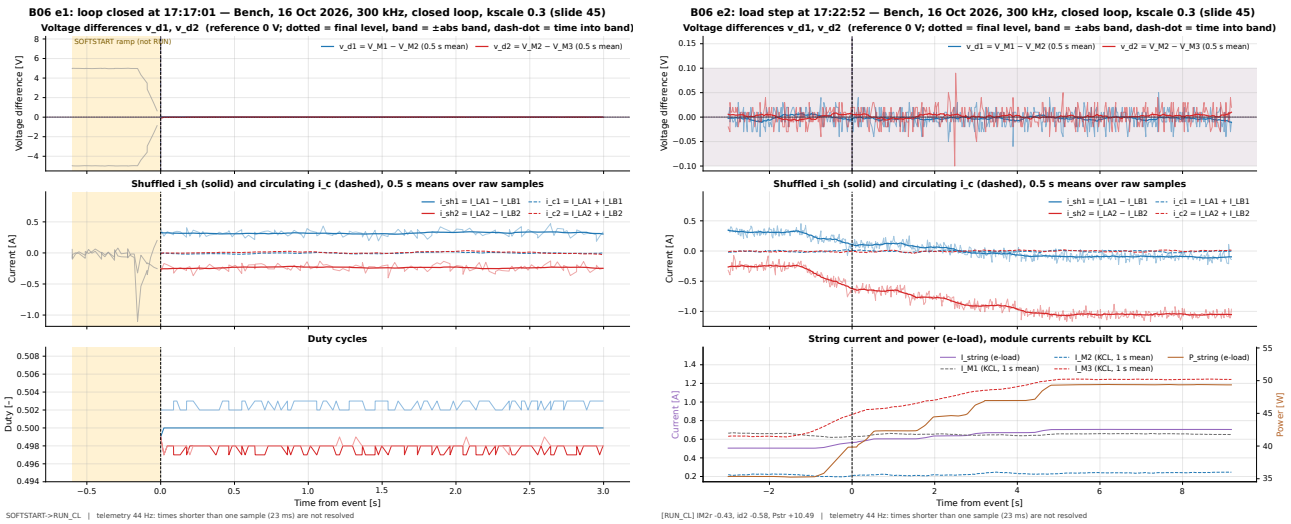


Figure 73: Event zooms for the window above (left: `B06_e1_close`, right: `B06_e2_load_step`).

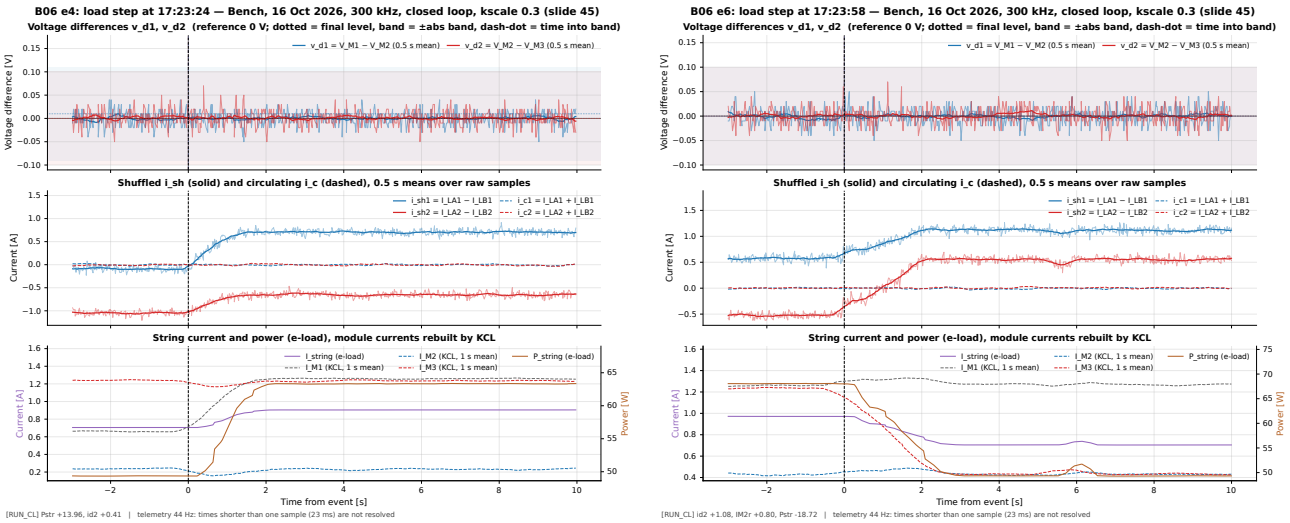
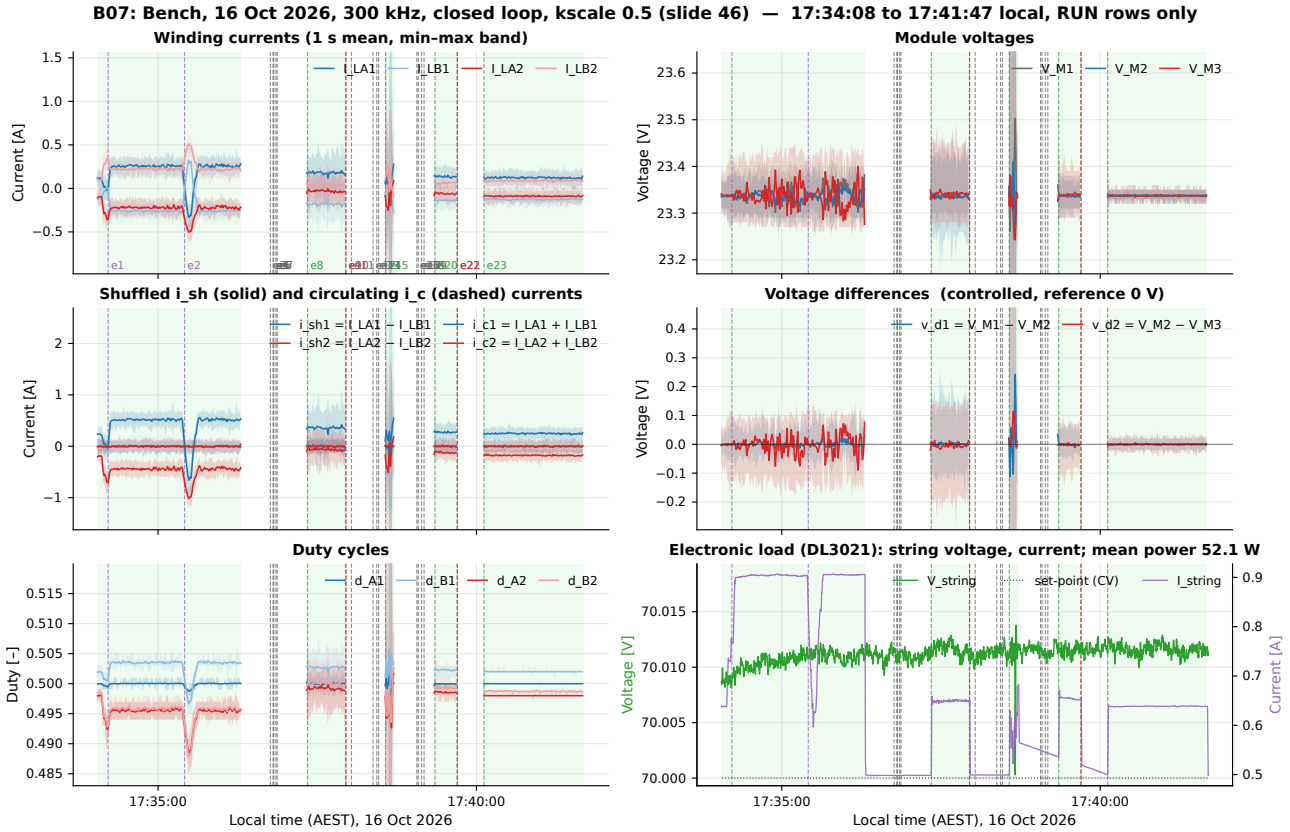


Figure 74: Event zooms for the window above (left: `B06_e4_load_step`, right: `B06_e6_load_step`).



13384 RUN frames at 44 Hz; dropped 0 frozen and 0 start-up/shutdown frames; VM1: telemetry (e-load V via Pi, 2 Hz); gain set 20261016-1628-3mod-qspice300-bench
e1 17:34:13 load step ([RUN_CL] Pstr +9.96) e2 17:35:25 load step ([RUN_CL] id2 -0.46, Pstr -16.09) e3 17:36:45 kscale applied (kscale 0.3)
e4 17:36:48 kscale applied (kscale 0.2) e5 17:36:49 kscale applied (kscale 0.3) e6 17:36:51 kscale applied (kscale 0.4)
e7 17:36:52 kscale applied (kscale 0.5) e8 17:37:20 loop closed e9 17:37:56 kscale applied (kscale 0.6)

Figure 75: B07: 300 kHz closed loop, $k = 0.3/0.4/0.5/0.6$ sweep; $k = 0.6$ oscillated (17:38:34) and two live gain changes tripped the main-loop stall fault (slide p46).

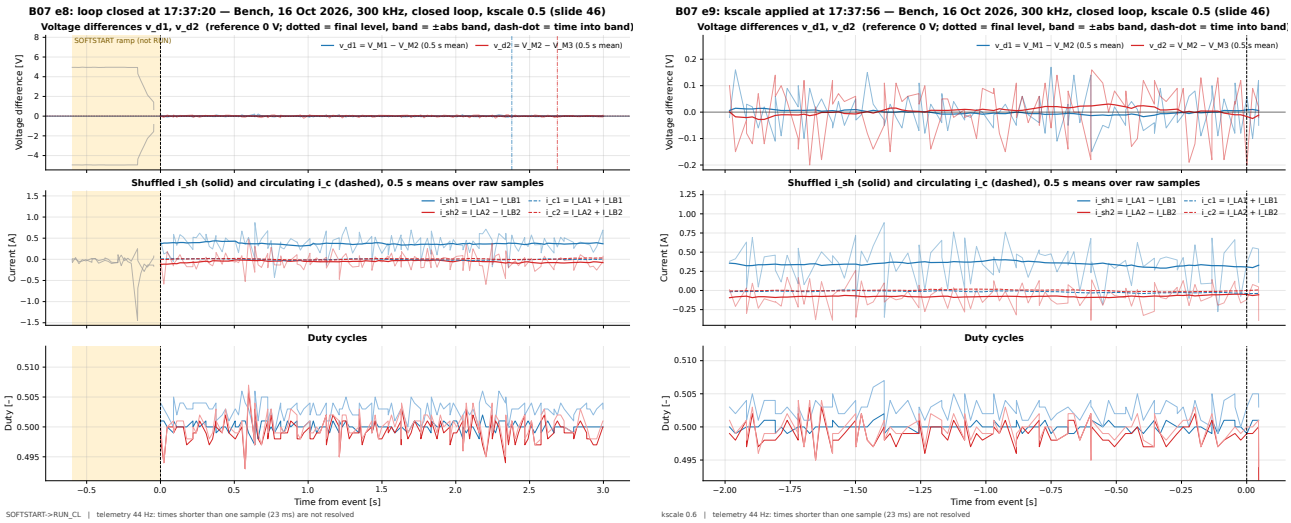


Figure 76: Event zooms for the window above (left: `B07_e8_close`, right: `B07_e9_kscale_apply`).

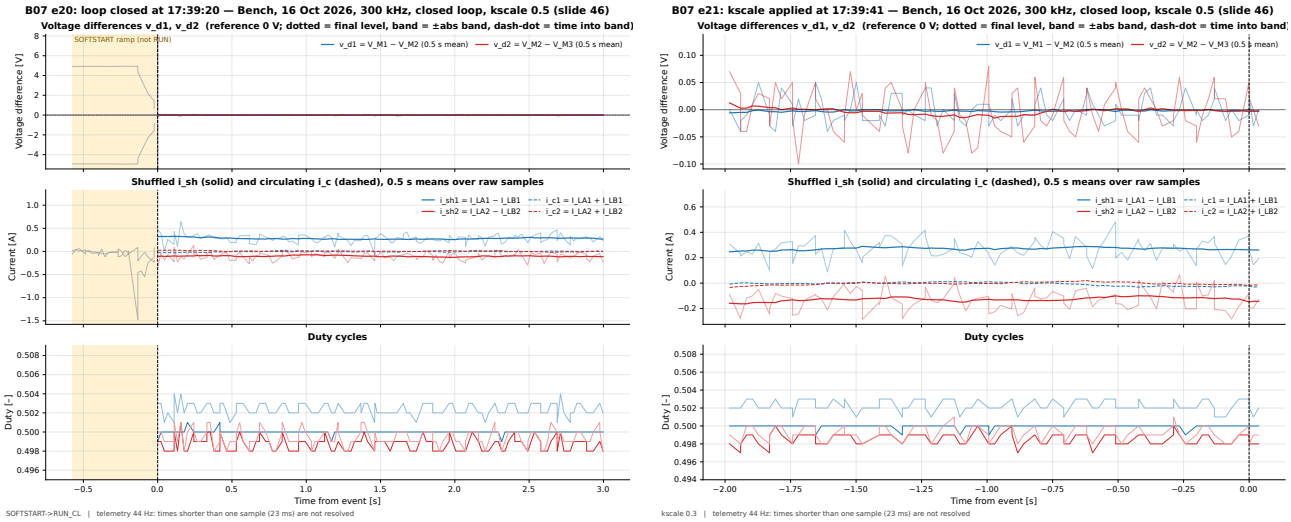


Figure 77: Event zooms for the window above (left: `B07_e20_close`, right: `B07_e21_kscale_apply`).

I.3 400 kHz

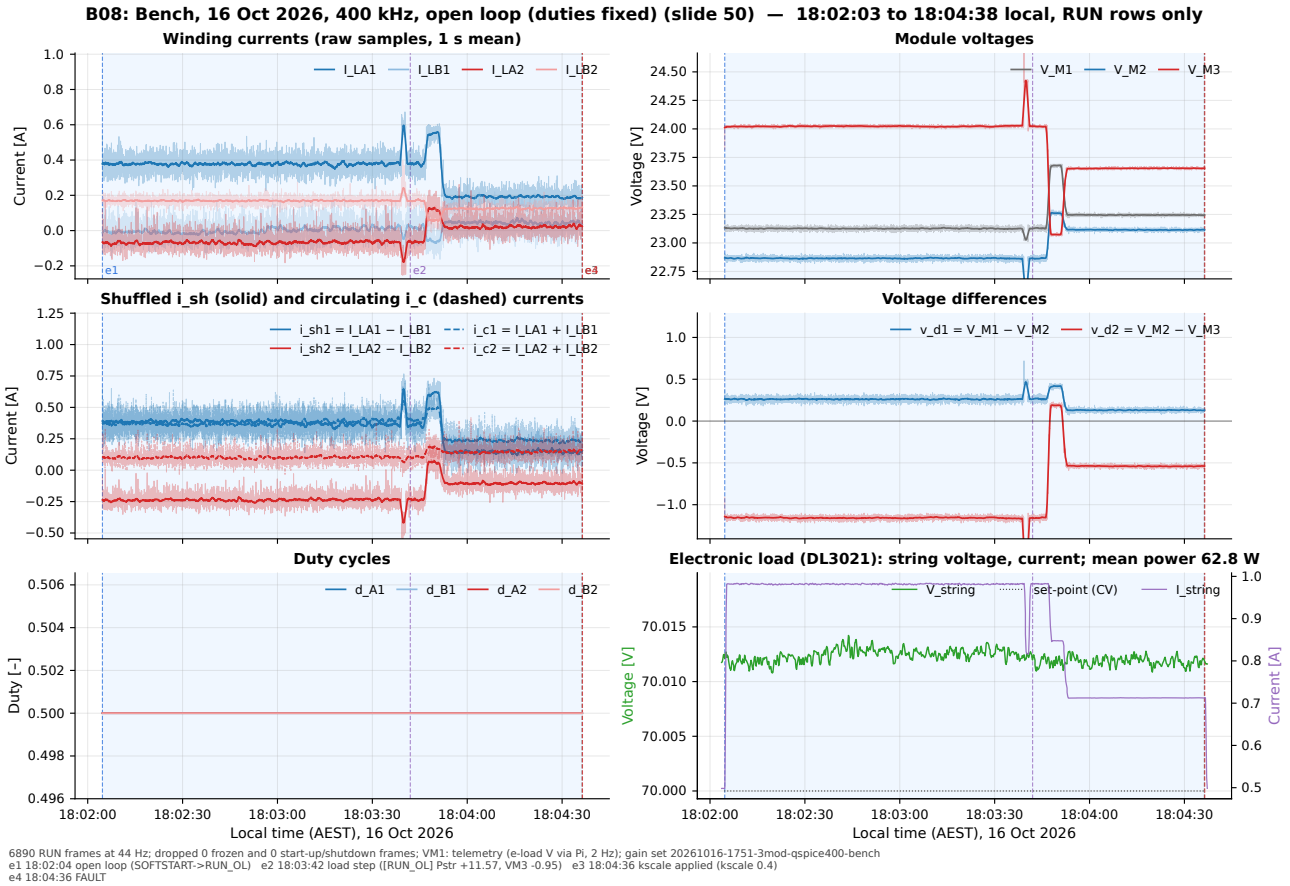


Figure 78: B08: 400 kHz open loop, PSU change at 18:03:40 (slide p50).

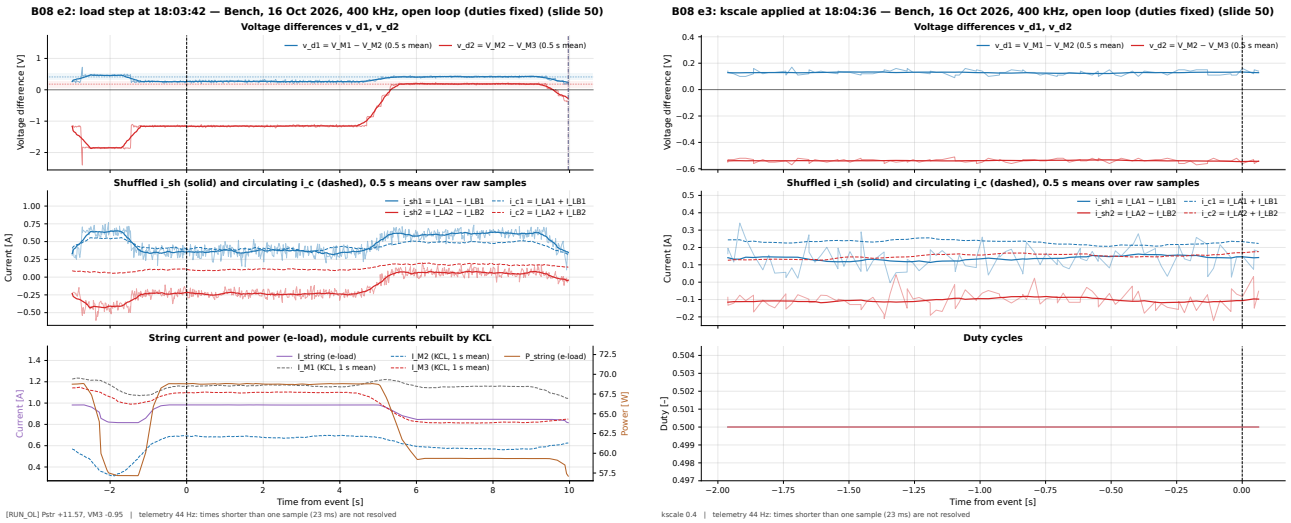


Figure 79: Event zooms for the window above (left: `B08_e2_load_step`, right: `B08_e3_kscale_apply`).

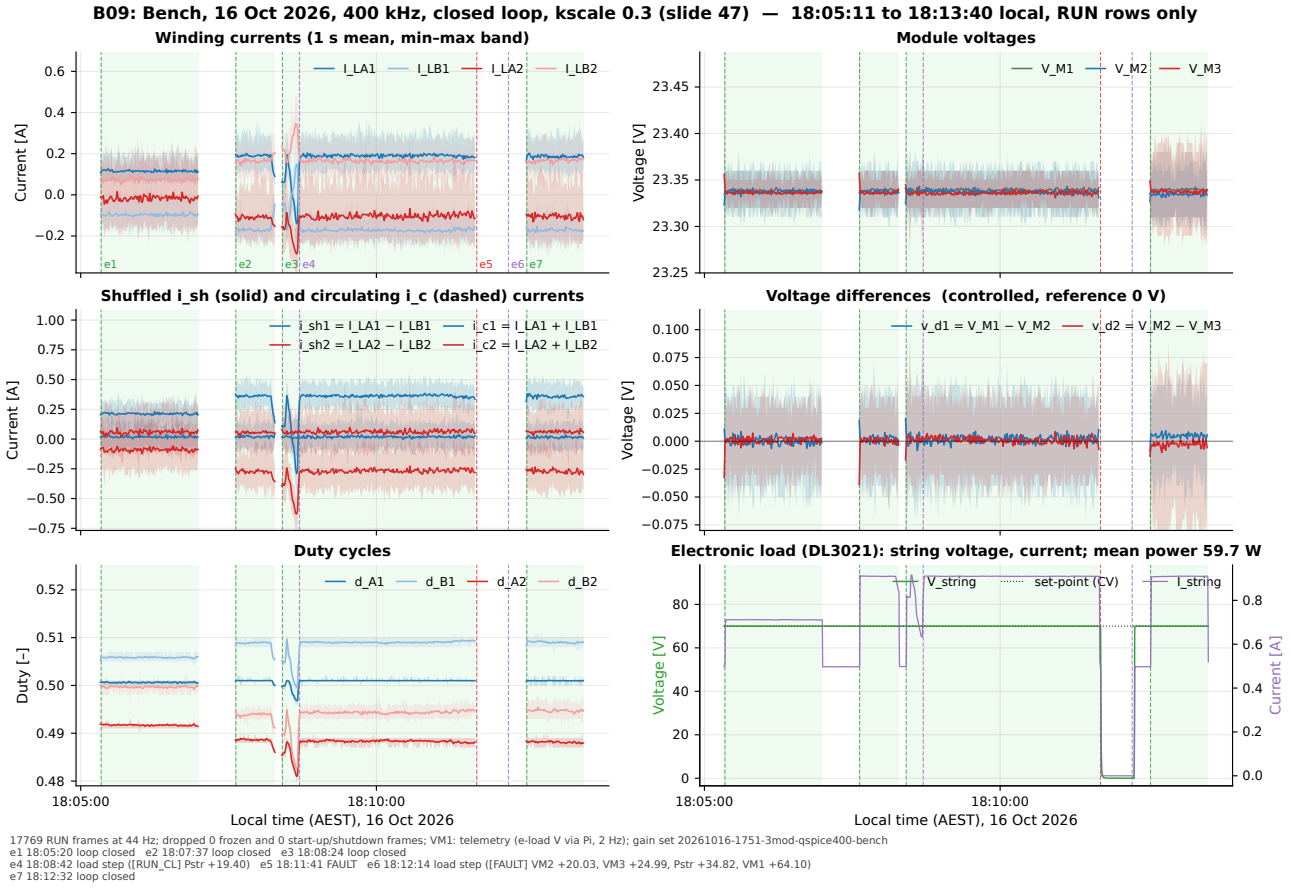


Figure 80: B09: 400 kHz closed loop, $k = 0.3$ (slide p47).

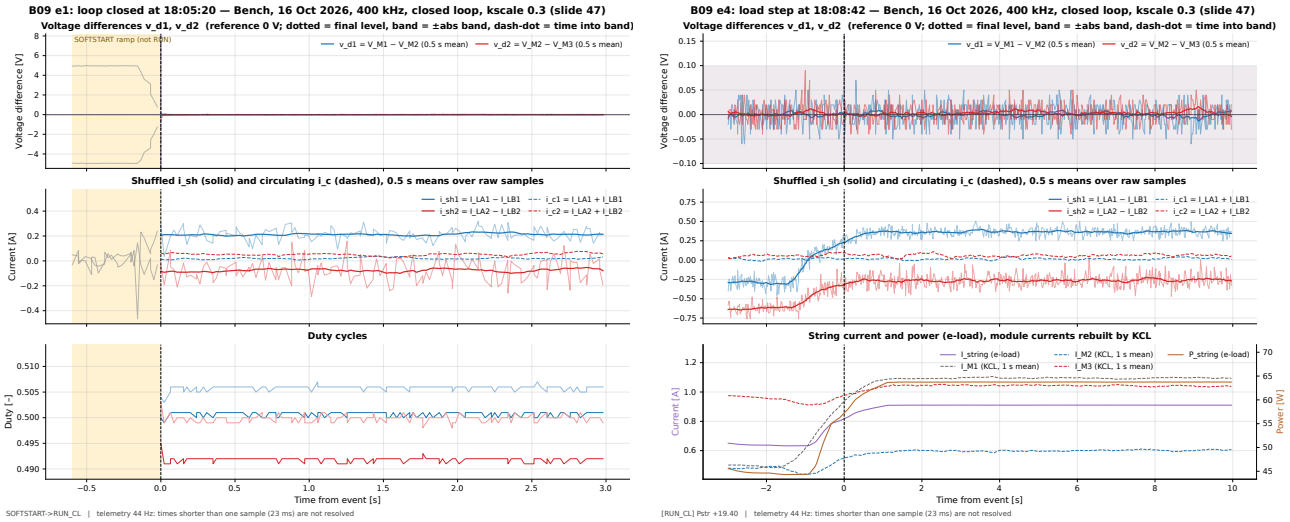


Figure 81: Event zooms for the window above (left: `B09_e1_close`, right: `B09_e4_load_step`).

I.4 Tables

Table 18: Open-loop steady state on hardware (duties 0.5): the residual $v_d = -R_m i_m$ and the circulating current the loop has to remove.

Case	f_{sw}	T [s]	$\bar{v}_{d1}$	$\bar{v}_{d2}$	$\bar{i}_{c1}$	$\bar{i}_{c2}$	$\bar{i}_{d1}$	$\bar{i}_{d2}$	V_{M1}	V_{M2}	V_{M3}	P_{load} [W]
B01	200	321	0.01	-0.17	0.66	3.54	0.47	-2.92	23.3	23.3	23.4	67.9
B01	200	380	0.00	-0.17	0.68	3.55	0.42	-2.92	23.3	23.3	23.5	66.9
B01s	200	409	0.24	0.05	0.46	2.62	0.77	-2.03	23.5	23.3	23.2	65.6
B05	300	280	0.08	-0.58	0.51	0.05	0.55	-0.57	23.2	23.1	23.7	69.1
B05	300	18	0.08	-0.58	0.51	0.05	0.54	-0.58	23.2	23.1	23.7	69.1
B08	400	149	0.23	-0.94	0.34	0.12	0.32	-0.19	23.2	22.9	23.9	62.9
S01	200	184	-0.25	-0.32	-0.75	0.96	-2.20	-0.88	20.1	20.3	20.6	67.7
S03	200	171	-0.04	-0.07	0.08	1.00	-0.38	-0.08	20.3	20.3	20.4	9.5

Table 19: Loop closures on hardware (IDLE → SOFTSTART → RUN_CL): the residual left by the 150 ms soft-start ramp and how the loop removes it. $t_{s,abs}$ is the time to stay inside ±0.1 V / ±0.2 A.

Case	Time	k	Signal	IDLE level	y at RUN_CL	y_∞	SSE	Rise	$t_{s,abs}$ [s]	Noise
B02	15:31:37	0.1	vd2	-4.97	-0.01	0.000	0.000	-	0.00	0.030
B02	15:31:37	0.1	ic2	0.01	1.58	-0.399	-0.399	$\leq$ 89 ms	2.84	0.980
B02	15:34:19	0.1	vd2	-4.95	0.06	0.000	0.000	-	0.00	0.015
B02	15:34:19	0.1	ic2	0.03	-0.98	-0.458	-0.458	$\leq$ 88 ms	2.91	1.629
B03	16:02:08	0.3	vd2	-4.95	-0.09	0.015	0.015	$\leq$ 71 ms	1.90	0.044
B03	16:02:08	0.3	ic2	0.01	0.10	0.158	0.158	-	2.99	0.528
B04	16:07:03	0.3	vd2	-4.95	-0.07	0.030	0.030	-	1.48	0.044
B04	16:07:03	0.3	ic2	-0.00	-0.07	-0.116	-0.116	-	2.99	0.586
B06	17:17:01	0.3	vd2	-4.98	-0.21	0.000	0.000	$\leq$ 23 ms	0.00	0.000
B06	17:17:01	0.3	ic2	-0.02	-0.04	-0.012	-0.012	-	0.00	0.056
B06	17:23:14	0.3	vd2	-4.98	-0.13	0.000	0.000	$\leq$ 23 ms	0.00	0.015
B06	17:23:14	0.3	ic2	-0.01	-0.01	-0.015	-0.015	-	0.00	0.066
B07	17:37:20	0.5	vd2	-4.96	-0.19	0.010	0.010	$\leq$ 23 ms	2.69	0.074
B07	17:37:20	0.5	ic2	-0.02	-0.04	0.007	0.007	-	0.00	0.063
B07	17:38:34	0.6	vd2	-4.96	-0.20	0.035	0.035	$\leq$ 23 ms	2.51	0.037
B07	17:38:34	0.6	ic2	-0.01	-0.04	-0.002	-0.002	-	1.26	0.044
B07	17:39:20	0.4	vd2	-4.94	-0.19	-0.010	-0.010	$\leq$ 23 ms	0.00	0.059
B07	17:39:20	0.4	ic2	-0.02	-0.05	-0.019	-0.019	-	0.00	0.062
B07	17:40:07	0.3	vd2	-4.95	-0.20	0.000	0.000	$\leq$ 23 ms	0.00	0.015
B07	17:40:07	0.3	ic2	-0.01	-0.02	0.008	0.008	-	0.00	0.061
B09	18:05:20	0.3	vd2	-4.96	-0.46	0.000	0.000	$\leq$ 23 ms	0.00	0.015
B09	18:05:20	0.3	ic2	0.02	0.06	0.082	0.082	-	0.73	0.091

Case	Time	k	Signal	IDLE level	y at RUN_CL	y_∞	SSE	Rise	$t_{s,abs}$ [s]	Noise
B09	18:07:37	0.3	vd2	-4.97	-0.80	0.000	0.000	$\leq$ 23 ms	0.00	0.015
B09	18:07:37	0.3	ic2	0.02	0.14	0.049	0.049	-	1.85	0.068
B09	18:08:24	0.3	vd2	-4.96	-0.17	0.010	0.010	$\leq$ 23 ms	0.00	0.015
B09	18:08:24	0.3	ic2	0.01	0.10	0.048	0.048	-	2.49	0.068
B09	18:12:32	0.3	vd2	-4.96	-0.80	-0.010	-0.010	$\leq$ 23 ms	0.18	0.015
B09	18:12:32	0.3	ic2	0.01	0.08	0.050	0.050	-	1.48	0.087
S02	15:08:45	0.1	vd2	0.00	-0.17	0.000	0.000	$\leq$ 43 ms	0.00	0.015
S02	15:08:45	0.1	ic2	-0.03	0.29	-0.049	-0.049	$\leq$ 40 ms	2.97	0.305
S04	15:52:05	0.1	vd2	-0.03	0.13	0.000	0.000	$\leq$ 23 ms	3.00	0.119
S04	15:52:05	0.1	ic2	-0.05	0.30	0.070	0.070	-	3.00	0.326
S05	15:56:54	0.2	vd2	-0.01	-0.26	-0.030	-0.030	$\leq$ 37 ms	2.99	0.119
S05	15:56:54	0.2	ic2	-0.04	0.34	-0.023	-0.023	$\leq$ 23 ms	2.97	0.245
S06	16:01:08	0.3	vd2	-0.02	-0.17	-0.015	-0.015	$\leq$ 23 ms	2.91	0.037
S06	16:01:08	0.3	ic2	-0.04	-0.03	-0.154	-0.154	-	2.99	0.302
S07	16:05:37	0.4	vd2	-0.03	-0.03	-0.030	-0.030	-	2.91	0.052
S07	16:05:37	0.4	ic2	-0.03	0.53	-0.226	-0.226	$\leq$ 23 ms	2.97	0.270
S08	16:09:44	0.5	vd2	-0.03	-0.13	0.010	0.010	$\leq$ 23 ms	2.66	0.059
S08	16:09:44	0.5	ic2	-0.04	0.06	-0.027	-0.027	-	2.97	0.412
S09	16:13:19	0.6	vd2	-0.03	-0.15	-0.010	-0.010	$\leq$ 23 ms	2.86	0.089
S09	16:13:19	0.6	ic2	-0.04	-0.01	-0.182	-0.182	-	2.99	0.386
S10	16:19:11	0.7	vd2	-0.03	0.07	-0.015	-0.015	-	2.77	0.104
S10	16:19:11	0.7	ic2	-0.05	-0.03	-0.006	-0.006	-	2.97	0.178
S11	16:23:11	0.8	vd2	-0.06	0.00	-0.010	-0.010	-	2.80	0.052
S11	16:23:11	0.8	ic2	-0.04	0.22	-0.099	-0.099	37.2 ms	3.00	0.279

Table 20: Gain changes applied while the loop was running (`C,APPLY` during RUN_CL) and the 300 kHz stability limit (from T1/T3 and the pod command log).

Case	Time	Action	Outcome
B04	16:07:50	kscale 0.3 $\rightarrow$ 0.4 applied in RUN_CL	fault 0x10 (module UV) within 1 s
B07	17:37:56	kscale 0.5 $\rightarrow$ 0.6 applied in RUN_CL	fault 0x2000 (main-loop stall) within 2 s
B07	17:38:34	kscale 0.6 run (6 s)	oscillation: $\sigma(v_{d2})$ 1.41 V, pk-pk 11.8 V, 3.2 Hz aliased
B07	17:39:41	kscale 0.6 $\rightarrow$ 0.3 applied in RUN_CL	fault 0x2000 (main-loop stall) within 2 s
B08	18:03:40	PSU change + gain set 400 kHz	OC fault 0x01, restarted

J Solar Test Raw Data (20 Oct 2026)

Carpark string of three KAUSSP80 modules into the DL3021 at 61 V CV, 200 kHz. Peak sun 15:04–15:13 (S01 open loop, S02 closed loop $k = 0.1$), minimal sun from 15:48 (S03 open loop, S04–S11 closed loop $k = 0.1 \ldots 0.8$). S09–S11 are not on a slide. Module currents are reconstructed by KCL from the winding currents, duties and the e-load current.

J.1 Peak sun

Table 21: Bench (16 Oct 2026, NGE103 three-channel PSU into a 70 V CV load): power processed through the shufflers and reconstructed channel currents (indicative before the ADC fix).

Case	Loop	f_{sw}	T [s]	$\bar{P}_{load}$ [W]	$\bar{P}_{shuf}$ [W]	I_{M1}	I_{M2}	I_{M3}	E_{load} [Wh]	E_{shuf} [Wh]	η [%]
B01	OL	200	707	67.2	79.0	1.18	-0.72	2.42	13.20	15.52	99.6
B01s	OL	200	413	65.5	65.4	1.32	-0.47	1.95	7.52	7.50	99.8
B02	CL	200	162	44.5	16.5	0.81	0.41	0.69	2.00	0.74	100.0
B03	CL	200	144	40.6	5.0	0.63	0.51	0.60	1.62	0.20	100.0
B04	CL	200	48	40.6	6.6	0.67	0.51	0.56	0.54	0.09	100.0
B05	OL	300	306	68.9	26.1	1.26	0.42	1.27	5.86	2.22	99.7
B06	CL	300	513	43.3	18.5	0.83	0.24	0.79	6.17	2.64	99.7
B07	CL	300	298	52.0	15.4	0.91	0.43	0.89	4.31	1.27	99.8
B08	OL	400	153	62.7	11.9	1.05	0.64	0.99	2.66	0.50	99.8
B09	CL	400	397	59.6	12.7	1.01	0.58	0.97	6.57	1.40	99.7

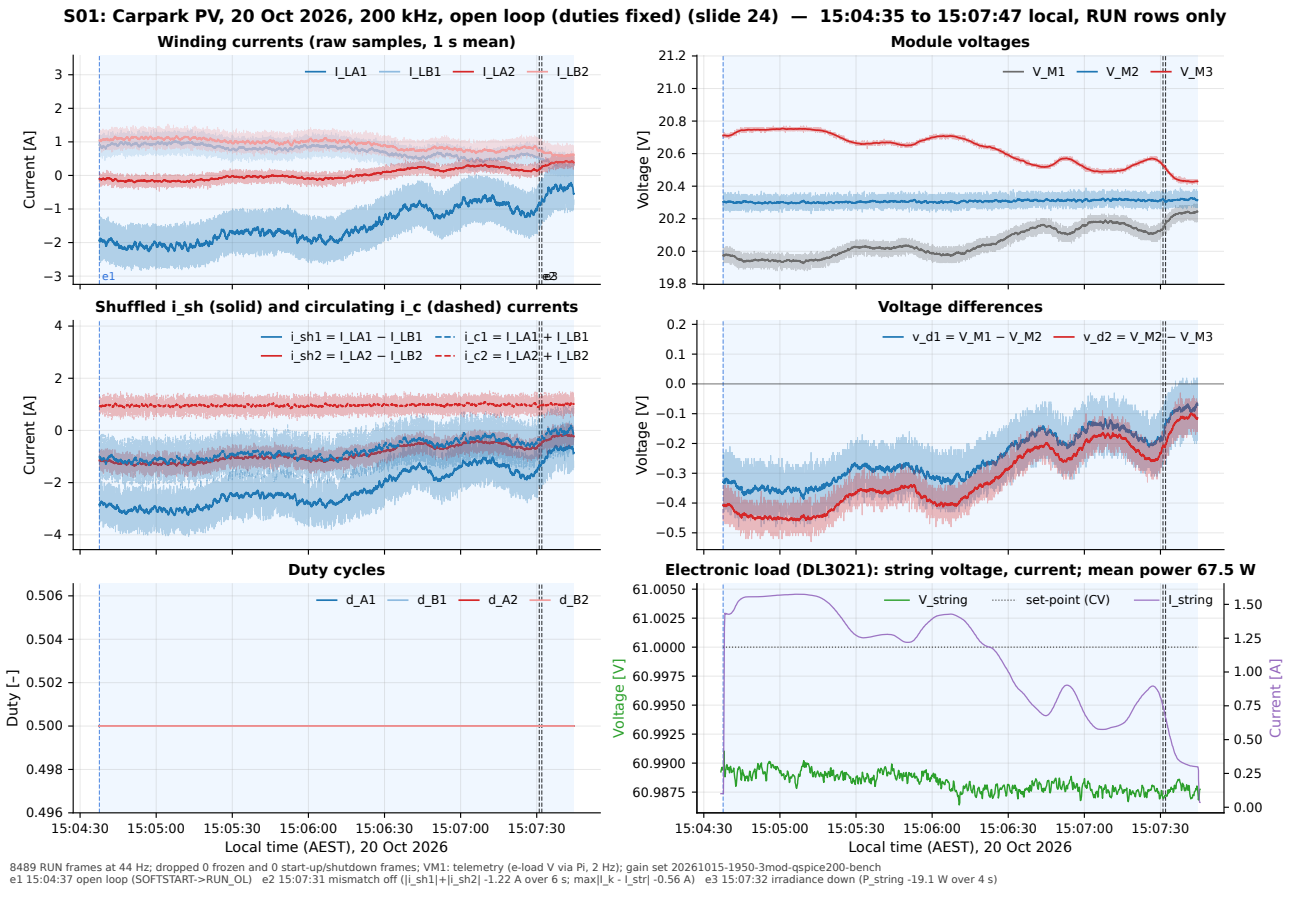


Figure 82: S01: open loop, peak sun (slide p24).

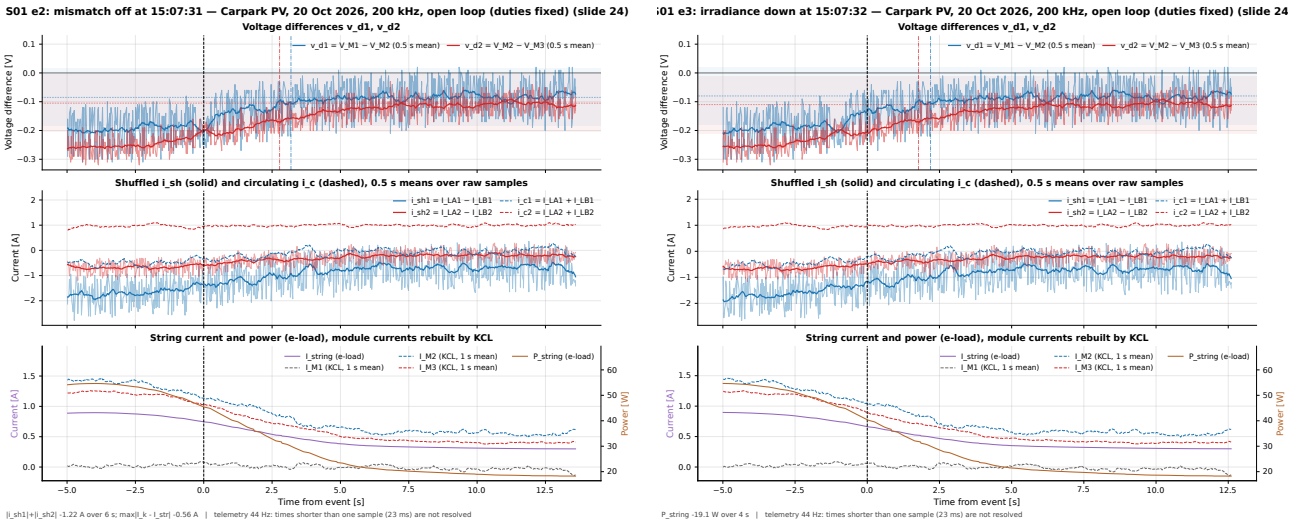


Figure 83: Event zooms for the window above (left: `S01_e2_mismatch_off`, right: `S01_e3_irradiance_down`).

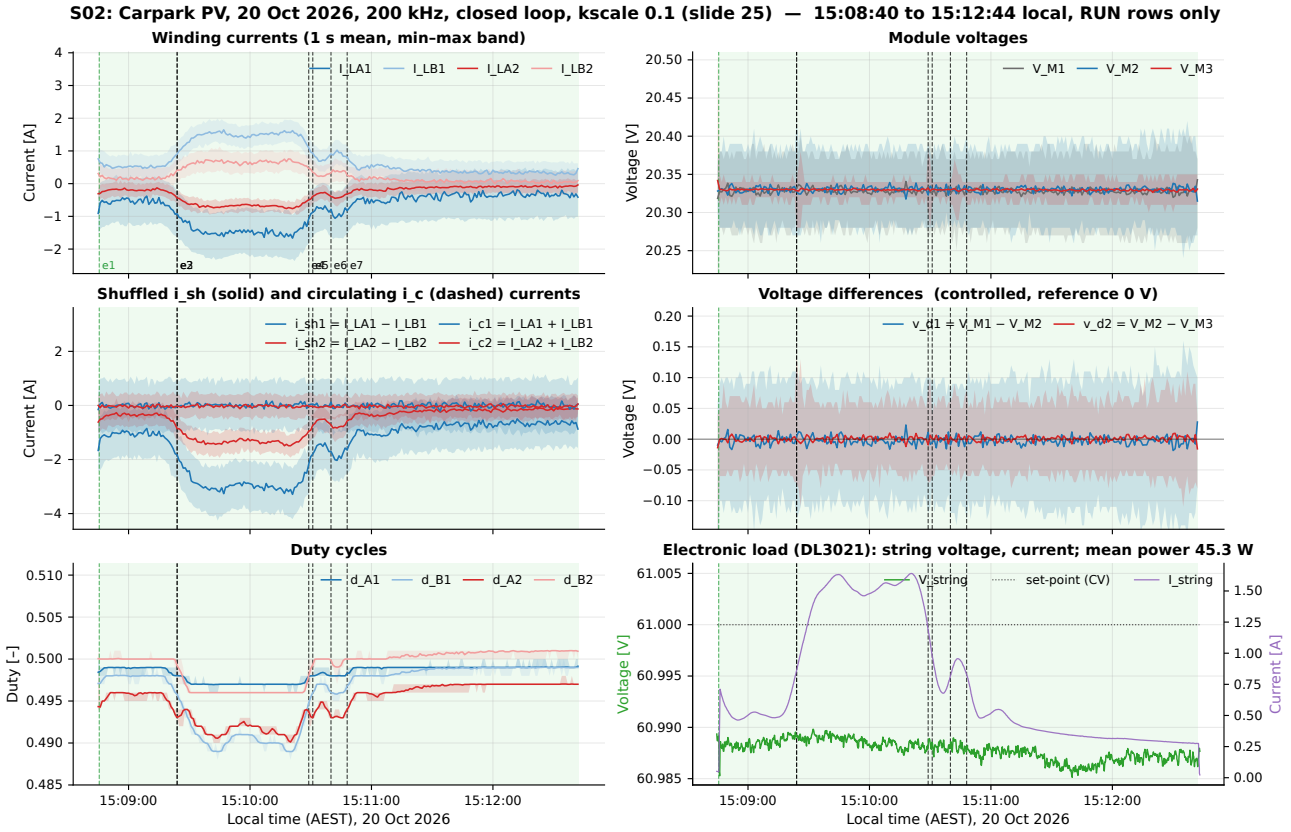


Figure 84: S02: closed loop $k = 0.1$, peak sun; shade event about 15:09:20–15:10:40 (slide p25).

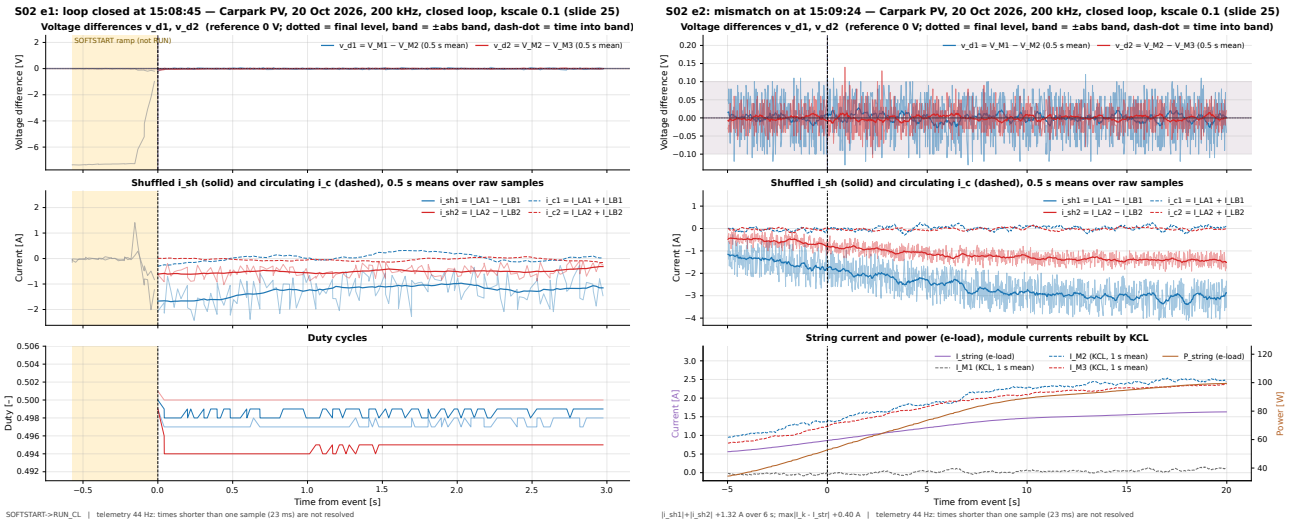


Figure 85: Event zooms for the window above (left: `S02_e1_close`, right: `S02_e2_mismatch_on`).

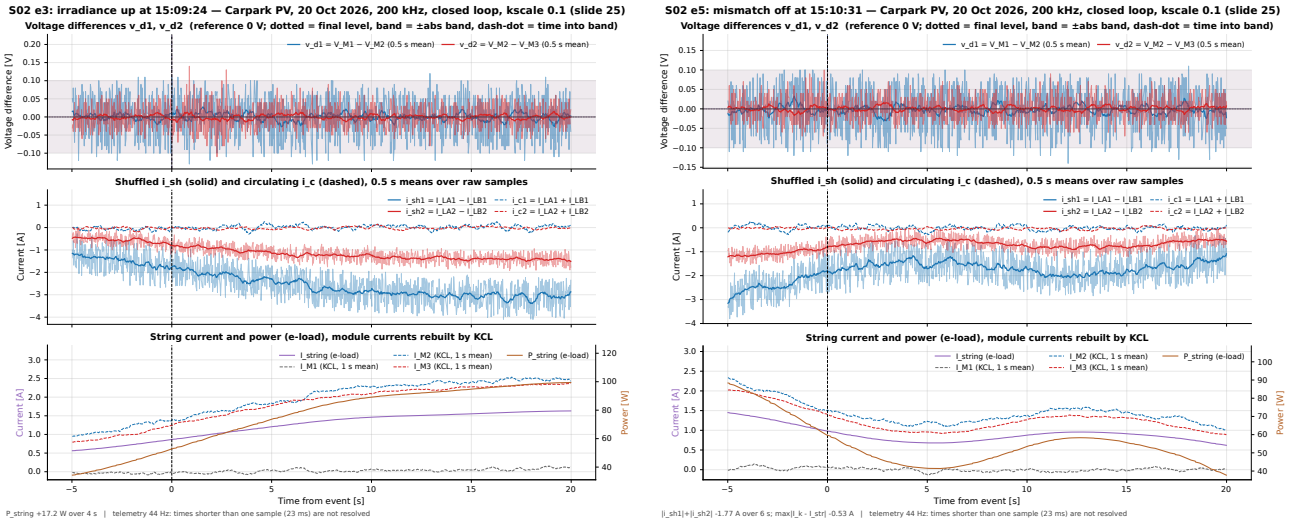


Figure 86: Event zooms for the window above (left: `S02_e3_irradiance_up`, right: `S02_e5_mismatch_off`).

J.2 Minimal sun

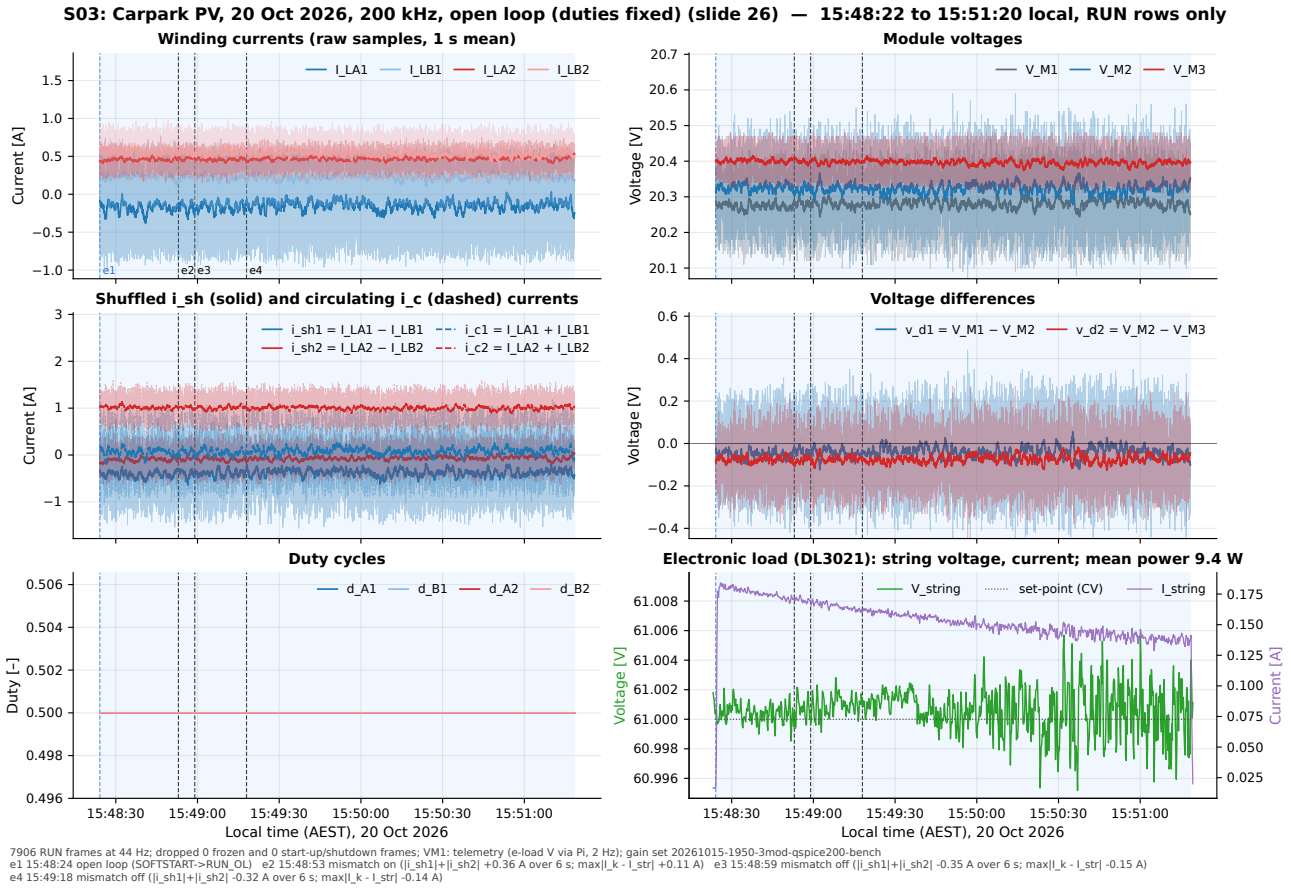


Figure 87: S03: open loop, minimal sun (slide p26).

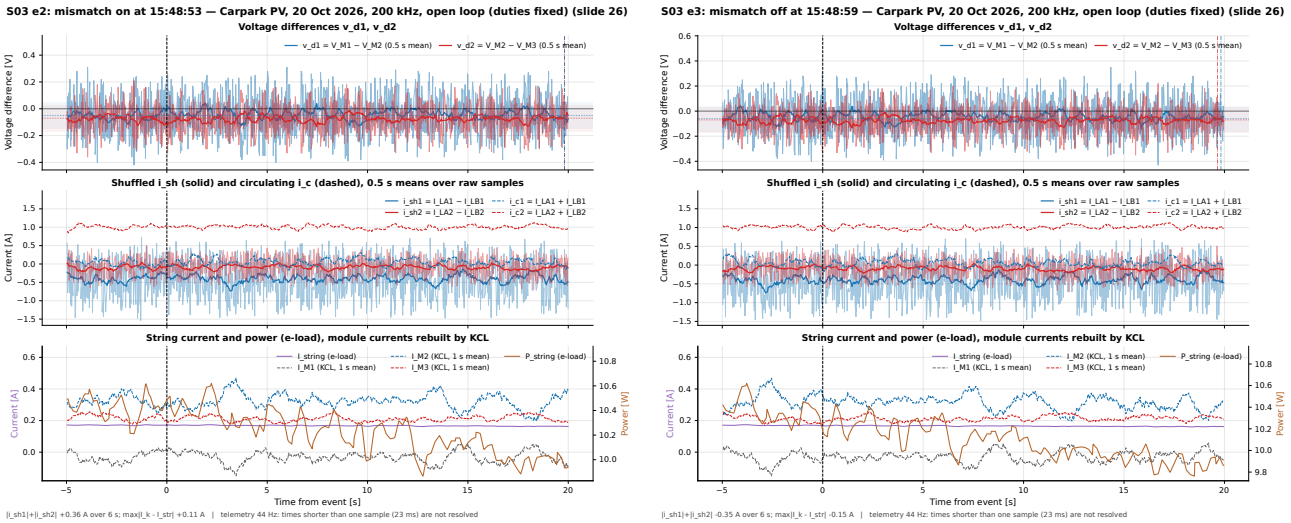


Figure 88: Event zooms for the window above (left: `S03_e2_mismatch_on`, right: `S03_e3_mismatch_off`).

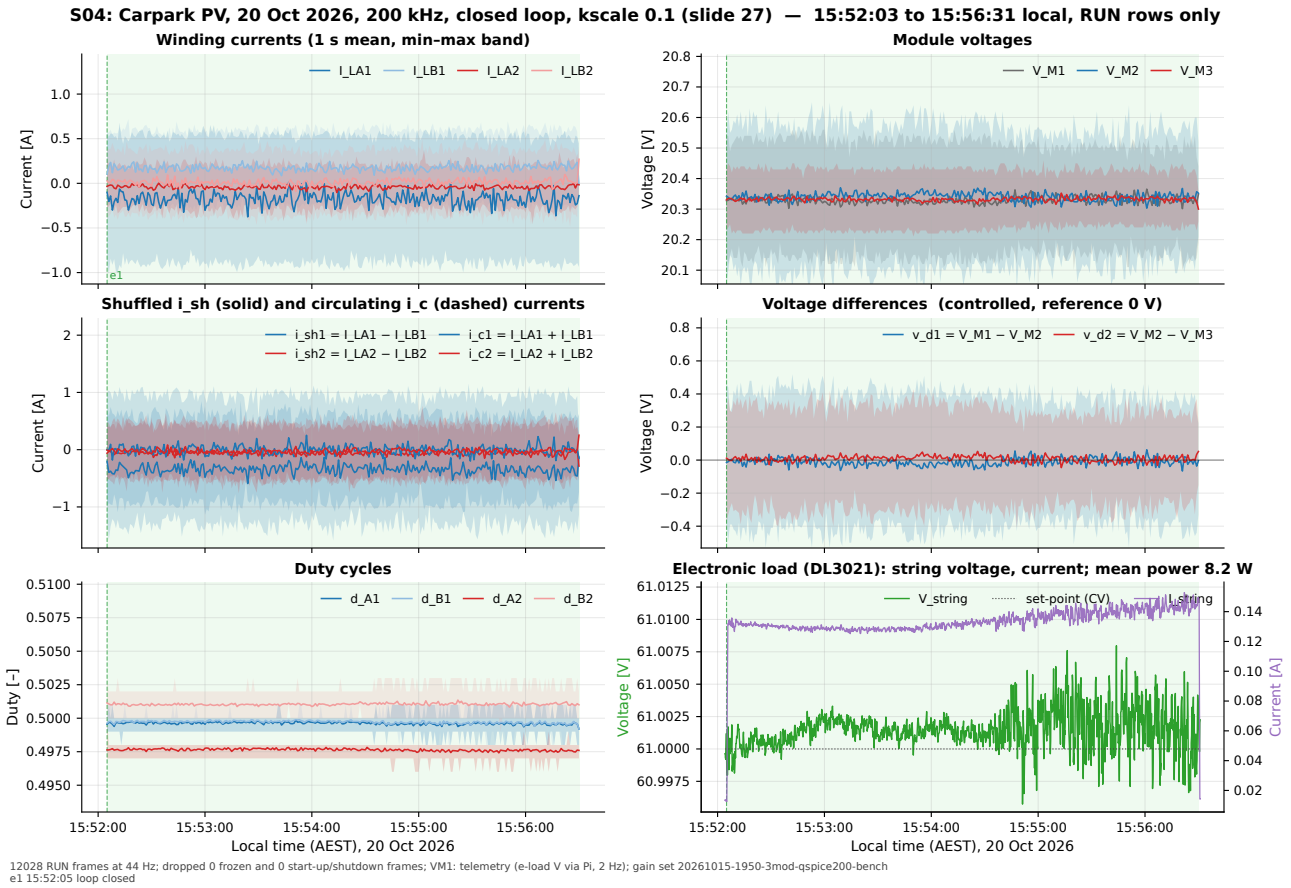


Figure 89: S04: closed loop $k = 0.1$ (slide p27).

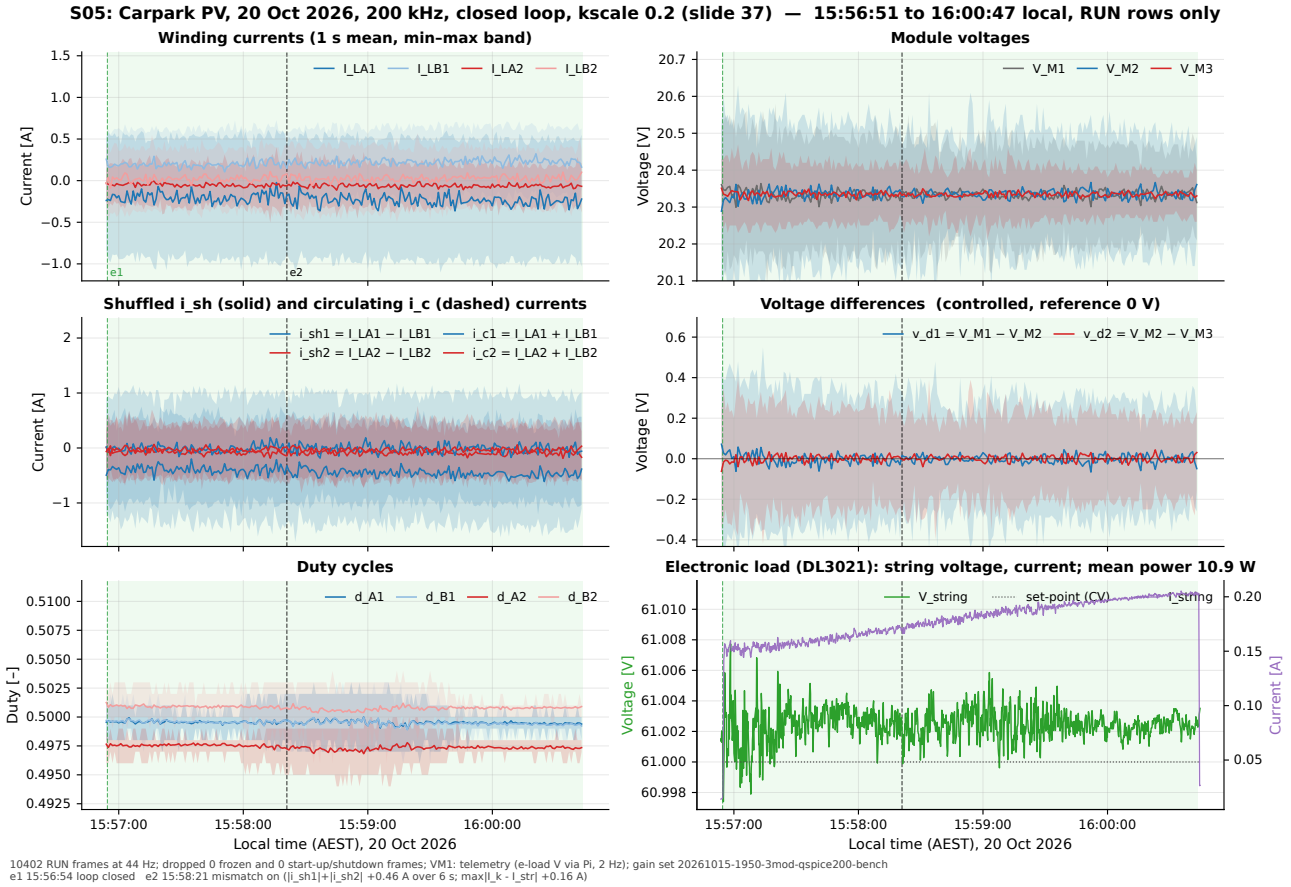


Figure 90: S05: closed loop $k = 0.2$ (slide p37).

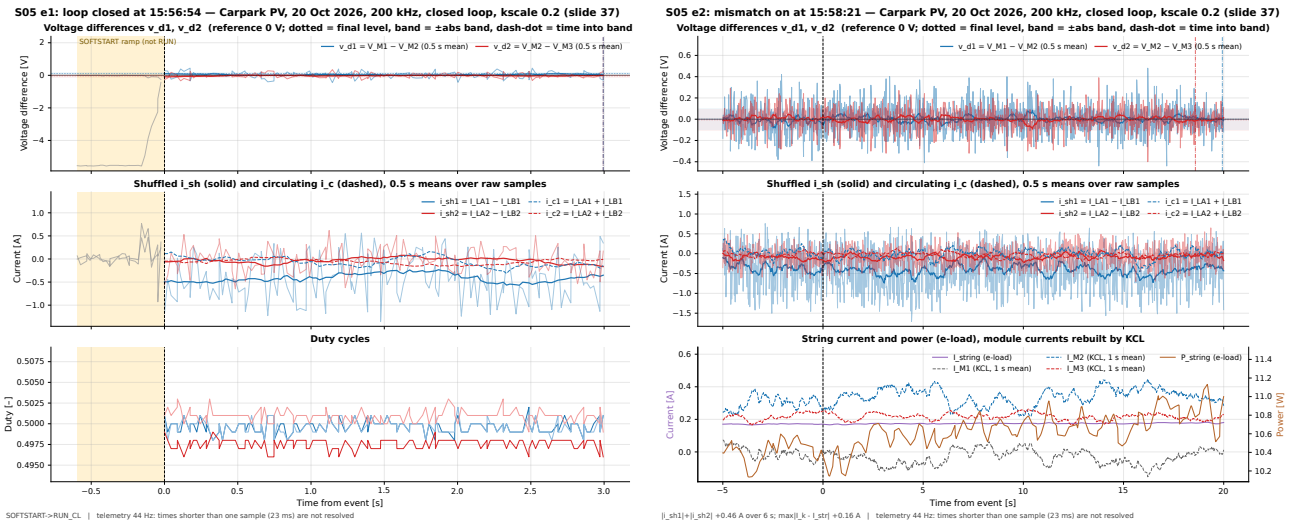


Figure 91: Event zooms for the window above (left: `S05_e1_close`, right: `S05_e2_mismatch_on`).

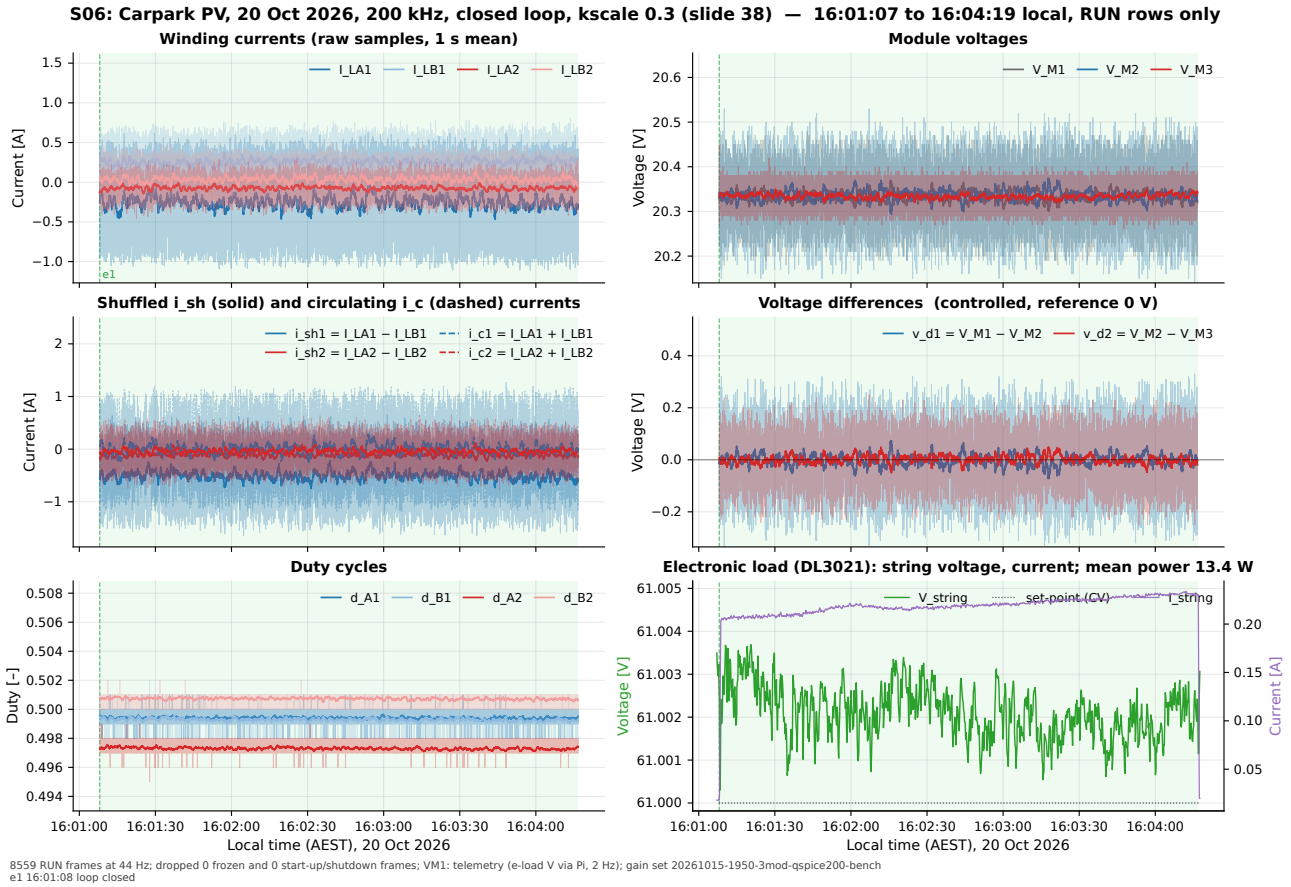


Figure 92: S06: closed loop $k = 0.3$ (slide p38).

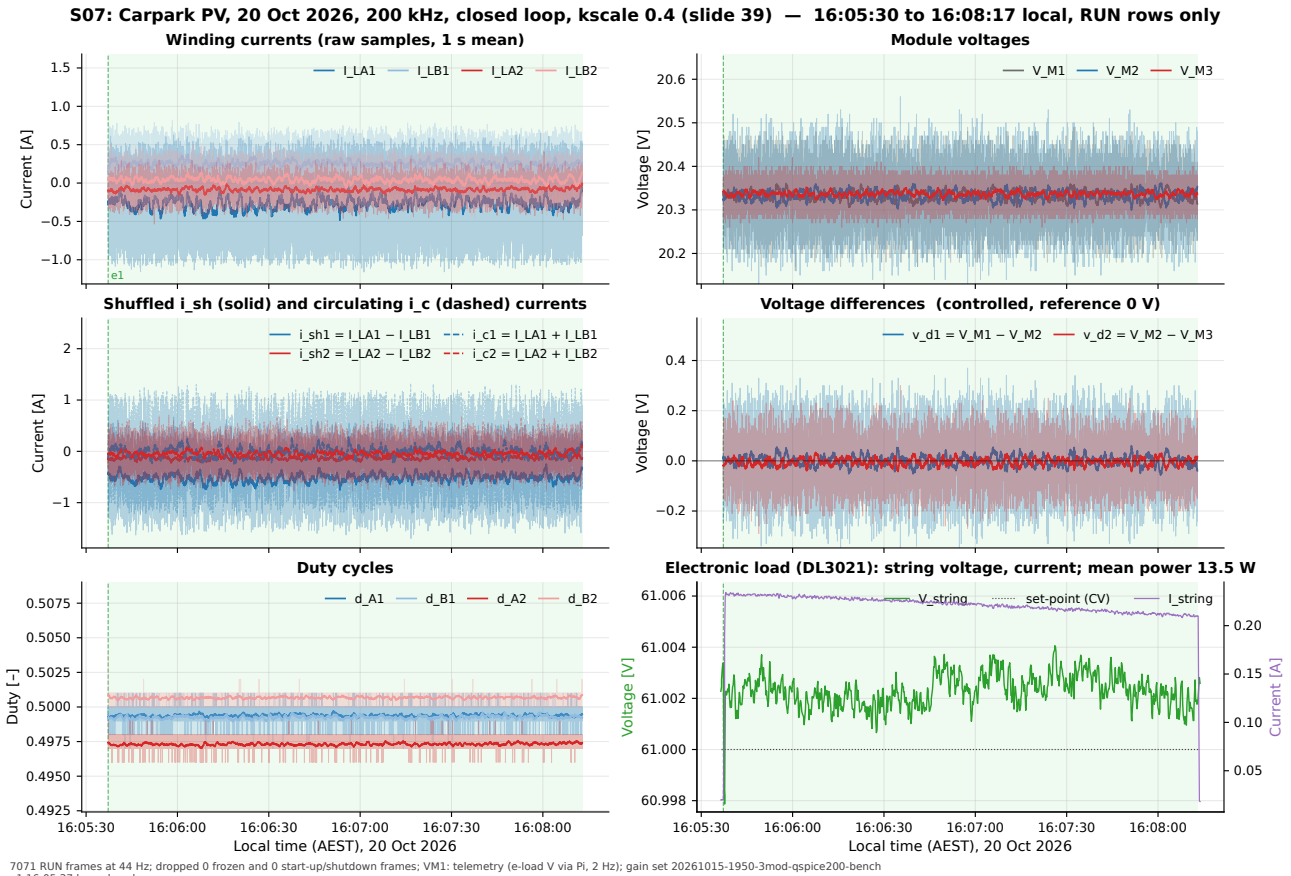


Figure 93: S07: closed loop $k = 0.4$ (slide p39).

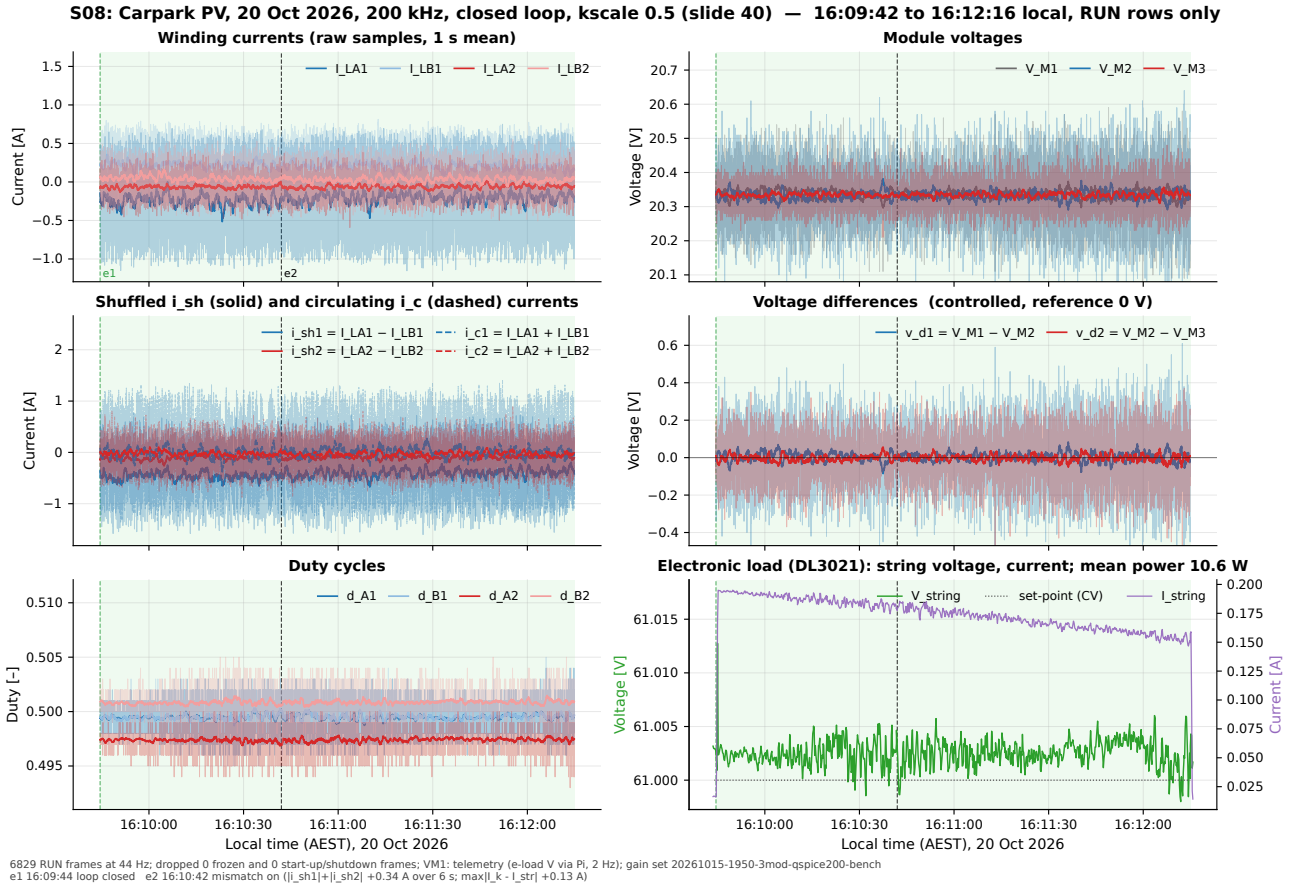


Figure 94: S08: closed loop $k = 0.5$ (slide p40).

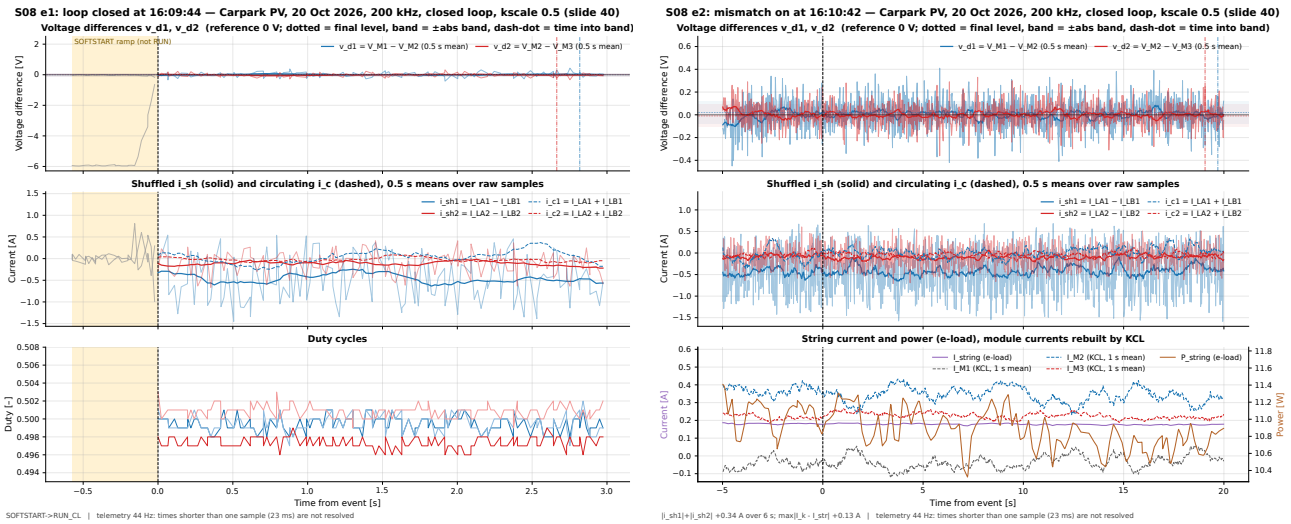


Figure 95: Event zooms for the window above (left: `S08_e1_close`, right: `S08_e2_mismatch_on`).

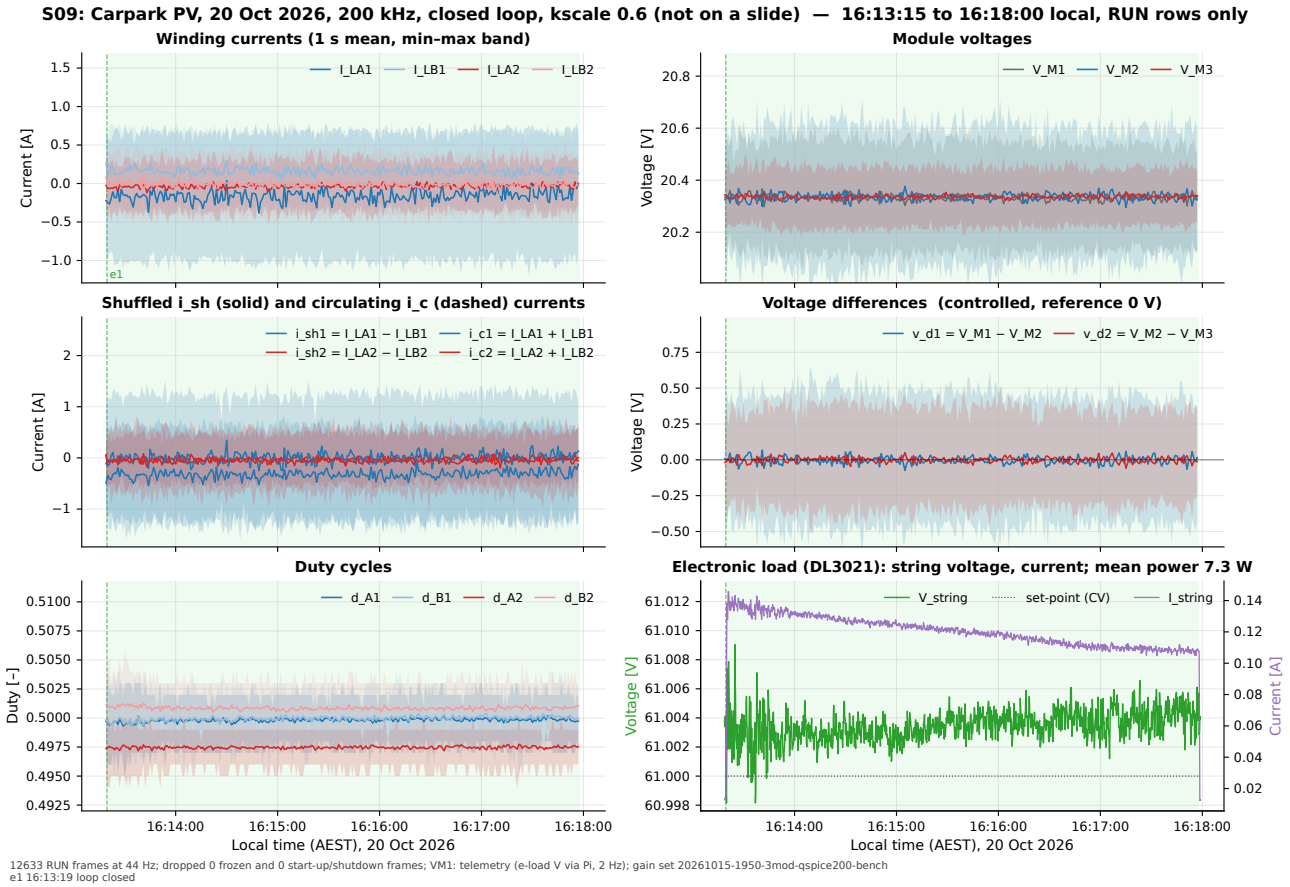


Figure 96: S09: closed loop $k = 0.6$ (slide not on a slide).

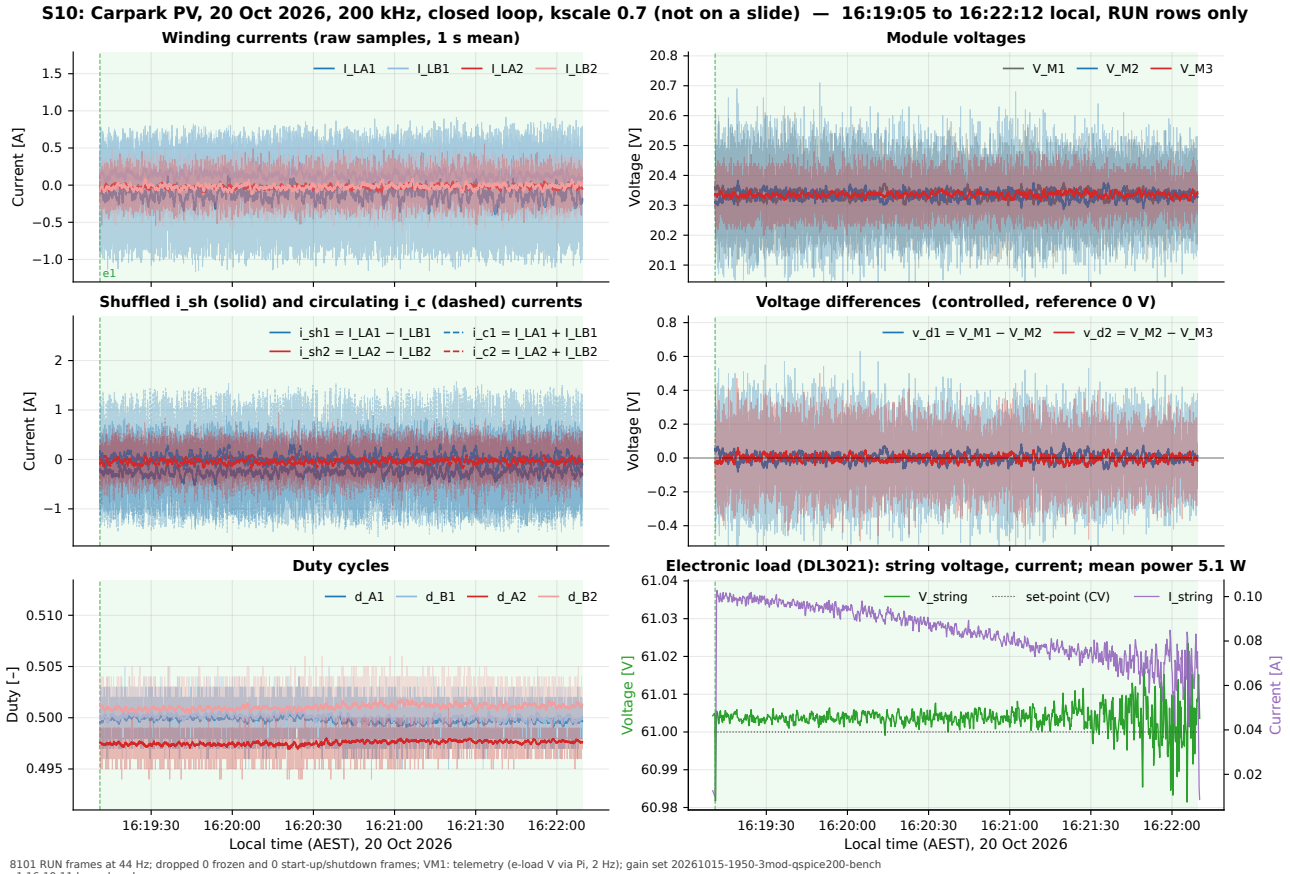


Figure 97: S10: closed loop $k = 0.7$ (slide not on a slide).

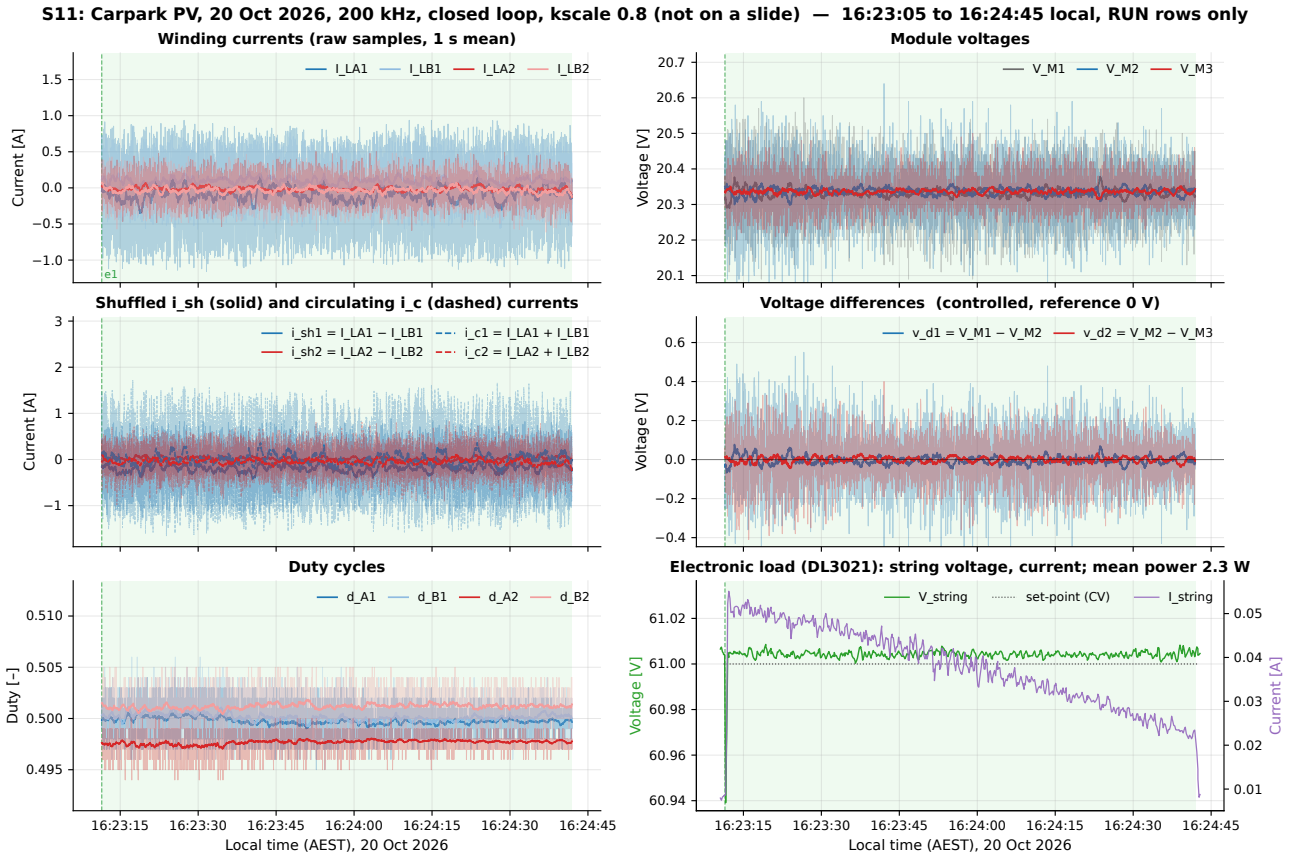


Figure 98: S11: closed loop $k = 0.8$ (slide not on a slide).

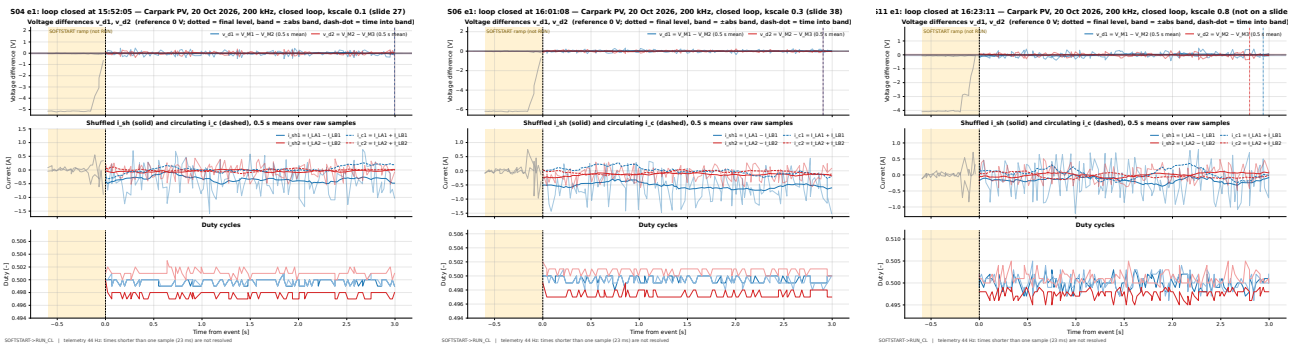


Figure 99: Loop closures at $k = 0.1$, 0.3 and 0.8 in minimal sun (S04, S06, S11).

J.3 Tables

Table 22: Detected shade / irradiance events in the 20 Oct solar windows (current-mismatch and string-power level shifts; irradiance was not logged).

Case	Event	Time	Signal	y_0	y_∞	Δ	Peak dev.	$t_{s,2\%}$ [s]
S01	mismatch_off	15:07:31	id2	-0.62	-0.20	0.42	–	0.66
S01	mismatch_off	15:07:31	vd2	-0.25	-0.11	0.14	–	2.93
S01	mismatch_off	15:07:31	Pstr_i	53.35	18.40	-34.95	–	4.89
S01	mismatch_off	15:07:31	IM3	1.18	0.41	-0.77	–	4.14
S01	irradiance_down	15:07:32	Pstr_i	51.23	18.39	-32.83	–	1.84
S01	irradiance_down	15:07:32	id2	-0.61	-0.21	0.39	–	0.00
S01	irradiance_down	15:07:32	vd2	-0.24	-0.11	0.13	–	1.93
S02	mismatch_on	15:09:24	id2	-0.57	-1.44	-0.87	–	4.56
S02	mismatch_on	15:09:24	vd2	0.00	0.00	0.00	0.06	0.00
S02	mismatch_on	15:09:24	Pstr_i	42.20	98.24	56.04	–	5.92
S02	mismatch_on	15:09:24	IM3	0.97	2.33	1.35	–	7.51
S02	irradiance_up	15:09:24	Pstr_i	42.20	98.24	56.04	–	5.92
S02	irradiance_up	15:09:24	id2	-0.57	-1.44	-0.87	–	4.56
S02	irradiance_up	15:09:24	vd2	0.00	0.00	0.00	0.06	0.00
S02	irradiance_down	15:10:29	Pstr_i	86.29	53.38	-32.91	–	0.00
S02	irradiance_down	15:10:29	id2	-1.20	-0.71	0.50	–	0.00
S02	irradiance_down	15:10:29	vd2	0.00	0.00	0.00	-0.05	0.00
S02	mismatch_off	15:10:31	id2	-1.04	-0.60	0.44	–	11.40
S02	mismatch_off	15:10:31	vd2	0.00	0.00	0.00	0.05	0.00
S02	mismatch_off	15:10:31	Pstr_i	75.64	47.10	-28.54	–	0.00
S02	mismatch_off	15:10:31	IM3	1.78	1.05	-0.73	–	0.00
S02	mismatch_on	15:10:40	id2	-0.55	-0.33	0.22	-1.02	7.31
S02	mismatch_on	15:10:40	vd2	0.00	0.00	0.00	0.05	0.00
S02	mismatch_on	15:10:40	Pstr_i	42.57	30.44	-12.13	–	11.70
S02	mismatch_on	15:10:40	IM3	0.98	0.67	-0.31	–	9.95
S02	mismatch_off	15:10:48	id2	-0.79	-0.38	0.42	–	0.00
S02	mismatch_off	15:10:48	vd2	0.00	0.00	0.00	0.05	0.00
S02	mismatch_off	15:10:48	Pstr_i	56.71	32.50	-24.21	–	2.69
S02	mismatch_off	15:10:48	IM3	1.30	0.71	-0.59	–	1.93
S03	mismatch_on	15:48:53	id2	-0.10	-0.10	-0.00	-0.61	0.00
S03	mismatch_on	15:48:53	vd2	-0.07	-0.07	0.00	-0.26	18.03
S03	mismatch_on	15:48:53	Pstr_i	10.43	10.00	-0.43	0.62	0.00
S03	mismatch_on	15:48:53	IM3	0.22	0.22	-0.01	0.30	0.00
S03	mismatch_off	15:48:59	id2	-0.11	-0.12	-0.01	0.63	0.00
S03	mismatch_off	15:48:59	vd2	-0.07	-0.07	0.00	-0.26	0.75
S03	mismatch_off	15:48:59	Pstr_i	10.35	9.88	-0.47	0.55	0.00
S03	mismatch_off	15:48:59	IM3	0.22	0.22	-0.00	-0.32	0.00
S03	mismatch_off	15:49:18	id2	-0.11	-0.11	0.00	0.62	0.00
S03	mismatch_off	15:49:18	vd2	-0.07	-0.07	0.00	0.24	8.93
S03	mismatch_off	15:49:18	Pstr_i	9.90	9.60	-0.30	0.36	0.00
S03	mismatch_off	15:49:18	IM3	0.22	0.21	-0.01	-0.31	0.00
S05	mismatch_on	15:58:21	id2	-0.10	-0.06	0.04	0.71	19.36
S05	mismatch_on	15:58:21	vd2	0.00	-0.00	-0.00	-0.25	14.98
S05	mismatch_on	15:58:21	Pstr_i	10.45	10.86	0.42	-0.72	0.00
S05	mismatch_on	15:58:21	IM3	0.22	0.21	-0.01	-0.36	19.36
S08	mismatch_on	16:10:42	id2	-0.08	-0.07	0.01	-0.64	0.00
S08	mismatch_on	16:10:42	vd2	0.01	-0.01	-0.02	0.22	17.23
S08	mismatch_on	16:10:42	Pstr_i	11.12	10.73	-0.39	0.57	0.00
S08	mismatch_on	16:10:42	IM3	0.22	0.21	-0.01	0.32	0.00

K Synthesis: Inventory, Metrics and Cross-Phase Overlays

All tables are generated from `8_Results/analysis/out/tables/*.csv` by `tools/make_tables.py`; `make all` in `8_Results/analysis` rebuilds the CSVs from the exported test windows (the full set, including every event and every signal, is in `REPORT.md` and the CSVs).

K.1 Test-case inventory and data rules

Table 23: Test-case inventory: the raw-data slide windows re-exported from the pod's InfluxDB (RUN_OL / RUN_CL rows only).

Case	Slide	Setup	f_{sw} [kHz]	Loop	k	Window (AEST)	RUN [s]	Rate [Hz]	Note
B01	p48	BENCH	200	OL	–	14:32:00-14:44:56	706	40	duties 0.5; e-load CV set-point steps 125 <→ 60 V; VM1 from
B01s		BENCH	200	OL	–	15:09:00-15:16:00	412	40	not on a slide: the only open-loop duty steps (15:11:42 S2 +
B02	p42	BENCH	200	CL	0.1	15:31:34-15:34:25	161	11	before the ADC fix; PSU unstable
B03	p43	BENCH	200	CL	0.3	16:02:05-16:04:33	143	14	slide says kscale 20 %, telemetry says 0.3 with pset 0 (boot
B04	p44	BENCH	200	CL	0.3/0.4	16:07:00-16:07:49	47	14	kscale 0.4 applied at 16:07:50 → OC fault within 1 s
B05	p49	BENCH	300	OL	–	16:31:36-16:37:04	304	36	nan
B06	p45	BENCH	300	CL	0.3	17:17:00-17:25:49	512	44	PSU / e-load set-point steps 17:22:30-17:25:30 while closed
B07	p46	BENCH	300	CL	0.3/0.4/0.5/0.6	17:34:08-17:41:47	295	44	nan
B08	p50	BENCH	400	OL	–	18:02:03-18:04:38	152	44	PSU change at 18:03:40
B09	p47	BENCH	400	CL	0.3	18:05:11-18:13:40	392	44	nan
S01	p24	PV80	200	OL	–	15:04:35-15:07:47	187	44	peak sun
S02	p25	PV80	200	CL	0.1	15:08:40-15:12:44	237	44	peak sun; shade event about 15:09:20-15:10:40
S03	p26	PV80	200	OL	–	15:48:22-15:51:20	174	44	minimal sun
S04	p27	PV80	200	CL	0.1	15:52:03-15:56:31	265	44	minimal sun
S05	p37	PV80	200	CL	0.2	15:56:51-16:00:47	229	44	minimal sun
S06	p38	PV80	200	CL	0.3	16:01:07-16:04:19	189	44	minimal sun
S07	p39	PV80	200	CL	0.4	16:05:30-16:08:17	156	44	minimal sun
S08	p40	PV80	200	CL	0.5	16:09:42-16:12:16	150	44	minimal sun
S09		PV80	200	CL	0.6	16:13:15-16:18:00	278	44	not on a slide (closed_loop_timeslots.pdf)
S10		PV80	200	CL	0.7	16:19:05-16:22:12	178	44	not on a slide (closed_loop_timeslots.pdf)
S11		PV80	200	CL	0.8	16:23:05-16:24:45	90	44	not on a slide (closed_loop_timeslots.pdf)

B01s and S09–S11 are not on a slide. B03 ran at $k = 0.3$ (telemetry), not the 20 % written on the slide.

- Only RUN_OL / RUN_CL frames are analysed; frames repeating all ten control fields are dropped; leading/trailing frames where V_{M2} or V_{M3} left the run median by > 2 V (supply switched off before the FAULT) are dropped.
- I_{M1r} / I_{M2r} are recomputed from I_{LA}, I_{LB}, the duties and the e-load current by the firmware's KCL (listing 3): $I_1 = I_{str} + d_{A1} I_{LA1} - d_{B1} I_{LB1}$, $I_3 = I_{str} - (i_{sh2} - d_{A2} I_{LA2} + d_{B2} I_{LB2})$, $I_2 = I_3 + I_{M2r}$.
- V_{M1}: where the telemetry V_{M1} disagrees with the 5 Hz e-load by > 2 V (200 kHz open-loop runs) it is replaced by $V_{eload} - V_{M2} - V_{M3}$; $v_{d,2}$ is the primary hardware voltage signal.
- Hardware times shorter than one telemetry frame (23 ms at 44 Hz; 70–90 ms at 11–14 Hz) are bounds, printed "≤".

K.2 Hardware closed-loop steady state

Table 24: Closed-loop steady state on hardware (settled RUN_CL segments ≥ 5 s): the references $v_d = 0$, $i_c = 0$ are held with no measurable offset; σ is sample-to-sample sensing noise at 44 Hz.

Case	f_{sw}	k	T [s]	$\bar{v}_{d2}$ [V]	$\sigma(v_{d2})$	pk-pk	$\bar{i}_{c2}$ [A]	$\sigma(i_{c2})$	$\bar{i}_{d1}$ [A]	$\bar{i}_{d2}$ [A]	$\bar{P}_{load}$ [W]
B02	200	0.1	153	0.005	0.040	0.21	-0.023	1.36	0.35	-0.10	44.5
B03	200	0.3	140	0.007	0.074	0.36	0.050	0.40	0.09	-0.05	40.6
B04	200	0.3	44	0.010	0.078	0.39	0.066	0.41	0.19	0.04	40.6
B06	300	0.3	357	0.001	0.016	0.08	-0.000	0.05	0.31	-0.26	35.6
B06	300	0.3	148	0.001	0.016	0.08	-0.000	0.05	0.70	-0.49	61.7
B07	300	0.4	132	-0.002	0.069	0.27	-0.001	0.05	0.41	-0.48	61.0
B07	300	0.5	33	-0.005	0.095	0.36	0.000	0.06	0.36	-0.07	45.5
B07	300	0.6	6	0.052	1.406	11.81	0.004	0.06	0.19	-0.21	40.6

Case	f_{sw}	k	T [s]	$\bar{v}_{d2}$ [V]	$\sigma(v_{d2})$	pk-pk	$\bar{i}_{c2}$ [A]	$\sigma(i_{c2})$	$\bar{i}_{d1}$ [A]	$\bar{i}_{d2}$ [A]	$\bar{P}_{load}$ [W]
B07	300	0.4	18	-0.002	0.044	0.19	0.001	0.05	0.27	-0.12	45.8
B07	300	0.3	91	-0.000	0.013	0.06	-0.001	0.05	0.25	-0.18	44.7
B09	400	0.3	96	0.001	0.015	0.08	0.057	0.09	0.21	-0.09	49.8
B09	400	0.3	37	0.001	0.017	0.08	0.054	0.08	0.36	-0.28	63.6
B09	400	0.3	193	0.001	0.018	0.09	0.060	0.08	0.34	-0.28	62.9
B09	400	0.3	55	-0.003	0.035	0.16	0.061	0.09	0.36	-0.27	63.7
S02	200	0.1	234	0.001	0.031	0.14	-0.032	0.23	-1.51	-0.58	45.5
S04	200	0.1	262	0.008	0.149	0.70	-0.029	0.26	-0.35	-0.06	8.2
S05	200	0.2	226	0.001	0.108	0.56	-0.033	0.27	-0.44	-0.08	10.9
S06	200	0.3	186	0.002	0.088	0.41	-0.039	0.27	-0.51	-0.11	13.4
S07	200	0.4	153	-0.004	0.092	0.43	-0.045	0.28	-0.52	-0.13	13.5
S08	200	0.5	147	-0.002	0.114	0.62	-0.030	0.29	-0.43	-0.10	10.6
S09	200	0.6	275	-0.000	0.172	0.80	-0.050	0.30	-0.32	-0.02	7.3
S10	200	0.7	175	-0.002	0.138	0.71	-0.046	0.31	-0.26	-0.00	5.1
S11	200	0.8	88	-0.000	0.099	0.59	-0.048	0.31	-0.19	0.02	2.3

K.3 Cross-phase step metrics

Table 25: Circulating-current channel i_{c2} during the $+1$ V v_{d2} reference step (S4, 200 kHz): peak deviation and recovery, i.e. the decoupling check (the current loop should barely move).

Setup	k	peak an. [A]	t_s an. [ms]	peak QSP [A]	t_s QSP [ms]	SSE QSP [A]	peak Sim [A]	t_s Sim [ms]	SSE Sim [A]
BENCH	0.1	-0.004	0.0	-0.038	0.0	0.005	-0.007	0.0	0.006
BENCH	0.2	-0.004	0.0	-0.033	0.0	0.005	-0.019	0.0	0.007
BENCH	0.3	-0.005	0.0	-0.022	0.0	0.006	-0.029	0.0	0.006
BENCH	0.4	-0.005	0.0	–	–	–	–	–	–
BENCH	0.5	-0.005	0.0	-0.030	0.0	0.007	-0.026	0.0	0.010
BENCH	1.0	-0.008	0.0	-0.025	0.0	0.005	-0.037	0.0	0.016
PV80	0.1	-0.005	0.0	-0.051	0.0	0.064	0.017	0.0	-0.010
PV80	0.2	-0.005	0.0	-0.049	0.0	0.057	0.015	0.0	-0.010
PV80	0.3	-0.005	0.0	0.072	0.0	0.041	0.019	0.0	-0.010
PV80	0.4	-0.005	0.0	–	–	–	–	–	–
PV80	0.5	-0.005	0.0	-0.106	7.0	0.075	0.018	0.0	-0.007
PV80	1.0	-0.007	0.0	0.120	0.9	0.027	-0.018	0.0	-0.002

Table 26: Shade disturbance (S3: module 3 loses 1/3 of its current, PV80 plant): simulated recovery (ms) against the detected hardware shade events of 20 Oct.

k	Signal	τ an.	t_s an.	τ QSP	t_s QSP	OS QSP	t_s Sim	HW Δ	HW peak	HW cases
0.1	Pstr	–	–	0.00	6.3	6.2	0.1	21.95	–	S02 S02
0.2	Pstr	–	–	0.00	0.1	3.9	0.1	-0.01	-0.05	S03 S05
0.3	Pstr	–	–	0.00	3.3	7.4	0.1	–	–	nan
0.5	Pstr	–	–	0.00	1.6	4.1	0.1	-0.39	0.57	S08
1.0	Pstr	–	–	0.00	0.7	6.1	0.1	–	–	nan
0.1	id2	0.08	18.9	0.07	15.6	0.0	0.2	-0.33	-1.02	S02 S02
0.2	id2	0.08	9.3	0.05	0.3	0.0	0.2	0.02	0.05	S03 S05
0.3	id2	0.08	6.1	0.07	7.5	0.0	0.2	–	–	nan
0.4	id2	0.08	4.5	–	–	–	–	–	–	nan
0.5	id2	0.08	3.6	0.06	3.0	0.0	0.2	0.01	-0.64	S08
1.0	id2	0.08	1.6	0.06	0.9	0.0	0.2	–	–	nan
0.1	vd2	–	–	–	–	–	–	0.00	0.05	S02 S02
0.2	vd2	–	–	0.01	2.0	20.5	–	-0.00	-0.25	S03 S05
0.5	vd2	–	–	–	–	–	–	-0.02	0.22	S08

K.4 Overlays

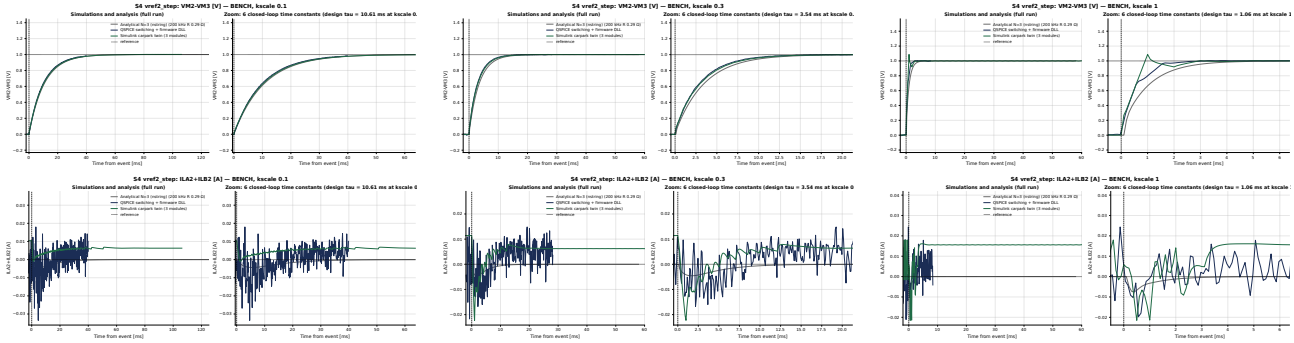


Figure 100: S4 $+1\,\mathrm{V}$ reference step, BENCH plant, $k = 0.1/0.3/1$: $v_{d,2}$ (top) and $i_{c,2}$ (bottom), analytical / QSPICE / Simulink overlaid; hardware closures as markers where they exist.

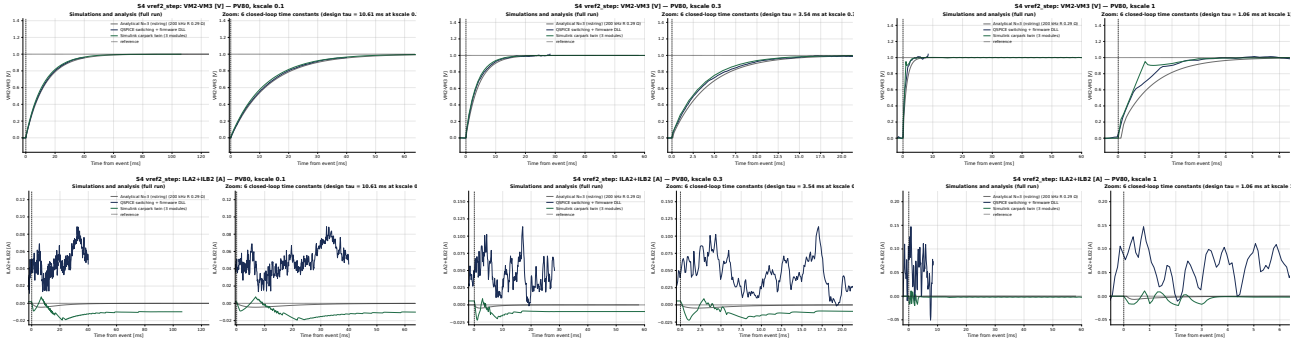


Figure 101: S4 $+1\,\mathrm{V}$ reference step, PV80 plant, $k = 0.1/0.3/1$: $v_{d,2}$ (top) and $i_{c,2}$ (bottom), analytical / QSPICE / Simulink overlaid; hardware closures as markers where they exist.

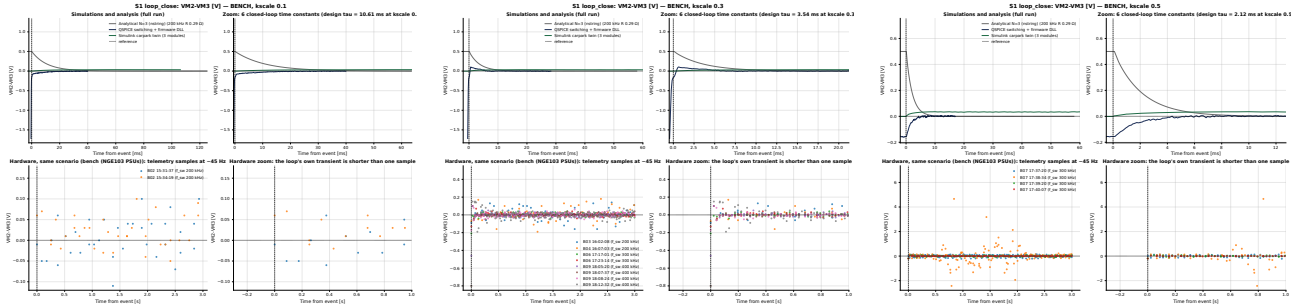


Figure 102: S1 loop closure on the open-loop residual, BENCH plant, $k = 0.1/0.3/0.5$: $v_{d,2}$.

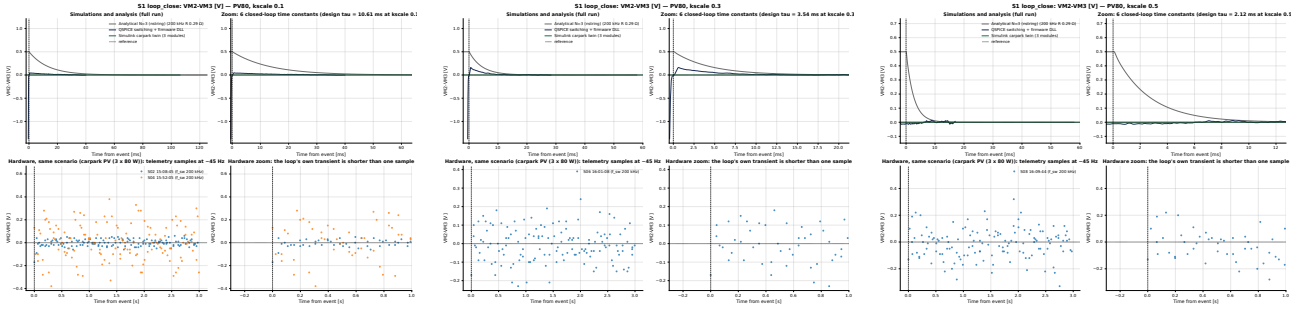


Figure 103: S1 loop closure on the open-loop residual, PV80 plant, $k = 0.1/0.3/0.5$: $v_{d,2}$.

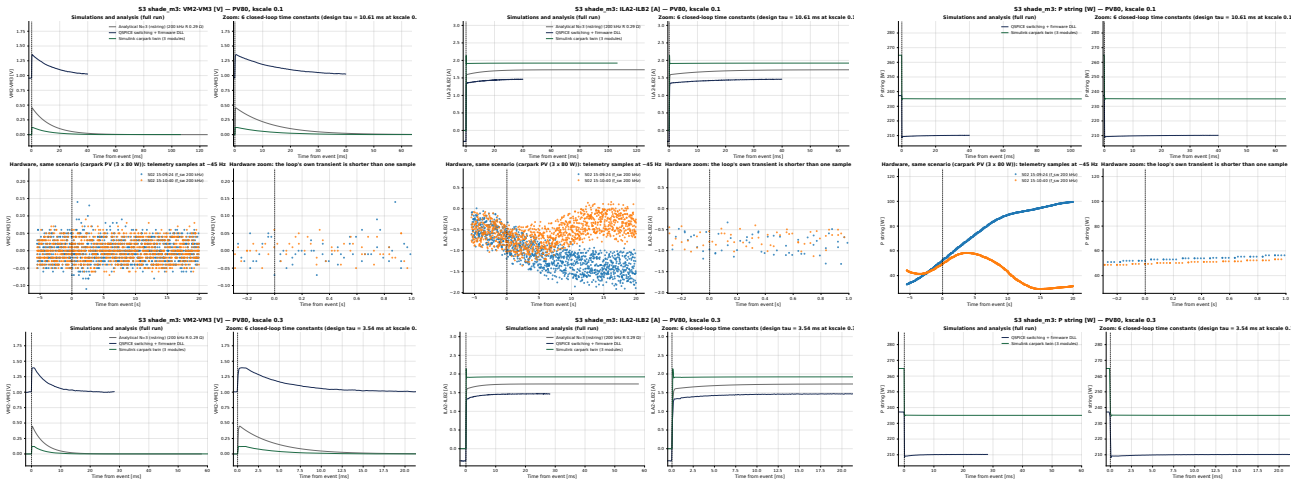


Figure 104: S3 shade on module 3, PV80, $k = 0.1$ (top) and 0.3 (bottom): $v_{d,2}$, $i_{m,2}$ and string power; hardware shade events of S02/S05/S08 as markers.

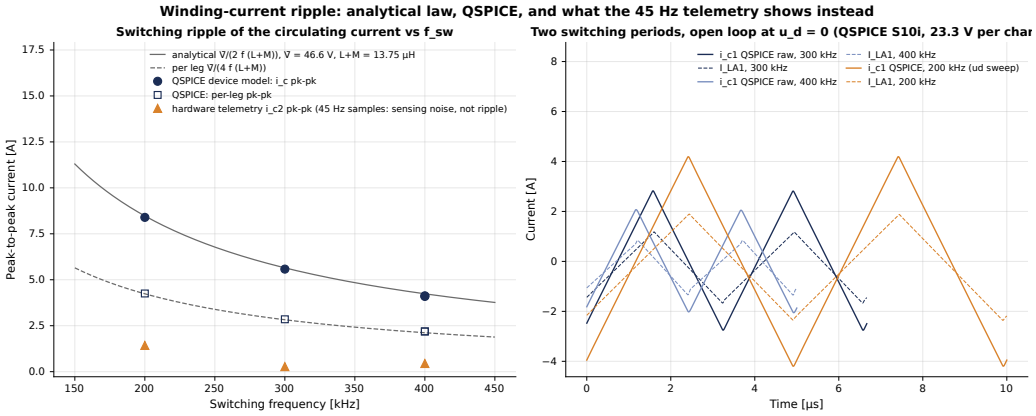


Figure 105: Winding-current ripple vs switching frequency: analytical law, QSPICE, and the 44 Hz telemetry noise (which is not the ripple).

L Thermal Data and Oscilloscope Captures

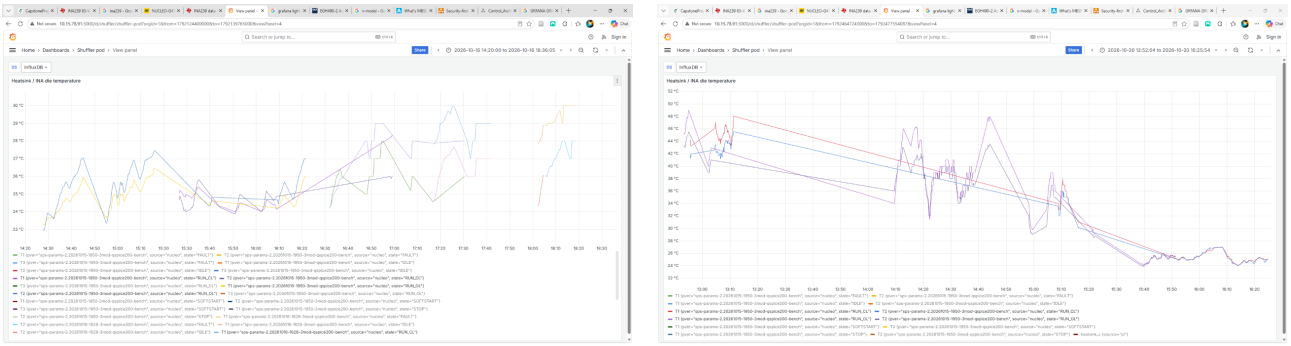


Figure 106: INA239 die temperatures (T_1–T_3) and the heatsink DS18B20 over the bench day (16 Oct, 14:20–18:36, left) and the solar day (20 Oct, 12:52–16:25, right); over-temperature trip at 85 °C (INA) and 80 °C (heatsink, Pi-side E-stop).

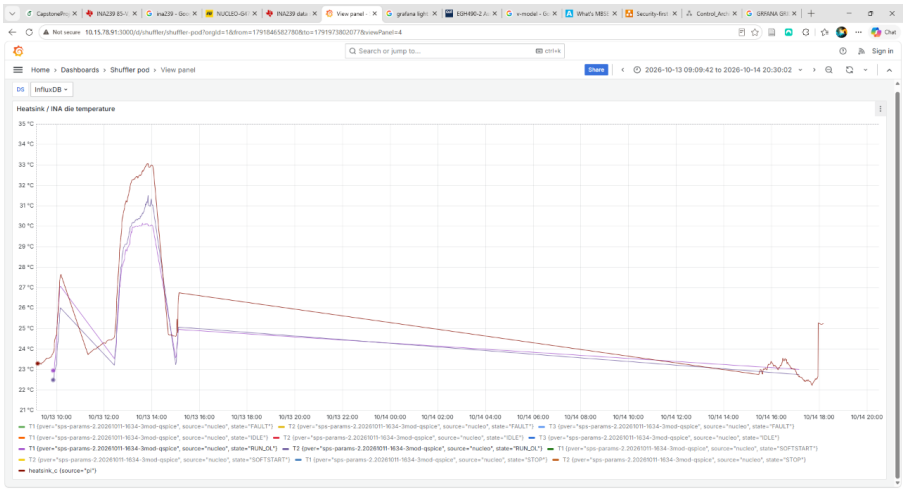


Figure 107: 100 kHz open-loop thermal run, 13–14 Oct (the "sort of working but bizarre" 100 kHz exploration).

▷ Trade-off statement for the body: temperatures rise with f_{sw} (dead-time commutation loss $R_{sw} \propto f_{sw}^3$) while the open-loop residual and the current-loop gain fall; 300 kHz balanced the two on the bench.

M The Journey

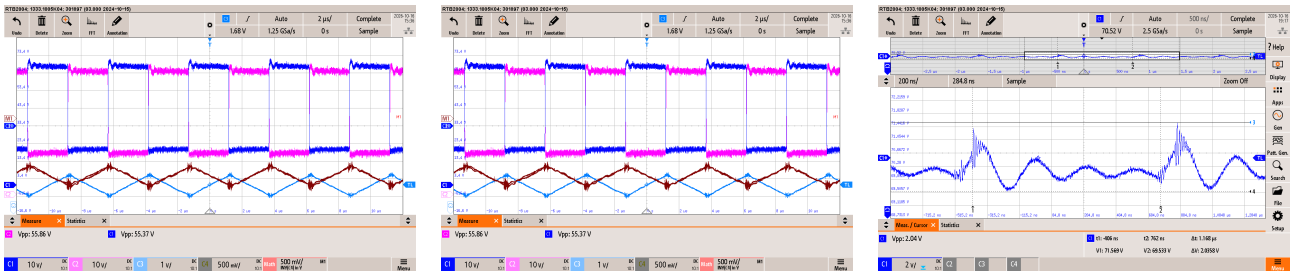


Figure 108: Oscilloscope captures of the switch-node waveforms at 200, 300 and 400 kHz (dead time and 180° interleave check).

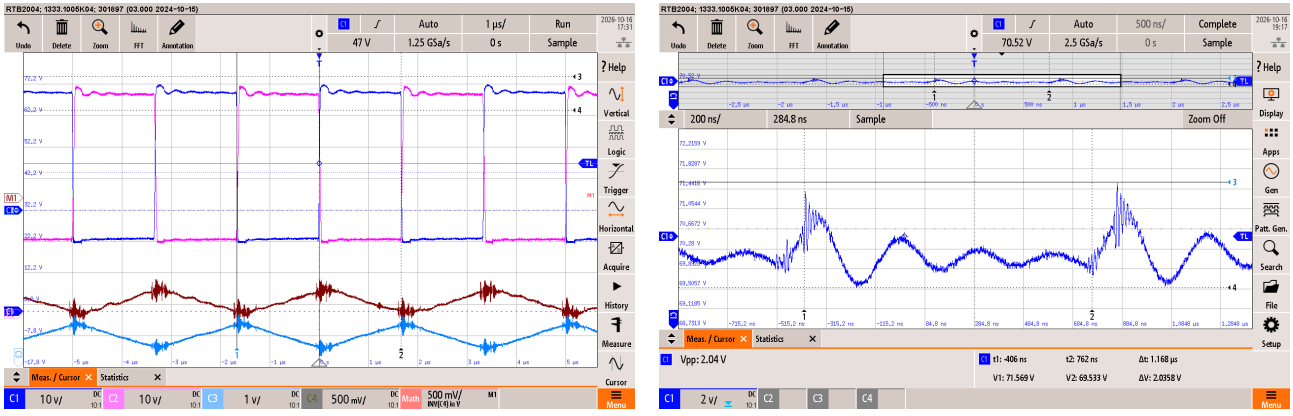
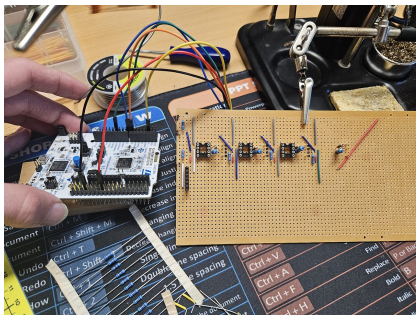


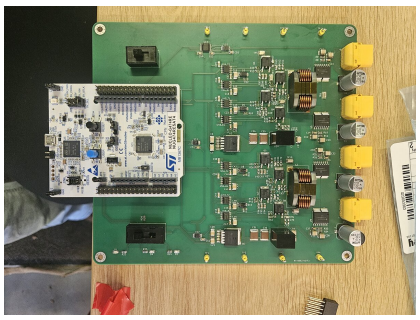
Figure 109: Further bench captures from 16 Oct (17:31 and 19:17).



(a) Aug: first proto-board with the Nucleo.

(b) Sep: the Version 0 PCB arrives.

(c) Oct 4: reflow in the PCB oven.

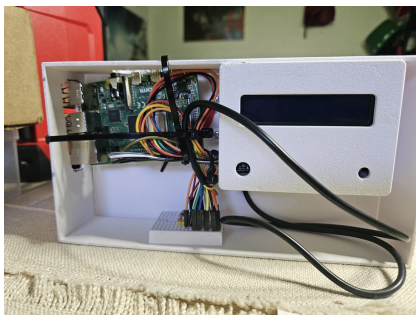


(d) Oct 4: assembled board with the Nucleo.

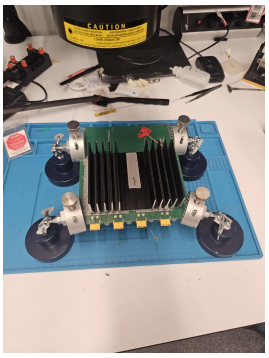
(e) Oct 4: the hardware teammate with the board.

(f) Oct 6: heatsink.

(g) Oct 7: enclosure work.



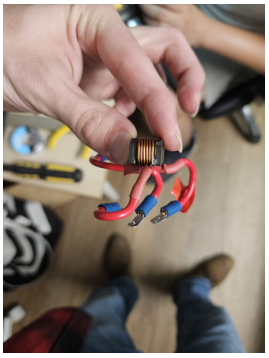
(h) Oct 11: pod with the Pi, LCD and IR receiver.



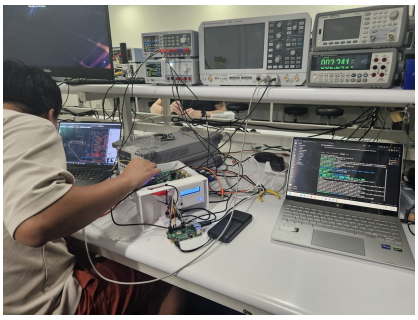
(i) Oct 14: heatsink mounted, thermal paste.

Figure 110: Build photographs (1 of 2).

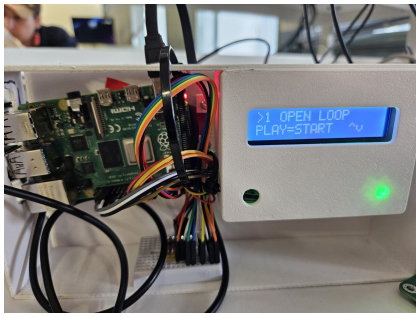
(a) Oct 14: U10, the top INA239 behind the reversed isolator.



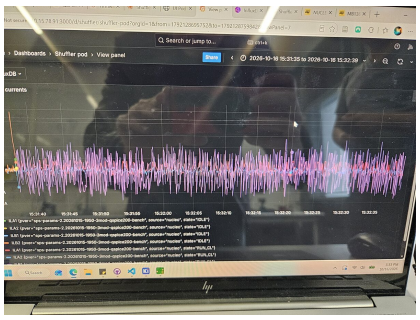
(b) Oct 14: ERUC23 coupled inductor rework.



(c) Oct 12: first bench runs.

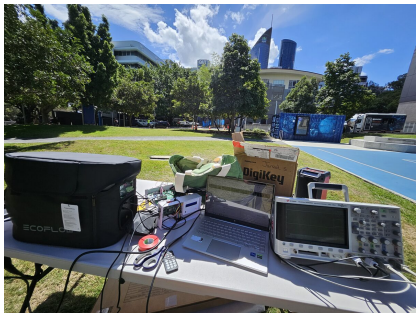


(d) Oct 16: pod LCD during the bench test.



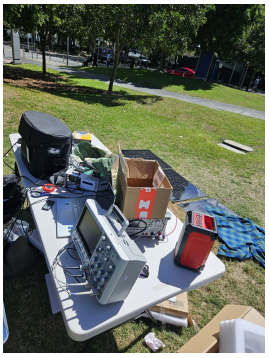
(e) Oct 16: Grafana live during closed loop.

(f) Oct 14: late-night bring-up.



(g) Oct 20: carpark set-up.

(h) Oct 20: it works in the sun.



(i) Oct 20: the string.

Figure 111: Build and test photographs (2 of 2). Videos of the bench runs are in `0_Documentation/Photos`.